Circuit tools and semiconductor physics

Practice questions · Day 01 · Sat 24 Oct · pack E1

5. Practice

From your course files

WS1 Q1 (course question). When there are more *electrons* than *protons* in a piece of wire, we say it has a ———— charge.

a) Negative
b) Positive
c) Neutral
d) Not enough information to answer

WS1 Q2 (course question). In Figure 1, i_1 is positive if positive charges are moving to the ———— or *negative* charges are moving to the ————.

a) Left, Right
b) Right, Right
c) Left, Left
d) Right, Left



WS1 Figure 1: conventional current direction (from the workshop sheet)

WS1 Q3 (course question). If our assumption about the higher potential end is wrong, the potential difference (voltage) will turn out to be ————.

a) Positive
b) Zero
c) Negative
d) Not enough information to answer.

WS1 Q4 (course question). Note the Passive sign convention for Ohm's law, Current flows from the higher potential to the lower potential end. Therefore, charge carriers ———— potential energy by moving from the ———— end of the voltage to the ———— end. The energy is ———— by the resistor.

a) Lose, Positive, Negative, Absorbed
b) Lose, Negative, Positive, Absorbed
c) Absorb, Positive, Negative, Released
d) Absorb, Negative, Positive, Released

WS1 Q5 (course question). In a resistor with current $i(t)$, the power is ————.

a) Always negative
b) Always positive
c) Negative or positive
d) Negative or positive or zero

WS1 Q6 (course question). In a capacitor, if the voltage is constant, current is zero, and the capacitor is ———— to DC.

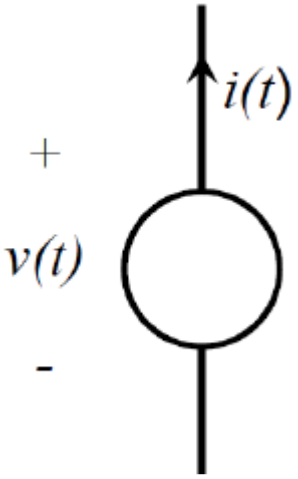
a) Short circuit
b) Open circuit
c) Constant voltage source
d) Either open or short circuit depending on the voltage magnitude

WS1 Q7 (course question). In an inductor, if the current is constant, voltage is zero, and the inductor is ———— to DC.

a) Short circuit
b) Open circuit
c) Constant current source
d) Either open or short circuit depending on the voltage magnitude

WS1 Q8 (course question). In the voltage source shown in Figure 5, current is assumed to flow from ———— potential end to the ———— potential end. If the source is supplying energy, $i(t)$ will be ————.

a) Low, High, Positive
b) High, Low, Negative
c) Low, High, Negative
d) High, Low, Positive

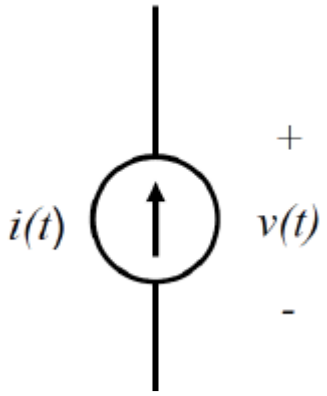


WS1 Figure 5: independent voltage source symbol, active notation (from the workshop sheet)

WS1 Q9 (course question). In the current source shown in Figure 6, if the source is supplying energy, $v(t)$ will be ————. Positive current leaves ———— end of the voltage.

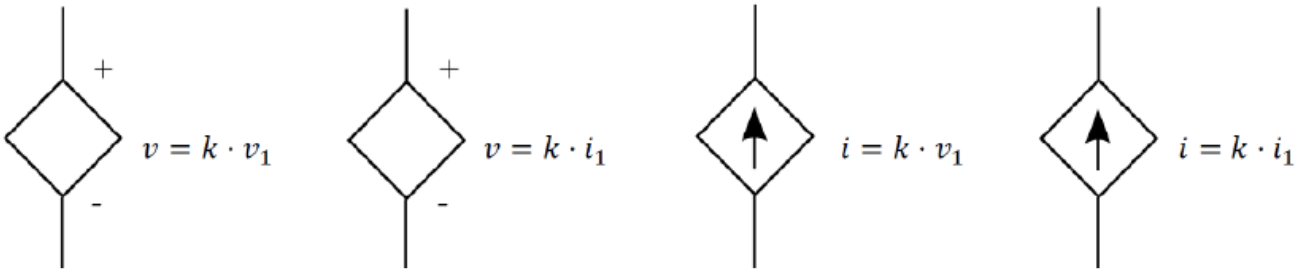
a) Positive, Negative
b) Positive, Positive

c) Negative, Positive
d) Negative, Negative



WS1 Figure 6: independent current source symbol, active notation (from the workshop sheet)

WS1 Q10 (course question). Identify the type of dependent/controlled sources from left to right in Figure 7.



WS1 Figure 7: dependent/controlled source symbols (from the workshop sheet)

WS1 Q11 (course question). What does short circuit mean? Select all correct options.
a) Zero potential difference
b) zero current
c) 0 Ω resistance
d) 0 S conductance
e) infinite resistance
f) infinite conductance

WS1 Q12 (course question). Which statement(s) is(are) correct about Passive Elements? Select all correct options.

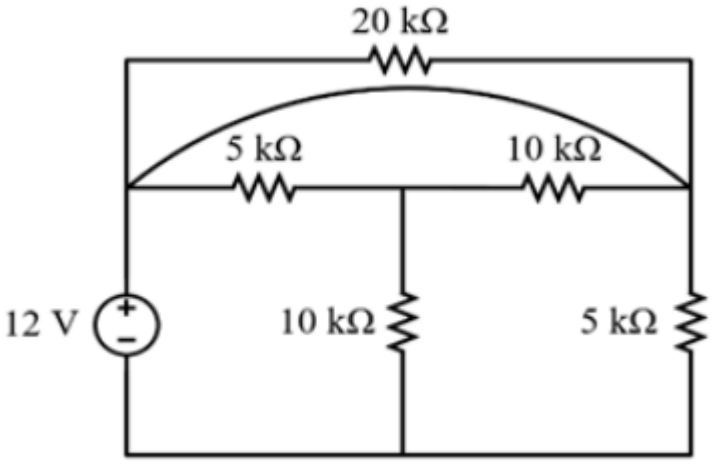
a) Deliver power when $p(t) = v(t) \times i(t) < 0$

b) Deliver power when $p(t) = v(t) \times i(t) > 0$

c) Absorb power when $p(t) = v(t) \times i(t) > 0$

d) Absorb power when $p(t) = v(t) \times i(t) < 0$

WS1 Q13 (course question). How many nodes and branches are there in the circuit given below:



WS1 Q13, Figure 9 (from the workshop sheet)

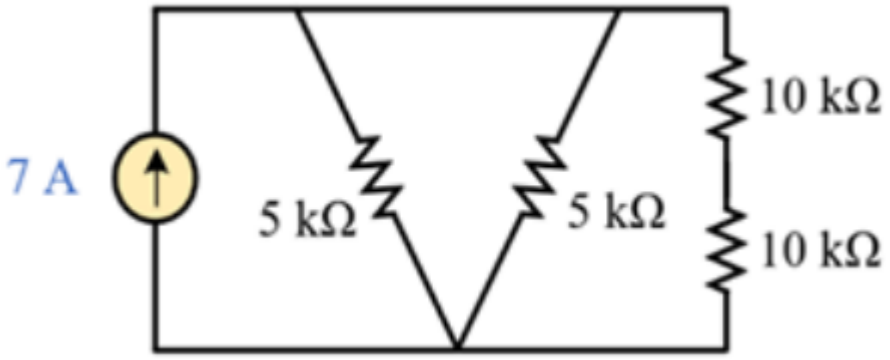
a) $N = 4,\ B = 6$

b) $N = 4,\ B = 7$

c) $N = 3,\ B = 5$

d) $N = 3,\ B = 7$

WS1 Q14 (course question). How many loops and meshes are there in the circuit given below:



WS1 Q14, Figure 10 (from the workshop sheet)

a) Loop=6, Mesh=3
b) Loop=3, Mesh=6
c) Loop=6, Mesh=6
d) Loop=6, Mesh=3

WS1 Q15 (course question). Determine currents I_1 and I_2 in the circuit shown in Figure 11.

WS1 Q15, Figure 11 (from the workshop sheet)

a) $I_1 = 0$ A, and $I_2 = 6$ A
b) $I_1 = 6$ A, and $I_2 = 3$ A
c) $I_1 = 0$ A, and $I_2 = 3$ A
d) $I_1 = 3$ A, and $I_2 = 3$ A

WS1 Q16 (course question). Which option shows the KVL equation for the loop shown in the circuit given in Figure 12?

WS1 Q16, Figure 12 (from the workshop sheet)

a) $-v_1 + v_4 - v_3 + v_2 = 0$
b) $-v_1 + v_4 - v_3 + v_2 + v_5 = 0$
c) $-v_1 + v_4 - v_3 + v_2 - v_5 = 0$
d) $v_1 + v_4 + v_3 + v_2 + v_5 = 0$

WS1 Q17 (course question). What is the equivalent voltage source in the circuit given in Figure 15? Consider the polarity of the 9 V source as the reference for the equivalent source

WS1 Q17, Figure 15 (from the workshop sheet)

a) 6 V
b) −16 V
c) 16 V
d) −6 V

WS1 Q18 (course question). In the circuit shown in Figure 18, what is the value of current I_1?

WS1 Q18–Q19, Figure 18 (from the workshop sheet)

a) 1.2 A
b) 4.8 A
c) 3.6 A
d) 4.5 A

WS1 Q19 (course question). In the circuit shown in Figure 18, what is the voltage value V_2?
a) 9.6 V

b) 2.4 V
c) 7.2 V
d) 9 V

WS1 Q20 (course question). Using Superposition technique, calculate I_x in the circuit shown in Figure 19.

WS1 Q20, Figure 19 (from the workshop sheet)

a) 0.8 A
b) 0.96 A
c) 0.66 A
d) 0.14 A

WS1 Q21 (course question). What is the Thevenin equivalent of the circuit seen from the AB terminals in the circuit shown in Figure 21?

WS1 Q21–Q22, Figure 21 (from the workshop sheet)

a) $V_{AB} = 45$ V, and $R_{AB} = 3\ \Omega$
b) $V_{AB} = 27$ V, and $R_{AB} = 3\ \Omega$
c) $V_{AB} = 27$ V, and $R_{AB} = 9\ \Omega$
d) $V_{AB} = 45$ V, and $R_{AB} = 9\ \Omega$

WS1 Q22 (course question). What is the Norton equivalent of the circuit seen from the AB terminals in the circuit shown in Figure 21?

a) $I_{AB} = 12$ A, and $R_{AB} = 9\ \Omega$
b) $I_{AB} = -15$ A, and $R_{AB} = 9\ \Omega$
c) $I_{AB} = -12$ A, and $R_{AB} = 3\ \Omega$
d) $I_{AB} = 15$ A, and $R_{AB} = 3\ \Omega$

WS1 Q23 (course question). Using source transformation, find the current I_x in the circuit shown in Figure 23?

WS1 Q23, Figure 23 (from the workshop sheet)

a) $I_x = 0.2319$ mA
b) $I_x = 0.1923$ mA
c) $I_x = 0.3129$ mA
d) $I_x = 0.9321$ mA

WS1 Q24 (course question). Using Nodal Analysis, determine voltage V_c in the circuit shown in Figure 24?

WS1 Q24–Q25, Figure 24 (from the workshop sheet; values are conductances in ℧ = S)

a) $V_c = 0.3556$ V
b) $V_c = 0.244$ V
c) $V_c = 0.2$ V
d) $V_c = 0.1$ V

WS1 Q25 (course question). Using Mesh Analysis, determine voltage V_{AB} in the circuit shown in Figure 24?
a) $V_c = 0.3556$ V
b) $V_c = 0.244$ V
c) $V_c = 0.2$ V
d) $V_c = 0.1$ V

WS2 Q1 (course question). Answer the following true or false questions by discussing them in your group: (please note that IX and X are related to diode):
I. At absolute zero temperature, the semiconductor behaves as a conductor.
II. No free electron or hole exists at absolute zero temperature.
III. At room temperature with thermal agitation, we have enough electrons and holes in a silicon wafer to have conductance.
IV. Holes are the majority charge carriers in a p-type semiconductor.
V. A hole is defined as the absence of an electron.
VI. A pure silicon has four valence electrons.
VII. A doped semiconductor is electrically charged at room temperature.
VIII. Diffusion current is the electric current caused by the charge carriers when they move from a higher concentration region to a lower concentration region.
IX. In an unbiased PN junction, the drift/saturation and diffusion currents are equal.
X. In the boundary of the two adjacent layers in a PN junction, we have the depletion layer and barrier voltage.

WS2 Q2 (course question). The cylinder shown here in Fig. 1 is made of copper, whose conductivity is $\sigma = 5.65 \times 10^5\ (\Omega\text{-cm})^{-1}$. Calculate the resistance of the cylinder when it conducts a current lengthwise.

WS2 Fig. 1: the cylinder, 8 mm long and 1 mm in diameter (from the workshop sheet)

WS2 Q3 (course question). The cylinder is now made of intrinsic silicon,
(a) Calculate its resistance at room temperature
(b) Compute the average drift velocities of holes and electrons when 5 volts is applied to this resistor. At room T, $n_i = 1.5 \times 10^{10}$ cm⁻³, $\mu_n = 1350$ cm²/Vs, $\mu_p = 500$ cm²/Vs.

WS2 Q4 (course question). The intrinsic concentration, n_i, of an n-type semiconductor is 1.5×10^{10} cm⁻³ at room temperature.
(a) Estimate the concentrations of free electrons and holes in this material at room temperature if $N_D = 10^{14}$ cm⁻³.
(b) Find the room temperature conductivity of the doped material if the electron and hole mobilities are 1350 and 500 cm²/Vs, respectively.

WS2 Q5 (course question). The intrinsic concentration of acceptor-type silicon has values of 1.5×10^{10} cm⁻³ and 10^{13} cm⁻³ at 300 K and 405 K, respectively, and the doping concentration is 2×10^{15} cm⁻³.
(a) Estimate the electron and hole concentrations at both temperatures.
(b) Use the exact equations to determine the concentrations at both temperatures.

WS2 Q6 (course question). At a temperature of 50 °C, for intrinsic silicon:
(a) Calculate the intrinsic carrier concentration n_i.
(b) Calculate the concentration of current carrying electrons (n) and holes (p).
(c) Calculate the resistivity of the silicon at this temperature.
(d) Calculate the resistance of a piece of silicon where the current is travelling a length of 5 µm over a cross-sectional area (perpendicular to the flow of current) of 3 µm × 3 µm.

(e) What is the difference (%) in resistance between what you calculated in (d) and the resistance of the silicon at room temperature (300 K)?

WS2 Q7 (course question). Find the resistivity and resistance according to the same constraints of Question 6 if the silicon is now doped with donors to a concentration of 5×10^{15} cm^{-3}.

PT Q1 (practice test, 5 marks). Circle the correct answer in the following multiple-choice questions (I to V).

I. When a PN junction is formed, why does a depletion region develop near the junction?

a) Electrons and holes recombine near the junction, leaving behind ionized dopants

b) The semiconductor becomes electrically insulating everywhere

c) All free electrons move from the N-region into the P-region permanently

d) The applied voltage removes all charge carriers from the semiconductor

II. The small-signal or dynamic resistance of a diode is related to:

a) The slope of the I–V characteristic at the operating point

b) The total length of the diode leads

c) The reverse breakdown voltage only

d) The DC supply resistance only

III. The main advantage of full-wave rectification over half-wave rectification is that full-wave rectification:

a) Uses only the negative half of the input waveform

b) Uses both half-cycles of the input waveform

c) Completely eliminates ripple without a capacitor

d) Requires no semiconductor devices

IV. Why is the base region of a BJT made relatively thin?

a) To allow most injected carriers to cross the base and reach the collector

b) To prevent carriers from reaching the collector

c) To increase the base resistance as much as possible

d) To eliminate the depletion region at the collector-base junction

V. Why are DC and AC analyses of a transistor amplifier often performed separately?

a) A transistor can operate with either DC or AC but never both

b) DC analysis establishes the operating point, while AC analysis examines small variations around that point

c) AC analysis determines the transistor's doping concentration
d) DC analysis is only needed when the transistor is damaged

New questions

N1. [5 marks] Circle the correct answer (I to V).

I. An n-type Si sample at 300 K has $N_D = 10^{16}\ \mathrm{cm}^{-3}$ ($n_i = 1.5 \times 10^{10}$). The hole concentration is:

a. 1.5×10^{10} b) 2.25×10^{4} c) 10^{16} d) 2.25×10^{20} cm⁻³

II. In a pn junction with no external bias:

a. No carriers cross the junction
b. The diffusion current equals the drift current, so the net current is zero
c. Only the drift current flows
d. The depletion region contains free electrons and holes

III. A junction has $N_A = 10^{17}$ and $N_D = 10^{15}$ cm⁻³. The depletion region:

a. Extends equally into both sides
b. Extends mostly into the p side
c. Extends mostly into the n side, with $x_n/x_p = 100$
d. Exists only on the p side

IV. A divider bias has R_1 from V_{CC} to the base node and R_2 from the base node to ground. The Thevenin resistance seen from the base is:

a. $R_1 + R_2$ b) $R_1 \parallel R_2$ c) R_2 only d) $R_1 R_2 / V_{CC}$

V. For an ideal op-amp with negative feedback, which pair is correct?

a. $v_+ = v_-$ and $i_+ = i_- = 0$
b. $v_o = 0$ and $i_+ = i_-$
c. $v_+ = 0$ and $v_- = v_o$
d. $i_+ = i_- = \infty$

N2. [6 marks] A p-type Si bar at 300 K has $N_A = 4 \times 10^{15}\ \text{cm}^{-3}$. Use $n_i = 1.5 \times 10^{10}$ cm⁻³, $\mu_n = 1350$ and $\mu_p = 500$ cm²/Vs. The bar is 2 mm long, with a rectangular cross-section of 0.5 mm × 0.2 mm. A voltage of 3 V is applied end to end.

a. Find the electron and hole concentrations. [1]
b. Find σ and ρ. Show the size of the minority-carrier term. [2]
c. Find the resistance and the current. [2]
d. Find the hole drift velocity. [1]

N3. [7 marks] (chained) A bias network has $V_{CC} = 15$ V, $R_1 = 68$ kΩ from V_{CC} to node B, and $R_2 = 12$ kΩ from node B to ground.

a. Find the Thevenin equivalent seen from node B. [3]
b. Node B is now connected to a branch made of an ideal 0.7 V drop in series with 101 kΩ to ground. (This is how a BJT base with $\beta = 100$ and $R_E = 1$ kΩ looks from the base: $(\beta + 1)R_E$.) Find the branch current I_B and the new voltage at B. [3]
c. A student ignores R_{Th} and uses $V_B = V_{Th}$. What percentage error does this give in I_B? [1]

Circuit for N3: the divider, with the 0.7 V + 101 kΩ branch connected in part (b).

N4. [7 marks] (chained) Node X is connected to +12 V through 20 kΩ, to −6 V through 30 kΩ, and to ground through 15 kΩ.

a. Find R_{Th} seen from X. [1]
b. Find V_{Th} by nodal analysis. Check your answer by superposition. [3]
c. A 5 kΩ load is connected from X to ground. Find V_X and the load current. [2]
d. Give the Norton equivalent. [1]

Circuit for N4: node X with three branches, and the 5 kΩ load added in part (c).

N5. [6 marks] A 12 V source connects through 2 kΩ to node 1. Node 1 has 4 kΩ to ground and 3 kΩ to node 2. Node 2 has 6 kΩ to ground. An independent 1 mA current source pushes current from ground into node 2.

a. Write the nodal equations and find V_1 and V_2. [4]
b. Find the power delivered by the current source, and state whether it supplies or absorbs power. [1]
c. Find the Thevenin resistance seen between node 2 and ground. [1]

Circuit for N5.

N6. [8 marks] (hard, chained) A lightly doped n-type Si sample has $N_D = 10^{13}\ \text{cm}^{-3}$. Use $B = 7.3 \times 10^{15}\ \text{cm}^{-3}\text{K}^{-3/2}$, $E_g = 1.12$ eV, $k = 8.62 \times 10^{-5}$ eV/K, $n_i(300\ \text{K}) = 1.5 \times 10^{10}$, $\mu_n = 1350$ and $\mu_p = 500$ cm²/Vs (assume constant).

a. Find n, p and σ at 300 K. [2]
b. Find n_i at 450 K. [1]
c. A student uses $n \approx N_D$ and $p = n_i^2/N_D$ at 450 K. Show why this is wrong. [1]
d. Find the correct n and p at 450 K. [2]
e. Find σ at 450 K, its ratio to the 300 K value, and V_T at 450 K. [2]

6. Quick recall check

1. At 27 °C, what is V_T?
2. Si at 300 K: what is n_i^2, and what is p in n-type material with $N_D = 10^{15}$?
3. When finding R_{Th}, what replaces a voltage source, and what replaces a current source?
4. Which side does the depletion region extend into more? Give the ratio formula.

5. For a divider bias (R_1 to V_{CC}, R_2 to ground), write V_{Th} and R_{Th}.

Diode models

Practice questions · Day 03 · Mon 26 Oct · pack E2

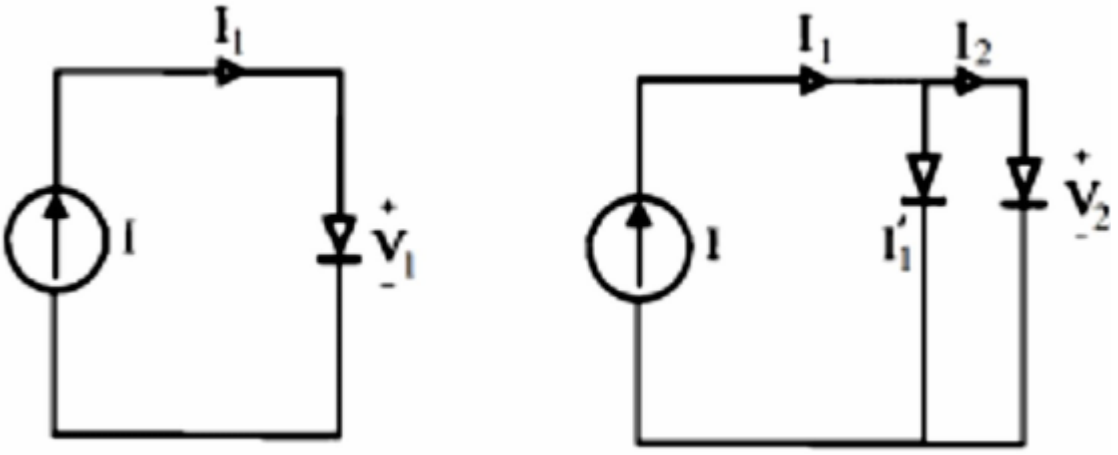
5. Practice

From your course files

WS2 Q8 (course question). A diode has a reverse saturation current of 0.2 pA, $n = 1$ and operates at room temperature. Find the voltage drop at the below currents:
i. 1 mA
ii. 10 mA
iii. 100 mA
iv. Explain how much the diode voltage drop changes for every order of magnitude increase in current. Discuss it in your group.

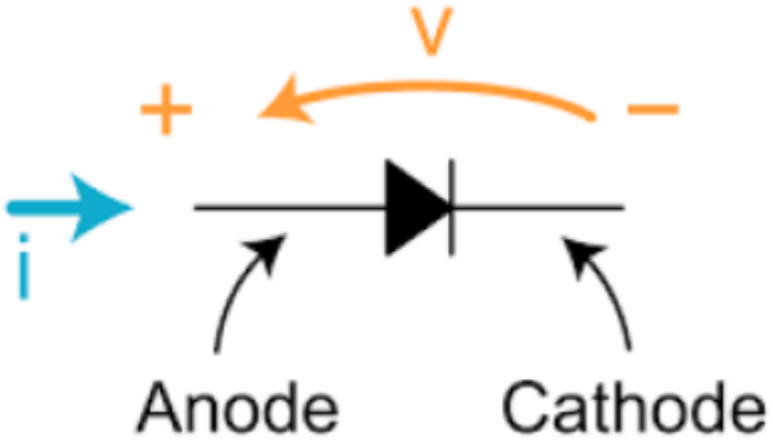
WS2 Q9 (course question). For a diode in the forward bias region at room temperature, we know that the current is 1 mA and 10 mA at 0.7 V and 0.8 V, respectively. What are the value of n and I_s for this diode?

WS2 Q10 (course question). The diode in the circuit below is energised by a current source I. If we add another similar diode parallel to the first one, how does the voltage change across the parallel diodes? Discuss it in your group.



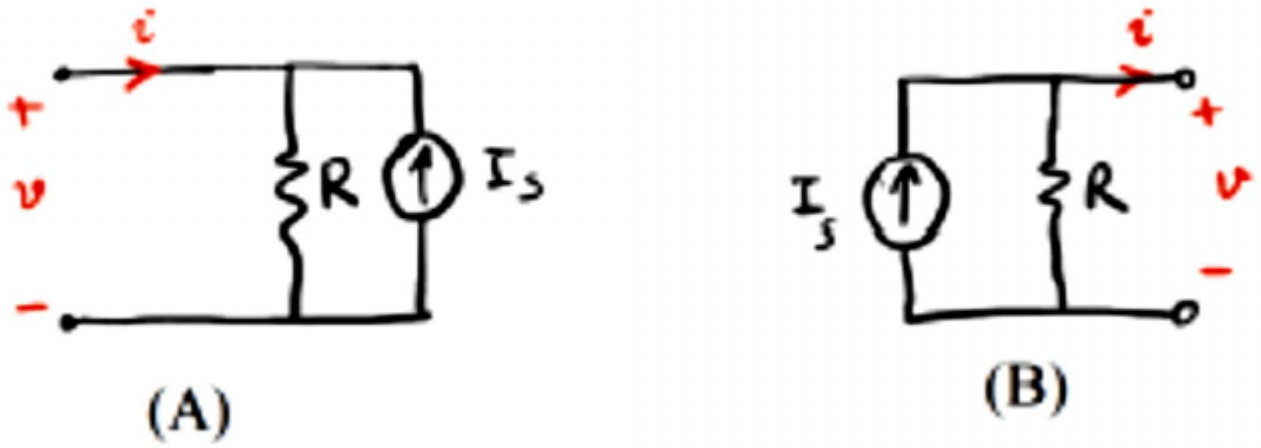
WS2 Fig. 2 (from the workshop sheet)

WS2 Q11 (course question). Consider a forward-biased diode. The end of the diode, which has the higher voltage, corresponds to [A] and its type is [B]. It is doped by [C]. The majority carriers are [D] and minority carriers are [E].



WS2 Fig. 3: diode symbol (from the workshop sheet)

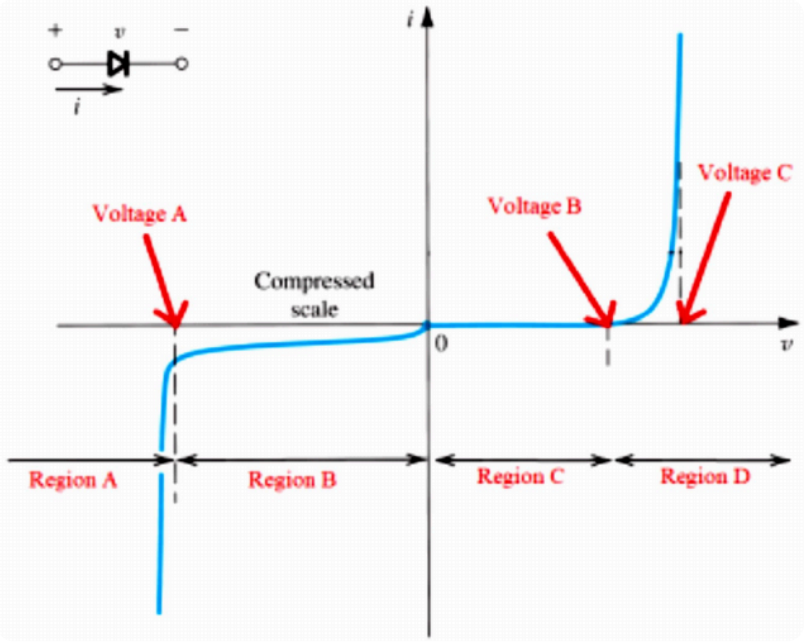
WS2 Q12 (course question). Do you expect a linear relationship between voltage and current in the two circuits below? Why? Discuss it in your group.



WS2 Fig. 4: circuits (A) and (B) (from the workshop sheet)

WS2 Q13 (course question). Now, plot the current (i)-voltage (v) characteristics curves for each of the two circuits in Fig. 4 based on the current direction and voltage polarity on the circuit and confirm your answer in Question 12.

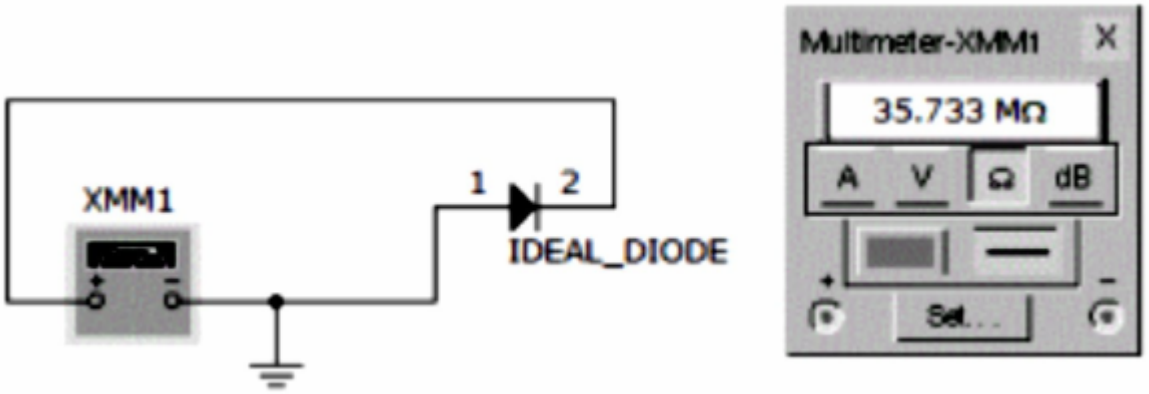
WS2 Q14 (course question). In the Fig. 5, full diode operational regions are shown according to the current direction and voltage polarities in the figure. Identify the regions and voltages mentioned in the figure. Discuss the diode's operation in each region in your group.



WS2 Fig. 5: diode operating regions (from the workshop sheet)

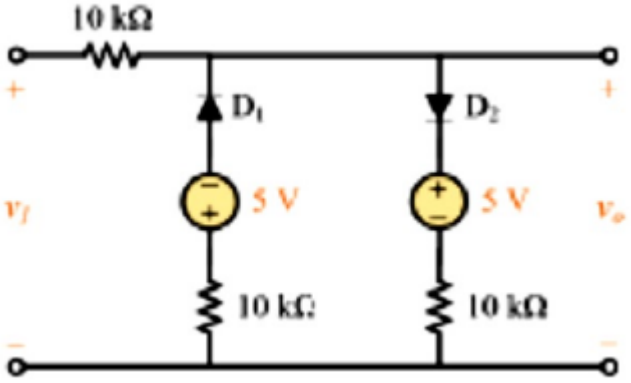
WS2 Q15 (course question). Considering the diode's operational regions in Question 14, we can linearize the diode's operation using the piece-wise linearization technique. How many pieces do you think we need? Discuss it in your group. *Note: Piecewise linear curve is a connected sequence of line segments. Piecewise linear function is a function whose domain can be decomposed into pieces on which the function is linear.*

WS2 Q16 (course question). In the circuit shown below, an ideal diode is connected to a multimeter. The device shows 35.733 MΩ. How do you explain this value? Is there anything wrong with the diode or our circuit? What do you expect to see if we rotate the diode connection? Discuss it in your group.



WS2 Fig. 6 (from the workshop sheet)

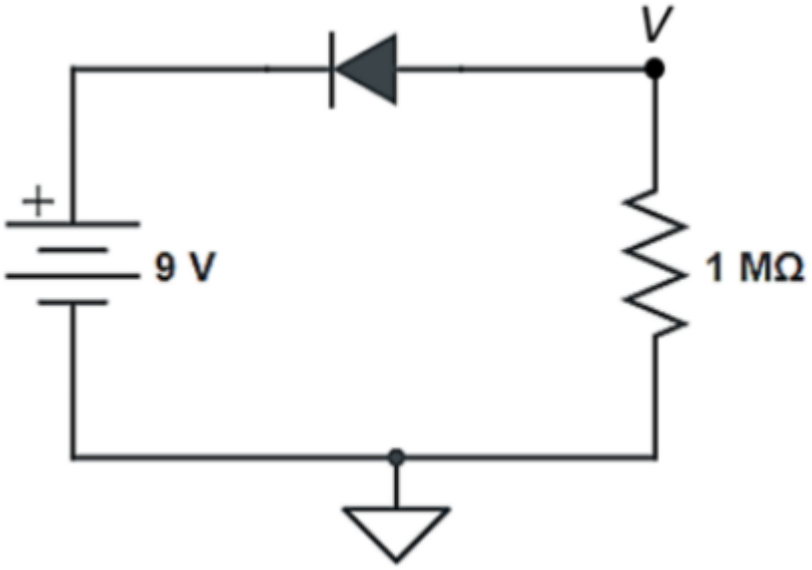
WS2 Q17 (course question). In the circuit below, replace the two diodes with their equivalent circuit model mentioned below in forward biasing (conducting) mode:
1. Ideal model
2. Voltage drop model, V_D
3. Voltage drop with resistor model, $V_D + R$



WS2 Fig. 8 (from the workshop sheet)

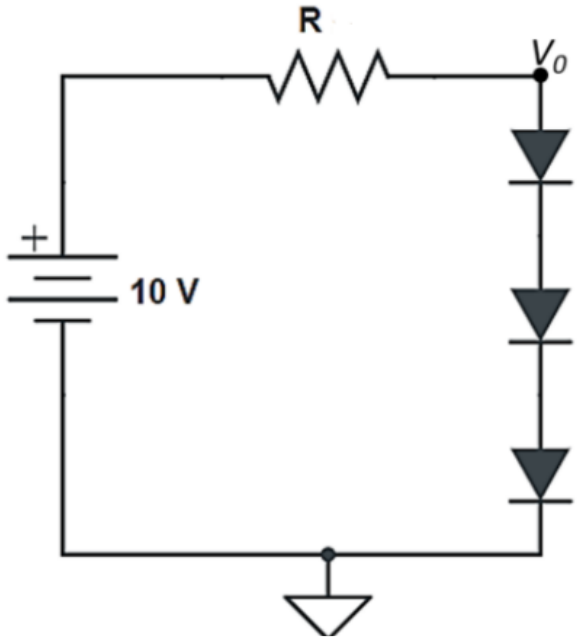
WS3 Q1 (course question). A silicon diode at 25 °C has a reverse saturation current of 1 µA. At what applied voltage will the diode current be 100 mA? Assume $n = 1$.

WS3 Q2 (course question). The diode circuit shown in Fig. 1 is a large, high-current device whose reverse current is reasonably independent of voltage, and doubles for every 10 °C rise in temperature. If $V = 1$ V at 20 °C, find the value of V at 40 °C and 0 °C.



WS3 Fig. 1 (from the workshop sheet)

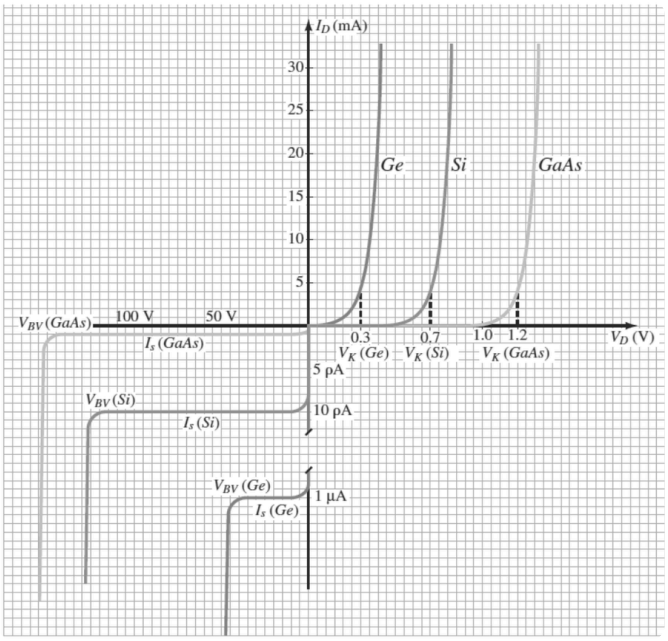
WS3 Q3 (course question). Design the circuit shown in Fig. 2 to provide an output voltage of 2.4 V. Assume that the diodes available have 0.7 V drop at 1 mA, and that such voltage varies by 0.1 V for every decade change in current.



WS3 Fig. 2 (from the workshop sheet)

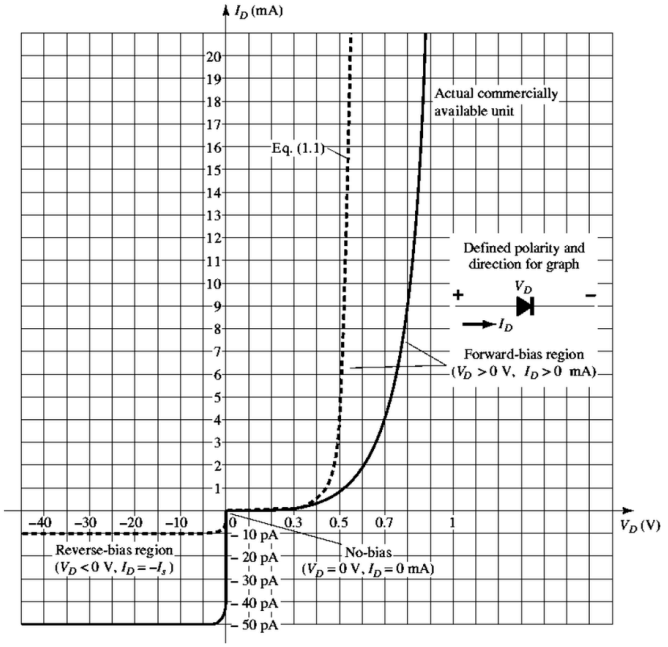
WS3 Q4 (course question). Determine the thermal voltage for a diode at a temperature of 20 °C. For the same diode, find the diode current if $I_s = 40$ nA, $n = 2$, and the applied bias voltage is 0.5 V.

WS3 Q5 (course question). Find the piecewise-linear equivalent circuit for the germanium and gallium arsenide diodes of Fig. 3.



WS3 Fig. 3 (from the workshop sheet)

WS3 Q6 (course question). The characteristics of a commercially available diode are shown in Fig. 4 below. Determine: (a) the static or dc resistance of the diode at a forward current of 4 mA. (b) the dynamic (ac) resistance at a forward current of 10 mA.



WS3 Fig. 4 (from the workshop sheet)

WS3 Q7 (course question). Determine the level of V_o for each network of Fig. 5 below:

WS3 Fig. 5 (a) and (b) (from the workshop sheet)

WS3 Q8 (course question). In the circuit shown in Fig. 6, find the voltages V_A and V_B using ideal diode model. (Assume $V_{threshold} = 0.7$ V)

WS3 Fig. 6 (from the workshop sheet)

PT Q2 (practice test, 5 marks). Consider a silicon diode with its exponential model and $n = 1.5$. Find the change in the voltage across it when the diode operates at 27 °C and the current changes from 0.1 mA to 10 mA.

PT Q3 (practice test, 8 marks). Consider the circuit below, which consists of the series combination of three identical diodes. Using the third piecewise linear model for the diodes with $V_D = 0.6$ V and $r_D = 0.20$ kΩ, answer the following questions. Assume that all diodes are **conducting**.
(a) Redraw the entire circuit using the third piecewise linear equivalent circuit model of diodes. You do not need to simplify the circuit. [2 marks]
(b) Verify whether all three diodes are **conducting** under the given circuit conditions. [4 marks]
(c) Find the output voltage, V_o, as shown on the circuit. [2 marks]

PT Q3 figure (from the practice-test paper)

New questions

N1 [4 marks] A diode with $n = 1.8$ operates at 50 °C and conducts 2 mA at 0.62 V.

a. Find V_T. [1]
b. Find V_D when the current falls to 0.2 mA. [2]
c. Find the small-signal resistance r_d at 2 mA. [1]

N2 [6 marks] At 27 °C a diode carries 0.5 mA at 0.60 V and 5 mA at 0.69 V.

a. Find n and I_S. [3]
b. Using your answer to (a), predict V_D at 20 mA. [1]
c. At 5 mA, find the dc resistance R_D and the small-signal resistance r_d, and say which one a 2 mV ac signal would see. [2]

N3 [4 marks] A silicon diode drops 0.65 V at 1 mA and has a reverse current of 10 nA, both at 25 °C. Use the course temperature rules.

a. Estimate V_D at 1 mA when the diode is at 75 °C. [1]
b. Find the reverse current at 75 °C and at 0 °C. [2]
c. The small-signal resistance at 1 mA ($n = 1$) changes between 25 °C and 75 °C. Give both values. [1]

N4 [7 marks] (multi-diode, iterate) A +10 V supply feeds a 1 kΩ resistor to node A. From A, diode D1 (anode at A) goes to node B, and B has a 2 kΩ resistor to ground. Also from A, diode D2 (anode at A) goes through a 1 kΩ resistor to the + terminal of an 8 V source whose − terminal is grounded. Use the constant-drop model, $V_D = 0.7$ V. V_o is the voltage at B.

a. Assume both diodes are ON. Solve, and show that the assumption fails. [3]
b. Correct the assumption, verify it, and find V_o. [4]

Circuit for N4.

N5 [8 marks] (two diodes, two input polarities, chained) V_i (+ at the top) feeds $R = 1$ kΩ to node A. V_o is the voltage at A. Three branches run from A to ground:

- **Branch 1:** D1 with its anode at A, then a 2 V battery with + facing D1, then 0.5 kΩ to ground.
- **Branch 2:** D2 with its **cathode** at A, its anode connected through 1 kΩ to ground.
- **Branch 3:** a 4 kΩ resistor.

Both diodes use the 3rd model with $V_D = 0.7$ V and $r_D = 0.1$ kΩ.

a. $V_i = +6$ V: state your assumptions, find V_o and the diode current, and verify. [3]
b. $V_i = -6$ V: repeat. [3]
c. Using the OFF-OFF circuit, find the range of V_i over which both diodes are OFF. [2]

Circuit for N5: three branches from A to ground.

N6 [8 marks] (OR gate with unequal inputs: the near-miss) $V_1 = 5$ V feeds 1 kΩ, then D1 (anode on the input side), to node V_o. V_2 feeds 1 kΩ, then D2, to the same node. V_o has 4 kΩ to ground. Both diodes use the 3rd model with $V_D = 0.7$ V and $r_D = 0.1$ kΩ.

a. Take $V_2 = 4$ V. Assume both diodes are ON, solve, and test the assumption. [3]
b. Correct the assumption, verify, and find V_o. [3]
c. Find the smallest V_2 at which D2 begins to conduct. [1]
d. Find V_o when $V_2 = 5$ V. [1]

Circuit for N6: diode OR gate with unequal inputs.

6. Quick recall check

1. With $n = 1$ and $V_T = 25$ mV, how much does V_D rise per decade of current? And with $n = 2$?
2. Two identical diodes are in parallel and driven by a current source. Does the voltage rise or fall compared with one diode, and by roughly how much?
3. What does each of the three piecewise-linear models become when the diode is OFF?
4. Find r_d at $I_D = 13$ mA for $n = 1$ and $V_T = 26$ mV.
5. A Si diode leaks 5 nA at 30 °C. What does it leak at 50 °C, and which way does its forward drop move?

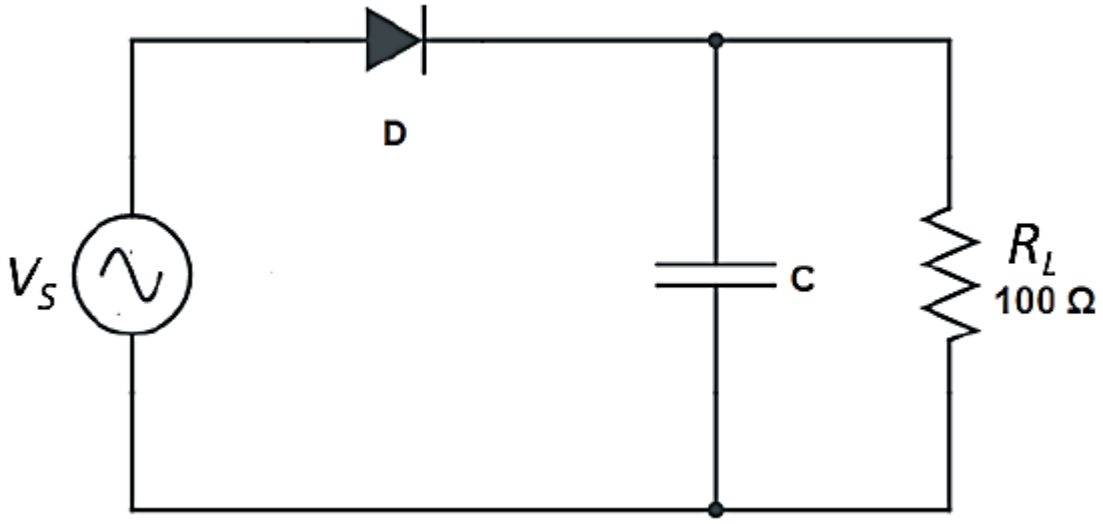
Rectifiers, Zener regulators, clippers

Practice questions · Day 05 · Wed 28 Oct · pack E3

5. Practice

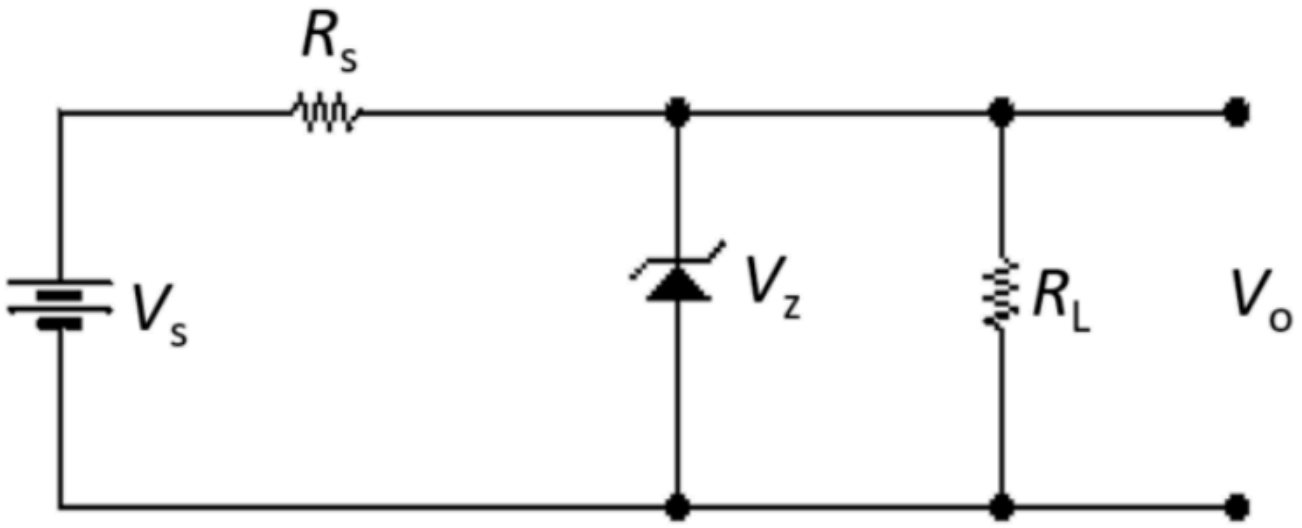
From your course files

WS3 Q9 (course question). Consider a half-wave rectifier with a filter capacitor C, where we have: $V_{S(rms)} = 12$ V, $f = 50$ Hz, $V_D = 0.8$ V, $R_L = 100\ \Omega$
a) Find the value of C which results in a ripple voltage V_{rpp} no larger than 1 V.
b) Find the following quantities: V_{Odc}: Output dc voltage, I_L: Load current, PIV: diode peak reverse voltage.



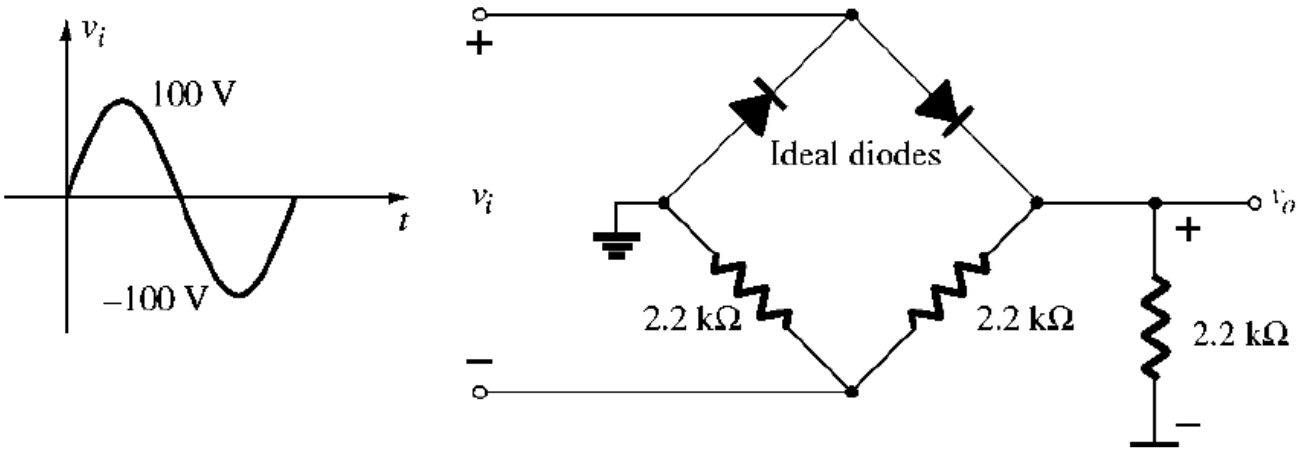
WS3 Fig. 7 (from the workshop sheet)

WS3 Q10 (course question). Consider the Zener diode circuit shows in Fig. 8 below, where: $V_{zk} = 10$ V and $r_z = 20\ \Omega$, $V_s = 15$ V, $R_s = 100\ \Omega$, and $R_L = 500\ \Omega$. Calculate V_O and I_s, then deduce I_L and I_Z. Calculate the power consumed in the Zener.



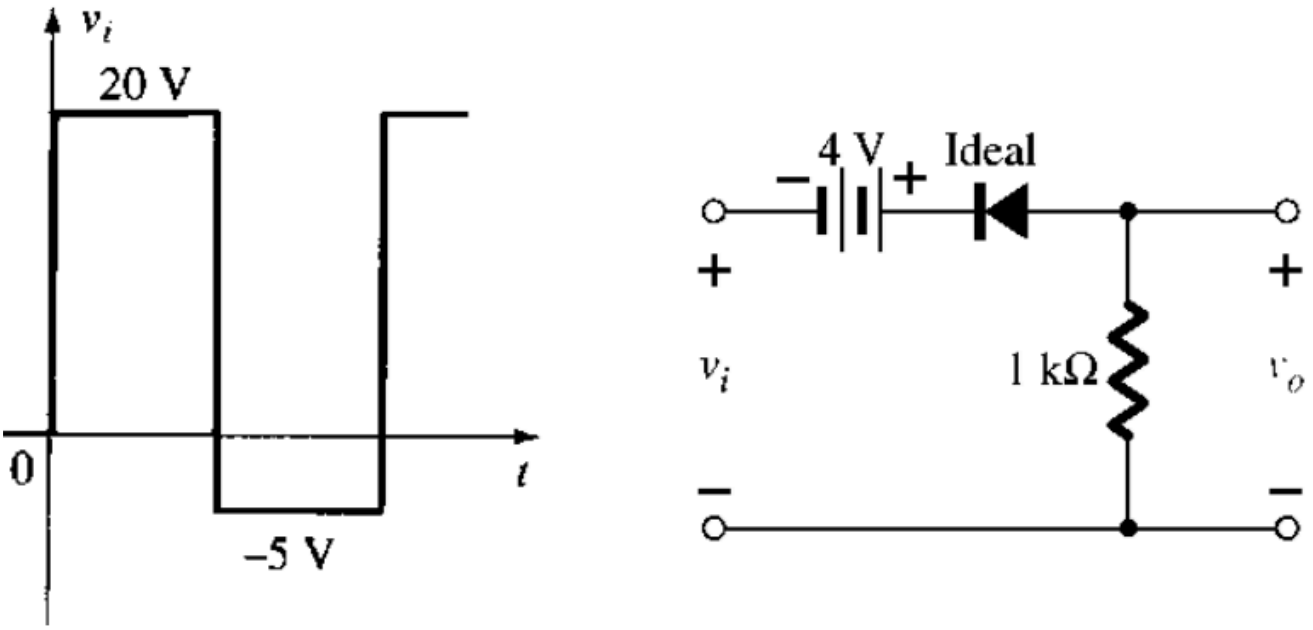
WS3 Fig. 8 (labelled “Fig. 4” on the sheet; from the workshop sheet)

WS3 Q11 (course question). Sketch v_o for the network of Fig. 9 and determine the dc voltage available.



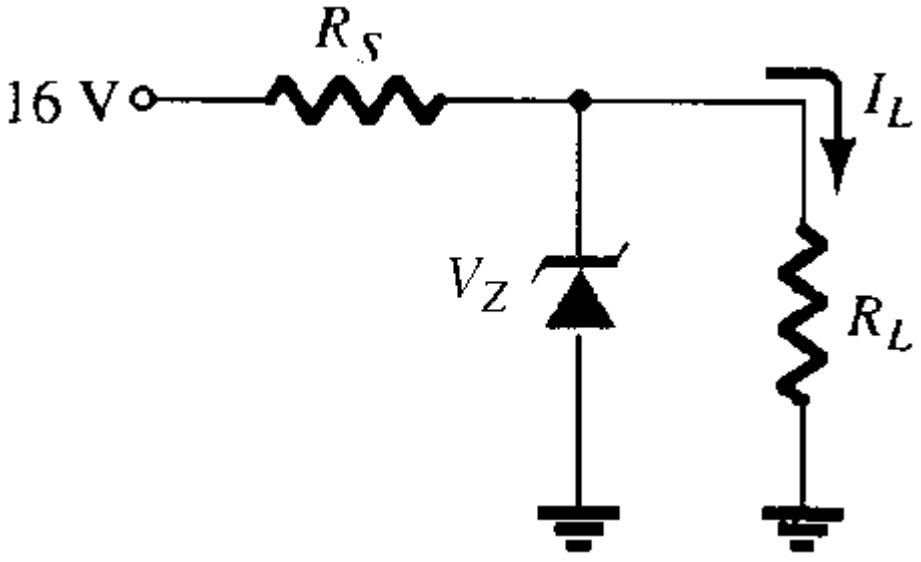
WS3 Fig. 9 (from the workshop sheet)

WS3 Q12 (course question). Determine v_o of the network of Fig. 10 for the input shown.



WS3 Fig. 10 (from the workshop sheet)

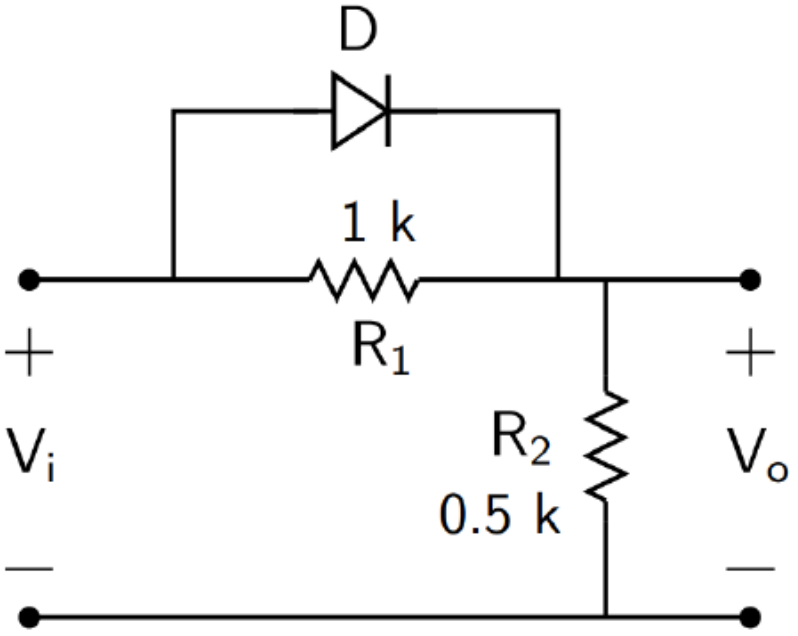
WS3 Q13 (course question). Design the network of Fig. 11 to maintain V_L at 12 V for a load variation (I_L) from 0 mA to 200 mA. That is, determine R_S and V_Z.



WS3 Fig. 11 (from the workshop sheet)

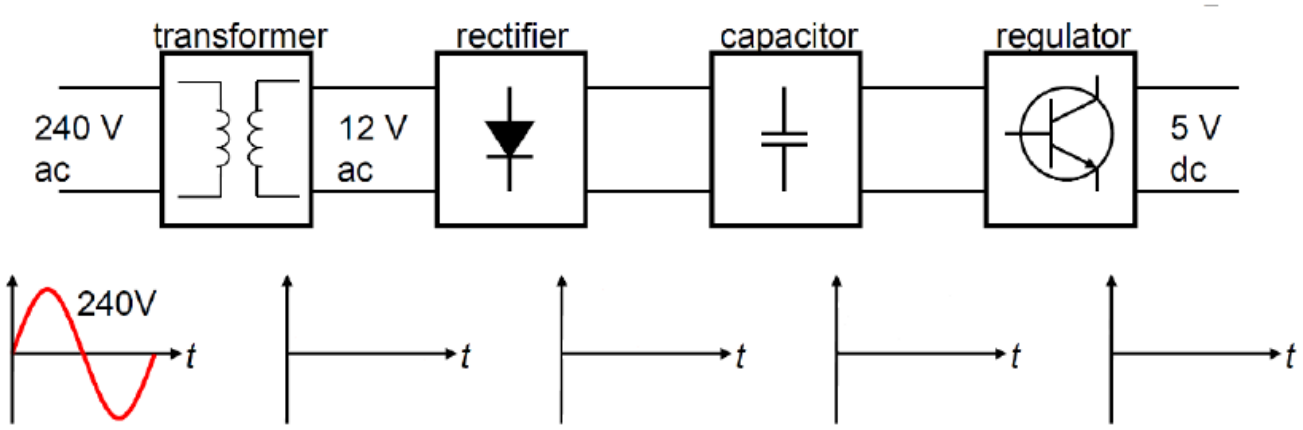
WS3 Q14 (course question). Consider a rectifier fed by a 60-Hz sinusoid having a peak value $V_P = 100$ V. Let the load resistance $R = 10$ kΩ. Find the value of the capacitance C that will result in a peak-to-peak ripple of 2 V. Also, calculate the fraction of the cycle during which the diode is conducting and the average and peak values of the diode current.

WS3 Q18 (course question). In the circuit below, the diode has a forward voltage drop $V_D = 0.7$ V. Plot V_o versus V_i for $-5 < V_i < 5$ V.



WS3 Fig. 15 (from the workshop sheet)

WS3 Q19 (course question). In the schematic below, we see the components of a typical linear DC power supply. Draw the signal at different stages of the circuit if we use half-wave or bridge rectifiers. Discuss the changes in the signal in your group. Assume that the forward voltage drop of the diode is zero.

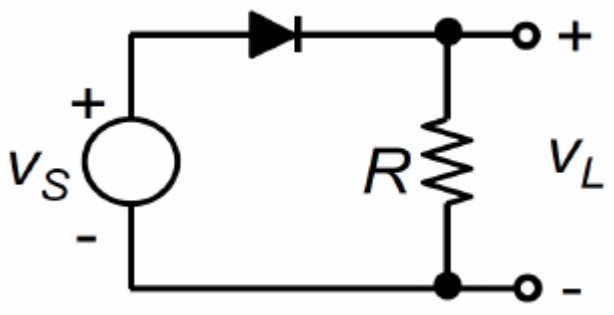


WS3 Fig. 16 (from the workshop sheet)

WS3 Q20 (course question). A diode having an internal resistance of $r_f = 20\ \Omega$ is used for half-wave rectification. If the diode forward voltage drop is zero, the supply voltage is $V_s = 50 \sin \omega t$ and load resistance is $R = 800\ \Omega$, find:

1. Maximum load current, I_m.
2. DC load current, I_{dc}.
3. Maximum voltage across the load, $V_{m,L}$.
4. DC voltage across the load, $V_{DC,L}$.
5. RMS value of the load current, I_{rms}.
6. RMS value of voltage across the load, $V_{L,rms}$.
7. DC voltage across the diode, $V_{D,DC}$.
8. Input AC power, $P_{AC,i}$.
9. Output DC power, $P_{DC,o}$.
10. The efficiency of the rectifier, η_e.

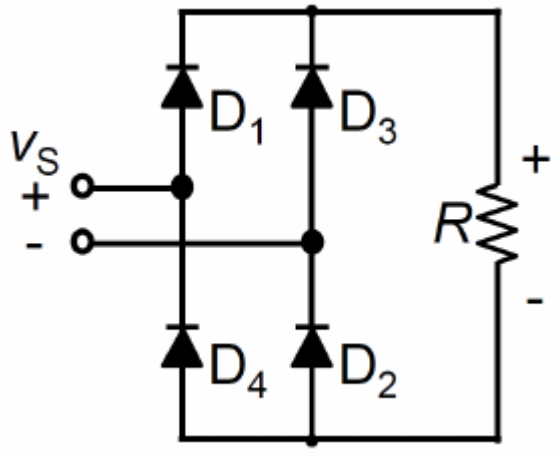
Equations that you may need: finding DC current from AC waveform, $I_{dc} = \dfrac{1}{2\pi} \displaystyle\int_0^{2\pi} I_m \sin(\omega t)\, d(\omega t)$; finding RMS current from AC waveform, $I_{RMS} = \sqrt{\dfrac{1}{2\pi} \displaystyle\int_0^{2\pi} [I_m \sin(\omega t)]^2 d(\omega t)}$.



WS3 Fig. 17 (from the workshop sheet)

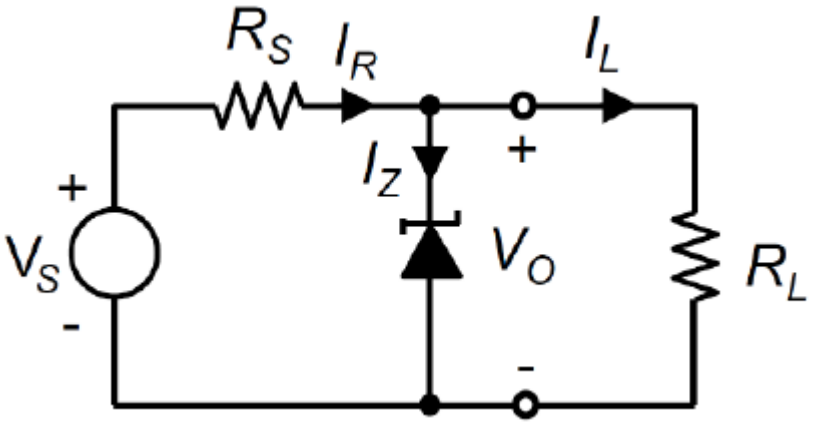
WS3 Q21 (course question). In the bridge type (full-wave rectifier) circuit shown below, the diodes have an internal resistance of $r_f = 20\ \Omega$. Assuming the diodes' forward bias voltage drop is zero, the supply voltage $V_s = 240\sin(314t)$ and the load is 460 Ω, find:

1. Maximum load current, I_m.
2. DC load current, I_{dc}.
3. Maximum voltage across the load, $V_{m,L}$.
4. DC voltage across the load, $V_{DC,L}$.
5. Peak inverse voltage (PIV) of each diode, V_{PIV}.
6. Output frequency, f_o.
7. Power dissipated by each diode, P_D.



WS3 Fig. 18 (from the workshop sheet)

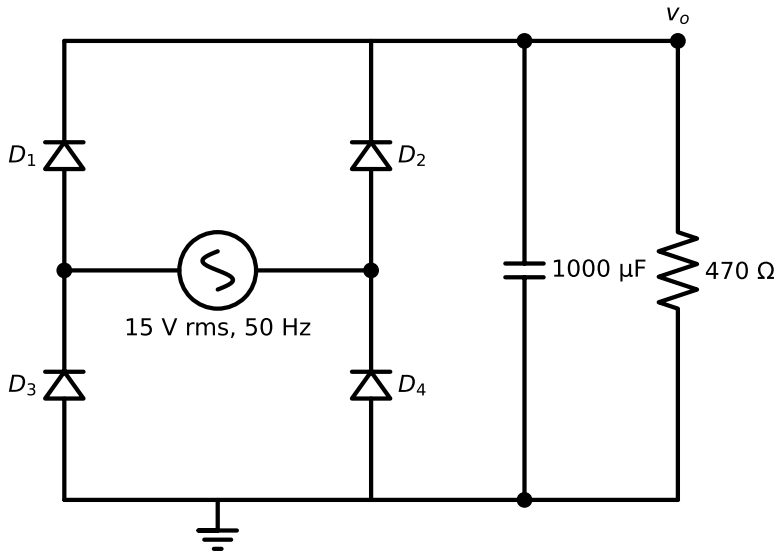
WS3 Q22 (course question). In the circuit below, we have a linear voltage regulator using a 4.8 V Zener diode. The Zener diode operates within a minimum and maximum current of $I_{Z,min} = 0.2$ mA and $I_{Z,max} = 7$ mA, respectively. Assuming that $R_s = 1$ kΩ and $V_s = 10 \pm 1$ V, obtain the maximum and minimum load, $R_{L,max}$ and $R_{L,min}$, in which the output voltage stays the same at 4.8 V.



WS3 Fig. 19 (from the workshop sheet)

New questions

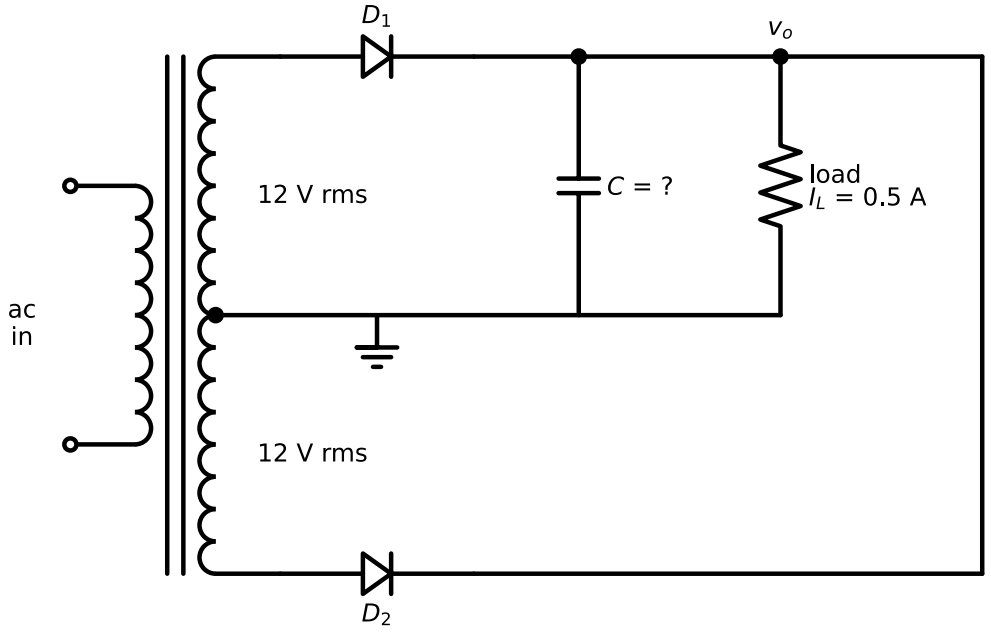
N1 (easy) [4 marks]. A bridge rectifier is fed from a 15 V rms, 50 Hz secondary. The silicon diodes each drop 0.7 V. The load is 470 Ω with a 1000 µF filter capacitor. Find (a) V_P [1], (b) the peak-to-peak ripple and the ripple frequency [2], (c) $V_{O(dc)}$ and the PIV of each diode [1].



Circuit for N1: bridge rectifier with filter.

N2 (medium) [6 marks]. A centre-tapped full-wave rectifier has 12 V rms on each half of the secondary at 50 Hz. Diode drop is 0.7 V. It must supply 0.5 A dc with no more than 1 V peak-to-peak ripple.

a. Choose C [2]. (b) Find $V_{O(dc)}$ and the PIV of each diode [2]. (c) Find the conduction angle and the peak diode current [2].



Circuit for N2: centre-tapped full-wave rectifier.

N3 (medium) [6 marks]. A Zener with $V_{Z0} = 6.0$ V and $r_z = 5\ \Omega$ is fed from $V_s = 12$ V through $R_S = 150\ \Omega$. $R_L = 1$ kΩ.

a. Find V_O, I_Z and P_Z [3]. (b) V_s rises to 13 V. Find the new V_O and the line regulation in mV/V [2]. (c) With $V_s = 12$ V, find the % regulation between no load and $R_L = 1$ kΩ [1].

Circuit for N3: the Zener drawn as its model, V_{Z0} in series with r_z.

N4 (hard, chained) [10 marks]. Design a 9.1 V supply for a load drawing 0–50 mA. Use a bridge rectifier from an 18 V rms, 50 Hz secondary (Si diodes, 0.7 V each), a filter capacitor allowing at most 2 V peak-to-peak ripple, and a 9.1 V Zener that needs $I_{Z,min} = 5$ mA.

a. Find V_P and the lowest voltage the capacitor reaches [2].
b. Find the largest R_S that keeps regulation, and choose a standard value [2].
c. Explain why I_S does not depend on the load. Using the average capacitor voltage, find I_S and choose C [3].
d. Find the worst-case Zener power and the diode PIV [2].
e. Estimate the peak diode current [1].

Circuit for N4: bridge, filter capacitor, series resistor and 9.1 V Zener.

N5 (medium) [5 marks]. A sine wave $v_i = 10 \sin \omega t$ V drives a 1 kΩ series resistor. v_o is taken across two parallel branches to ground. Branch 1 is diode D_1 (anode at the output) in series with a 3 V battery whose + terminal is at the cathode of D_1 and whose − terminal is at ground, so the cathode sits at +3 V. Branch 2 is diode D_2 (cathode at the output) in series with a 2 V battery whose − terminal is at the anode of D_2 and whose + terminal is at ground, so the anode sits at −2 V. Both diodes are Si, 0.7 V.

a. Find the transition voltages and write v_o in each region [3]. (b) Sketch v_o and find the peak current in the 1 kΩ [2].

Circuit for N5: double clipper.

N6 (easy) [4 marks]. (a) A clamper (series C, ideal diode from output to ground with its anode at ground) is driven by a ±10 V square wave. Give the output levels [2]. (b) A half-wave voltage

doubler runs from 12 V rms. Find the ideal output, the output with 0.7 V diodes, and the PIV of each diode [2].

Circuits for N6: (a) clamper, (b) half-wave voltage doubler.

6. Quick recall check

1. A bridge rectifier and a half-wave rectifier use the same C, R and V_P. What is the ratio of their ripple voltages?
2. Write the formula for C that gives a peak-to-peak ripple V_r in a half-wave supply with load current I_L.
3. In a Zener shunt regulator, at which load and supply conditions do you size R_S, and at which do you check P_Z?
4. What is the PIV of each diode in a centre-tapped rectifier with V_S peak per half-winding?
5. What happens to a varactor's capacitance when the reverse bias increases, and why?

BJT operation and DC bias I

Practice questions · Day 06 · Thu 29 Oct · pack E4

5. Practice

From your course files

WS4 Q6 (course question).
6.1. Which one of the following statements is correct for a BJT transistor?
A) BJT collector current can be controlled by the voltage between The Base and emitter
B) In a PNP transistor, the Emitter is an n-type semiconductor
C) A BJT has three PN junctions that form the three terminals
D) In a PNP transistor, the Collector is an n-type semiconductor
6.2. In a PNP BJT transistor,
A) the base-emitter junction is reverse biased in the cut-off mode
B) the collector-base junction is forward biased in the cut-off mode
C) the emitter-base junction is reverse biased in the cut-off mode
D) the emitter-collector junction is reverse biased in the cut-off mode
6.3. In an NPN BJT in the active mode of operation, the Base-Emitter PN junction is forward-biased so that
A) electrons flow from base to emitter
B) holes flow from emitter to base
C) electrons flow directly from emitter to collector
D) holes flow from emitter to base

WS4 Q7 (course question). For a BJT transistor, find the values of β for the values of α = {0.5, 0.8, 0.98, 0.999}.

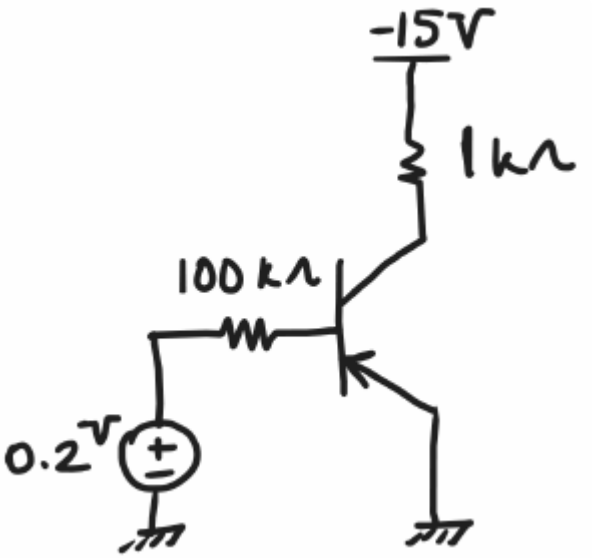
WS4 Q8 (course question). Consider a PNP transistor in active mode with a current gain of β = 50. Find,
1. Base (I_B) and emitter (I_E) currents when collector current is $I_C = 4$ mA.

2. Collector (I_C) and base (I_B) currents when emitter current is $I_E = 4$ mA.
3. Collector (I_C) and emitter (I_E) currents when base current is $I_B = 100$ µA.

WS4 Q9 (course question). Consider a transistor in the active mode region. Find,
1. Emitter current (I_E), β and α when collector and base currents are $I_C = 8$ mA and $I_B = 100$ µA.
2. Collector current (I_C), β and α when emitter and base currents are $I_E = 5$ mA and $I_B = 100$ µA.
3. Base current (I_B), β and α when collector and emitter currents are $I_C = 10$ mA and $I_E = 10.1$ mA.

WS4 Q10 (course question). Consider a transistor with a gain β = 50. For each of the following cases identify the operating mode (active, cut-off and saturation) of the transistor.
1. $I_B = 0$ µA and $I_C = 0$ mA.
2. $I_B = 100$ µA and $I_C = 5$ mA.
3. $I_B = 100$ µA and $I_C = 1$ mA.
4. $I_B = 200$ µA and $I_C = 10$ mA.

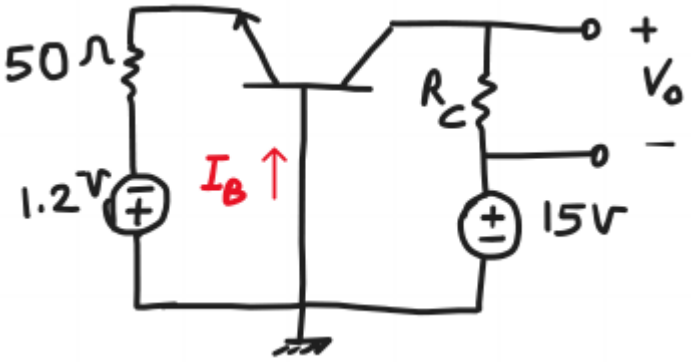
WS4 Q11 (course question). In the circuit below, replace the PNP transistor with its large-signal equivalent model if the transistor is in,
1. Active mode.
2. Saturation mode.
3. Cut-off mode.



WS4 Q11 circuit (from the workshop sheet)

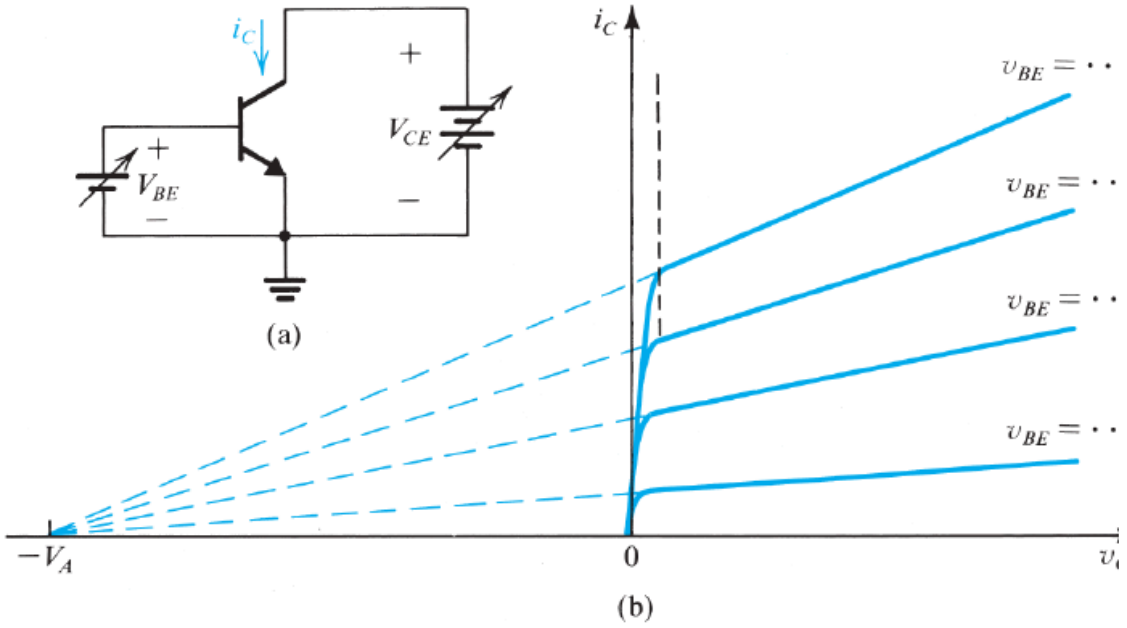
WS4 Q12 (course question). In the circuit below, replace the NPN transistor with its large-signal equivalent model if the transistor is in,

1. Active mode.
2. Saturation mode.
3. Cut-off mode.



WS4 Q12 circuit (from the workshop sheet)

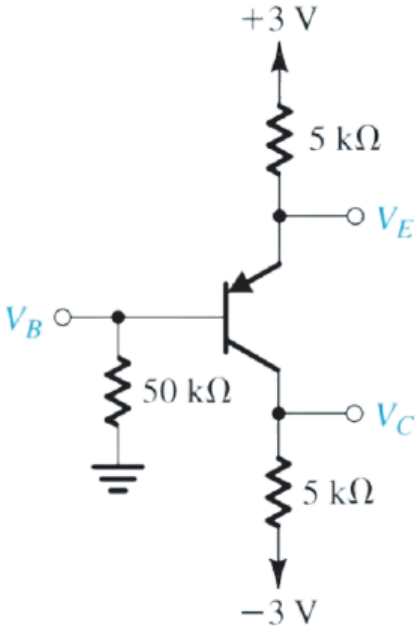
WS5 Q1 (course question). A practical $i_c - v_{CB}$ characteristics of a BJT are shown in the figure below. Identify active, saturation, cut-off regions, and early voltage. Also, explain what the non-zero slope means and how you expect the v_{BE} value to change from lower i_C to higher i_C.



WS5 Q1 figure (from the workshop sheet)

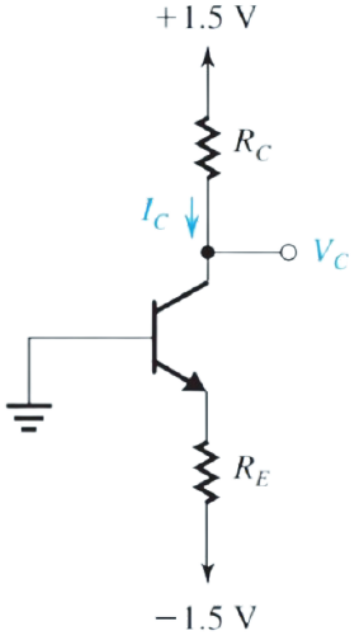
WS5 Q3 (course question). A single measurement indicates the emitter voltage of the transistor in the circuit below to be $V_E = 1$ V. Under the assumption that $V_{EB} = 0.7$ V and transistor is in active mode, find:

1. Emitter, collector and base currents.
2. Base and collector voltages.
3. α and β.



WS5 Q3 circuit (from the workshop sheet)

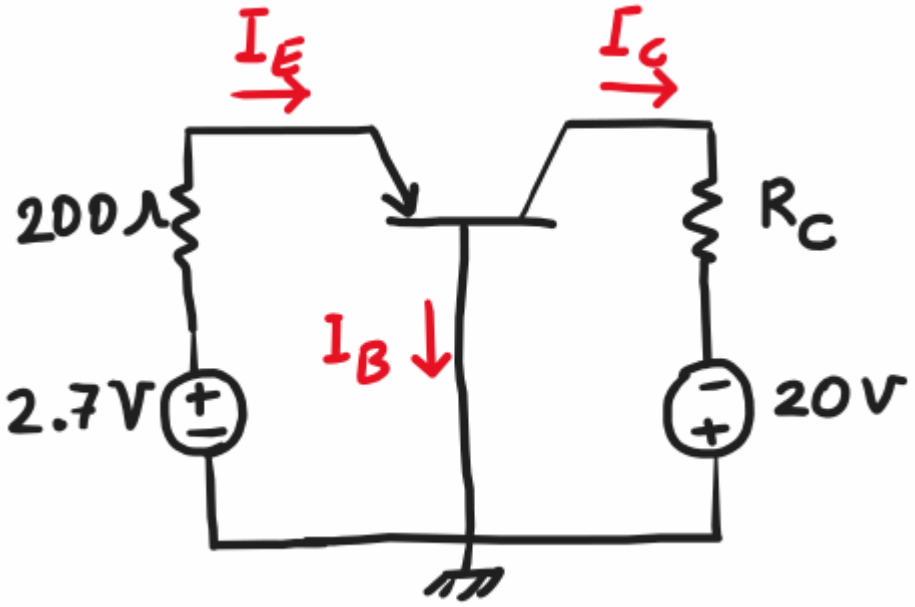
WS5 Q4 (course question). In the circuit below, the transistor is in active mode and has $V_{BE} = 0.7$ V and β = 100. Select resistors R_C and R_E such that $I_C = 0.2$ mA and $V_C = 0.5$ V.



WS5 Q4 circuit (from the workshop sheet)

WS5 Q5 (course question). In the circuit below, the PNP transistor has α = 0.9 and $V_{EB} = 0.7$ V (sometimes, we say $|V_{BE}|$ to avoid PNP and NPN voltage polarity confusion). Assume that the transistor is in active mode. Find,

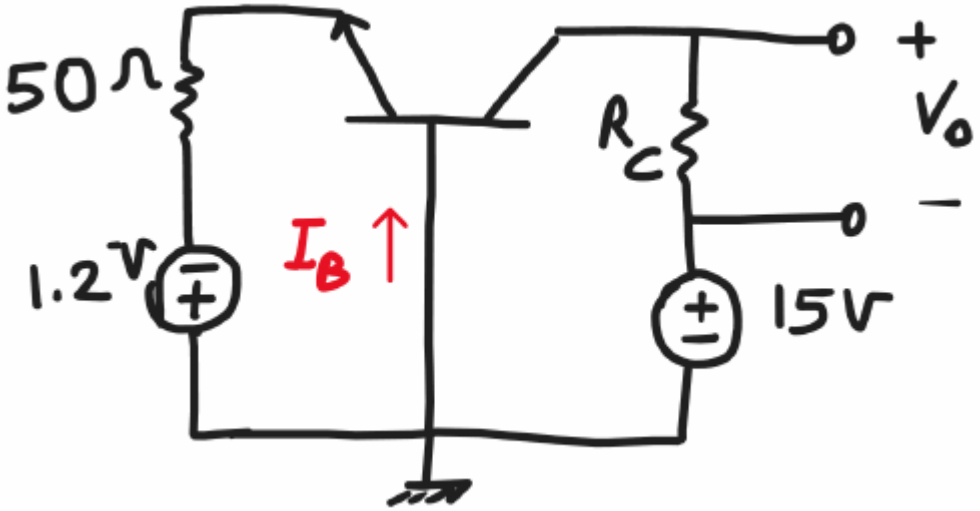
1. Emitter, collector and base currents.
2. What should be the minimum value of R_C if $|V_{CE,max}| = 20$ V?



WS5 Q5 circuit (from the workshop sheet)

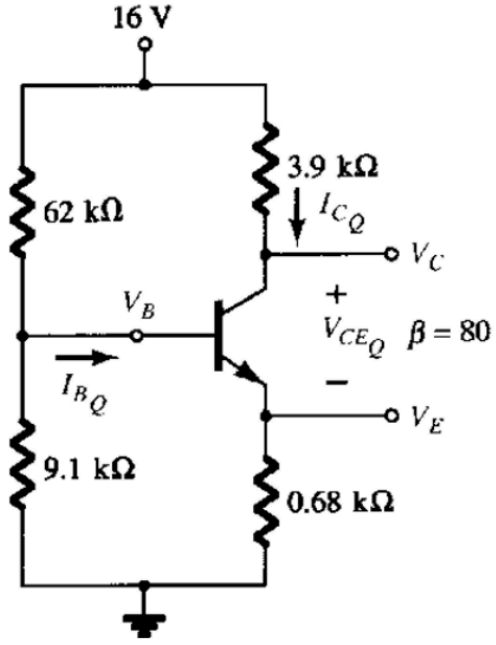
WS5 Q6 (course question). In the circuit below, the NPN transistor has α = 0.9 and $|V_{BE}| = 0.7$ V. Assume that the transistor is in active mode. Find,

1. the value of R_C if we want $|V_o| = 4.5$ V.
2. V_{CE} based on the value of R_C obtained previously.



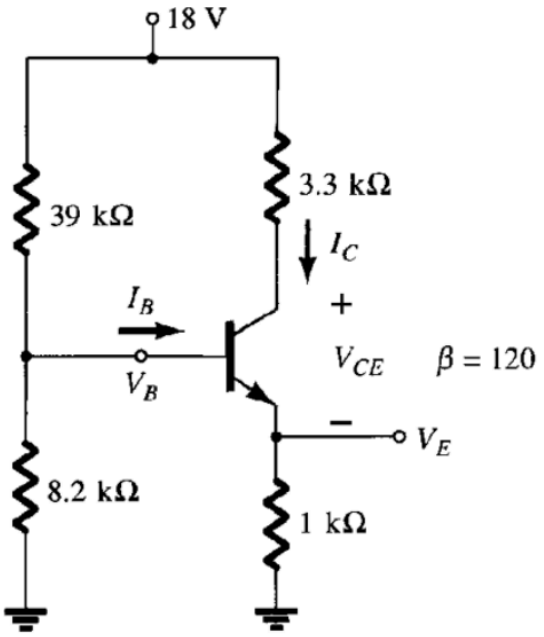
WS5 Q6 circuit (from the workshop sheet)

WS5 Q7 (course question). For the voltage-divider configuration in the circuit below, determine (a) I_{BQ}, (b) I_{CQ}, (c) V_{CEQ}, (d) V_C, (e) V_E and (f) V_B.



WS5 Q7 circuit (from the workshop sheet)

WS5 Q8 (course question). Determine the following for the voltage-divider configuration in the circuit below, using the approximate approach if the required condition is satisfied. (a) I_C, (b) V_{CE}, (c) I_B, (d) V_E and (e) V_B.



WS5 Q8 circuit (from the workshop sheet)

WS5 Q14 (course question). In the circuit below, assume that the BJT has β = 50, $V_{BE,on} = 0.7$ V, and $V_{CE,sat} = 0.2$ V. Find the base and collector currents and check the transistor operating mode.

WS5 Q14 circuit (from the workshop sheet)

PT Q6 (practice test, 6 marks). Draw the load line for the network below on the characteristics.

a. Find the Q-point and resulting values I_{C_Q} and V_{CE_Q}.

b. Find the value of β at the Q-point.

PT Q6 circuit (from the practice-test paper)

PT Q6 output characteristics (from the practice-test paper)

PT Q7 (practice test, 10 marks). For the circuit shown on the right, β = 50, $V_{BE,on} = 0.7$ V, and $V_{CE,sat} = 0.2$ V. Answer the following questions assuming that the BJT operates in the **active state**.

a) Find the Thevenin equivalent circuit (i.e., V_{Th} and R_{Th}) seen from the base of the transistor. [4 marks]

b) Redraw the circuit by replacing the transistor with its **DC equivalent circuit** model and Thevenin equivalent model in the base. [4 marks]

c) Determine collector current (I_C) and voltage (V_C). [2 marks]

PT Q7 circuit (from the practice-test paper)

New questions

New question 1 [4 marks, easy]. A transistor has β = 99 and $I_{CBO} = 10$ nA.

a. Find α. [1]
b. Find I_{CEO}. [1]
c. With $I_B = 20$ μA, find I_C and I_E including leakage, and comment on whether leakage matters here. [2]

New question 2 [5 marks, easy]. Fixed bias: $V_{CC} = 15$ V, $R_B = 470$ kΩ, $R_C = 3.3$ kΩ, emitter grounded, β = 100, $V_{BE} = 0.7$ V.

a. Find I_{BQ}, I_{CQ} and V_{CEQ}. [3]
b. Give the load-line intercepts and confirm the mode using V_{BC}. [2]

Circuit for New question 2: fixed bias.

New question 3 [6 marks, medium]. Emitter-stabilised bias: $V_{CC} = 18$ V, $R_B = 510$ kΩ, $R_C = 2.4$ kΩ, $R_E = 1.5$ kΩ, β = 100.

a. Find I_B, I_C and V_{CE}. [3]
b. Find V_E, V_B, V_C and V_{BC}. [2]
c. Find $I_{C,sat}$. [1]

Circuit for New question 3: emitter-stabilised bias.

New question 4 [8 marks, medium]. Voltage divider: $V_{CC} = 15$ V, $R_1 = 47$ kΩ, $R_2 = 10$ kΩ, $R_C = 2.7$ kΩ, $R_E = 1.2$ kΩ, β = 150.

a. Test whether the approximate method is valid. [1]
b. Find I_C, V_{CE} and I_B by the approximate method. [3]
c. Repeat by the exact method and give the % error in I_C of the approximation. [4]

Circuit for New question 4: voltage-divider bias.

New question 5 [10 marks, hard, chained]. Voltage divider: $V_{CC} = 12$ V, $R_1 = 33$ kΩ, $R_2 = 12$ kΩ, $R_C = 2.2$ kΩ, $R_E = 0.56$ kΩ, β = 60.

a. Test the approximation. [1]
b. A student ignores the test and uses the approximate method anyway. What I_C and V_{CE} do they get, and what is physically wrong with the result? [3]
c. Do the exact analysis: I_B, I_C, V_{CE}, V_E, V_B, V_C. Confirm the mode. [4]
d. Find $I_{C,sat}$ and state where the Q-point sits on the load line. [2]

Circuit for New question 5: voltage-divider bias.

New question 6 [6 marks, hard, reading characteristics and design]. On a set of CE output characteristics, a load line runs from $V_{CE} = 24$ V to $I_C = 8$ mA. It crosses the $I_B = 30$ μA curve at $I_C = 4.2$ mA. The circuit is emitter-stabilised with $R_C = 2.4$ kΩ.

a. Find V_{CC}, $R_C + R_E$ and R_E. [2]
b. Find V_{CEQ} and β_{dc} at the Q-point. [2]
c. Find R_B, choose the nearest standard value (E12), and find the resulting I_B. [2]

Circuit for New question 6: emitter-stabilised bias, V_CC, R_E and R_B to be found from the load line.

6. Quick recall check

1. Convert α = 0.995 to β, and β = 150 to α.
2. An NPN has $V_B = 2.7$ V, $V_E = 2.0$ V and $V_C = 2.2$ V. Which mode is it in?
3. Write the base-loop equation for I_B in emitter-stabilised bias.
4. What condition allows the approximate voltage-divider method, and why?
5. What are the two intercepts of the DC load line for a circuit with R_C and R_E?

BJT DC bias II

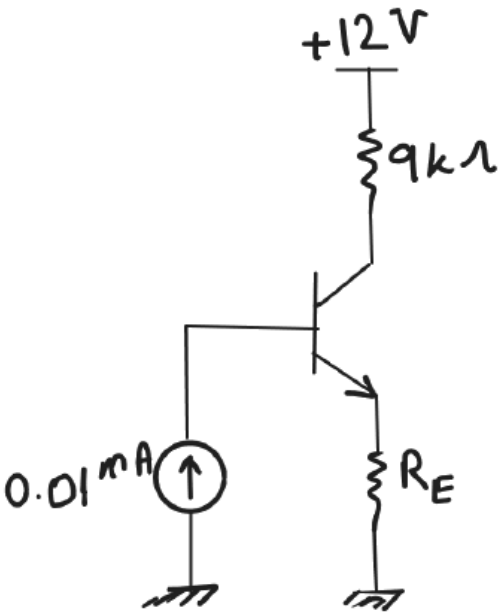
Practice questions · Day 08 · Sat 31 Oct · pack E5

5. Practice

From your course files

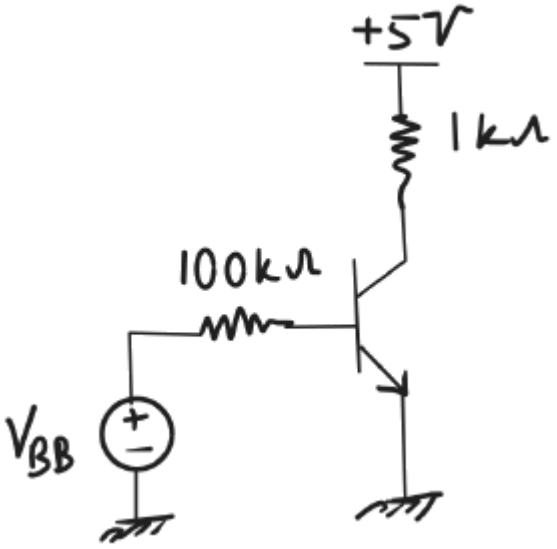
WS5 Q12 (course question). In the circuit below, assume that the BJT has $\beta = 150$, $V_{BE,on} = 0.7$ V, and $V_{CE,sat} = 0.2$ V. Answer the following questions:

1. For $R_E = 1\ \text{k}\Omega$, find the operating mode of the BJT.
2. Assuming that $100 \leq \beta \leq 200$, for what values of R_E, the transistor will always be in **Saturation** mode?



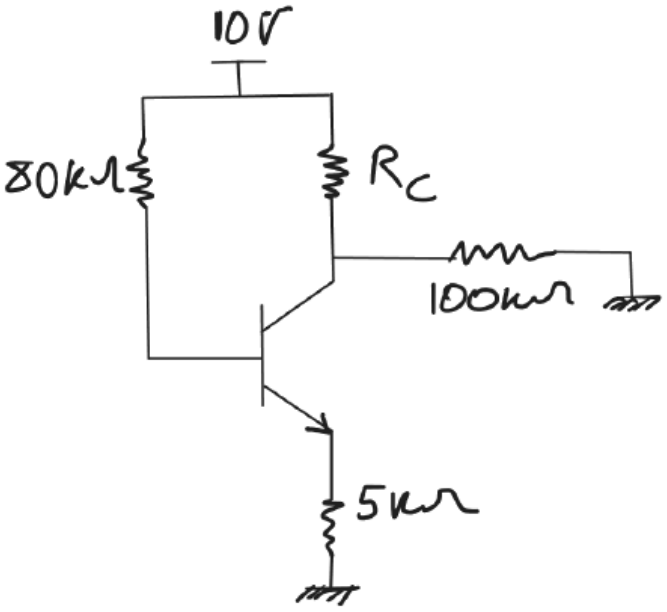
WS5 Q12 figure (from the workshop sheet)

WS5 Q13 (course question). In the circuit below, assume that the BJT has $100 \leq \beta \leq 200$, $V_{BE,on} = 0.7$ V, and $V_{CE,sat} = 0.2$ V. What is the minimum base current and base voltage (V_{BB}) in which the transistor will go into **saturation** mode?



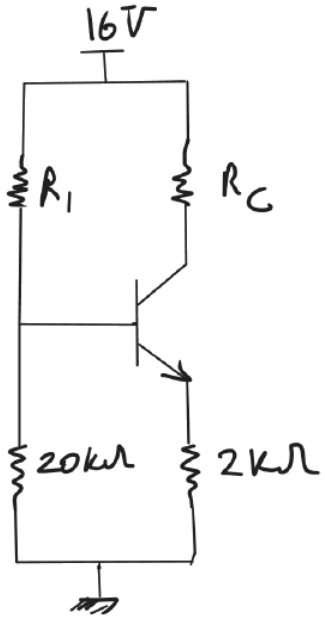
WS5 Q13 figure (from the workshop sheet)

WS5 Q17 (course question). In the circuit below, assume that the BJT has $\beta = 100$, $V_{BE,on} = 0.7$ V, $V_{CE,sat} = 0.2$ V. Find the value of R_C in which $V_{CE} = 1$ V.



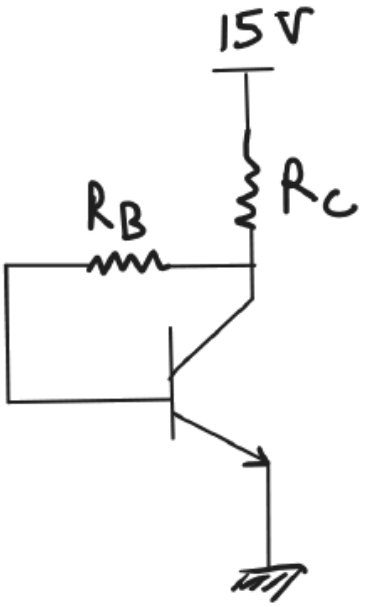
WS5 Q17 figure (from the workshop sheet)

WS5 Q18 (course question). In the circuit below, assume that the BJT has $\alpha = 0.985$, $V_{BE,on} = 0.3$ V, and $V_{CE,sat} = 0.2$ V. Calculate R_1 and R_C to ensure that $I_C = 2$ mA and $V_{CE} = 6$ V.



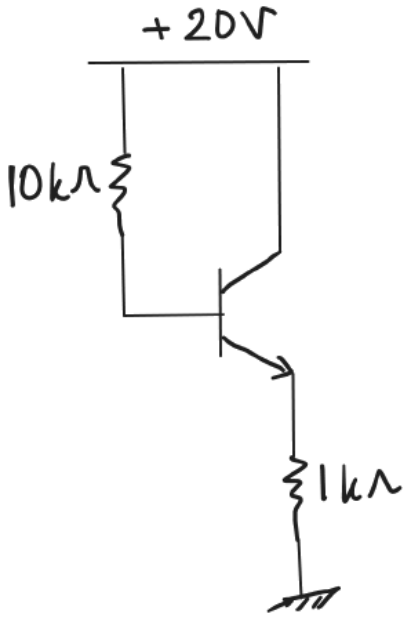
WS5 Q18 figure (from the workshop sheet)

WS5 Q19 (course question). In the circuit below, assume that the BJT has $\beta = 100$ and $V_{BE,on} = 0.3$ V. Calculate R_B to ensure that $I_C = 1$ mA and $V_{CE} = 2$ V.



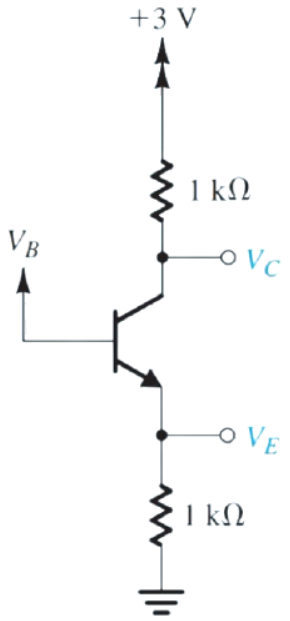
WS5 Q19 figure (from the workshop sheet)

WS5 Q9 (course question, *). In the circuit below, assume that the BJT has $\beta = 100$, $V_{BE,on} = 0.7$ V, and $V_{CE,sat} = 0.2$ V. Find the BJT operating mode in this condition.



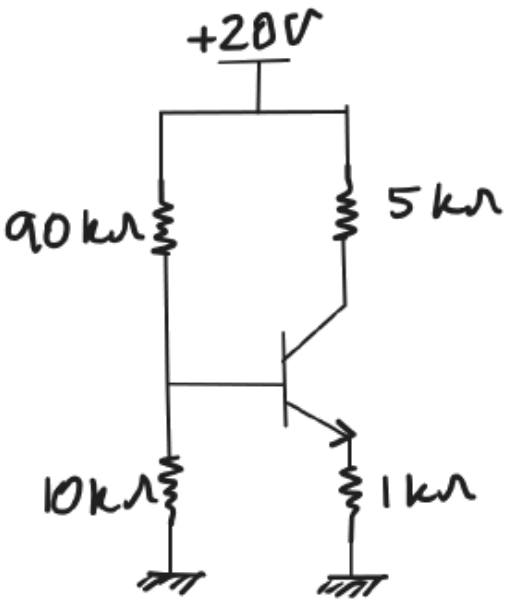
WS5 Q9 figure (from the workshop sheet)

WS5 Q11 (course question, *). In the circuit below, assume that the BJT has a very high β, $V_{BE,on} = 0.7$ V, and $V_{CE,sat} = 0.2$ V. When $V_B = 0$ V, find the BJT operating mode.



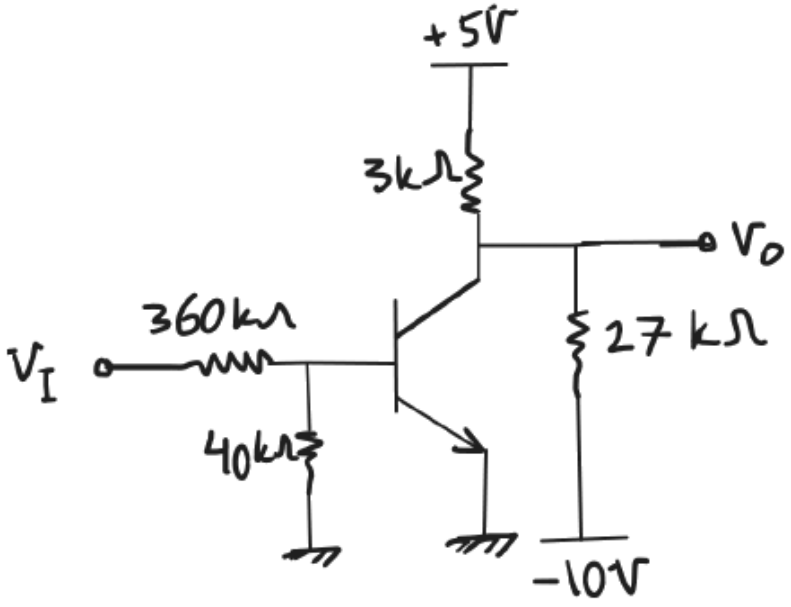
WS5 Q11 figure (from the workshop sheet)

WS5 Q14 (course question, *). In the circuit below, assume that the BJT has $\beta = 50$, $V_{BE,on} = 0.7$ V, and $V_{CE,sat} = 0.2$ V. Find the base and collector currents and check the transistor operating mode.



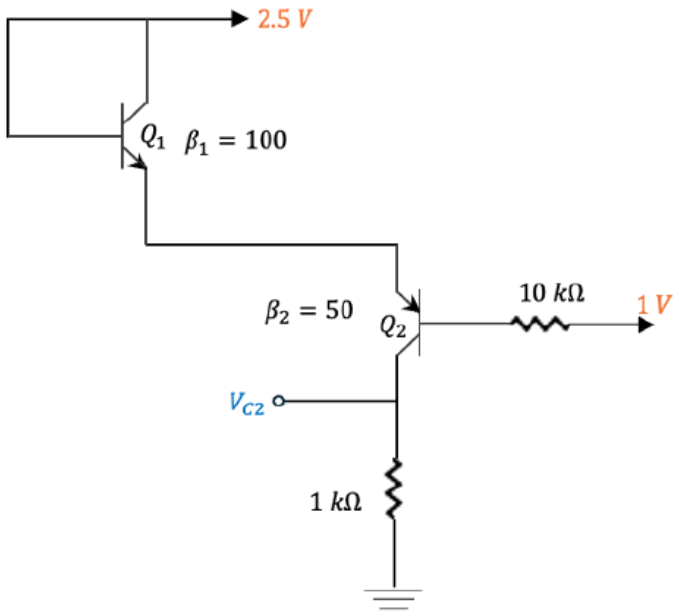
WS5 Q14 figure (from the workshop sheet)

WS5 Q15 (course question, *). In the circuit shown in Figure 1, assume that the BJT has $\beta = 40$, $V_{BE,on} = 0.7$ V, and $V_{CE,sat} = 0.2$ V. Find the output voltage, V_o, when $V_I = 15$ V.



WS5 Q15 figure (Figure 1 of the workshop sheet)

PT Q8 (practice test, 6 marks). Consider the circuit below where Q_1 and Q_2 have different current gain, $\beta_1 = 100$ and $\beta_2 = 50$. You can assume $V_{BE,on}$ = 0.7 V and $V_{CE,sat} = 0.2$ V for both BJTs and both BJTs are in **active state**. Find the output voltage, V_{C2} by showing your work.

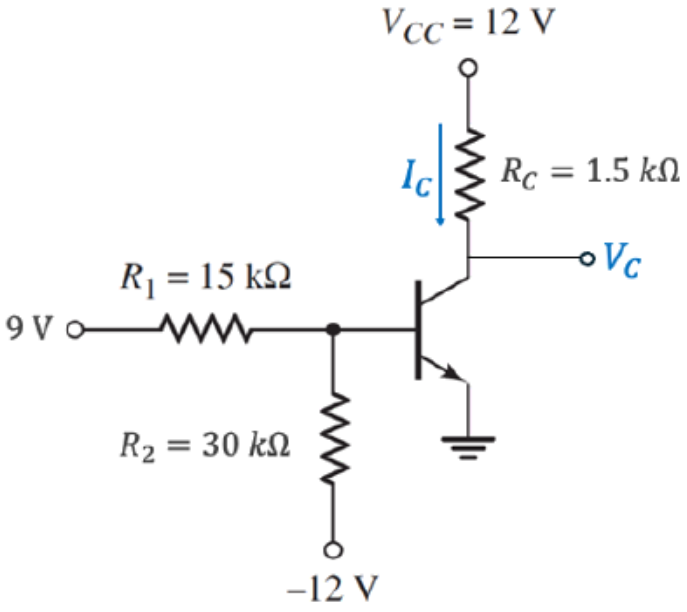


Practice test Q8 figure (from the practice test)

PT Q7 (practice test, 10 marks). For the circuit shown on the right, $\beta = 50$, $V_{BE,on} = 0.7$ V, and $V_{CE,sat} = 0.2$ V. Answer the following questions assuming that the BJT operates in the **active state**.

a. Find the Thevenin equivalent circuit (i.e., V_{Th} and R_{Th}) seen from the base of the transistor. [4 marks]
b. Redraw the circuit by replacing the transistor with its **DC equivalent circuit** model and Thevenin equivalent model in the base. [4 marks]
c. Determine collector current (I_C) and voltage (V_C). [2 marks]

Then (for this pack): try the approximate divider method on the same circuit and explain what happens.

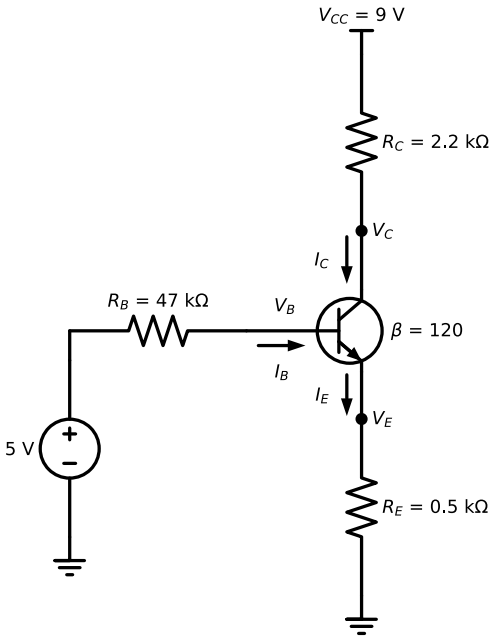


Practice test Q7 figure (from the practice test)

New questions

New Q1 [5 marks]. An npn transistor has its collector connected to V_{CC} = 9 V through R_C = 2.2 kΩ and its emitter to ground through R_E = 0.5 kΩ. Its base is driven from a separate 5 V source through R_B = 47 kΩ. β = 120, $V_{BE,on}$ = 0.7 V, $V_{CE,sat}$ = 0.2 V.

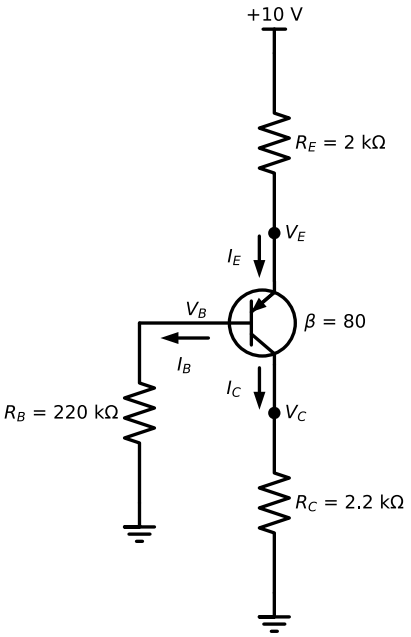
a. Assume active and find I_B, I_C and V_{CE}. [2]
b. Find the true state, I_B and I_C, and the forced β. [3]



Circuit for New Q1: separate 5 V base source with emitter resistor.

New Q2 [6 marks] (pnp). A pnp transistor has its emitter connected to +10 V through R_E = 2 kΩ. Its base is connected to ground through R_B = 220 kΩ. Its collector is connected to ground through R_C = 2.2 kΩ. β = 80, $|V_{EB,on}|$ = 0.7 V, $|V_{EC,sat}|$ = 0.2 V.

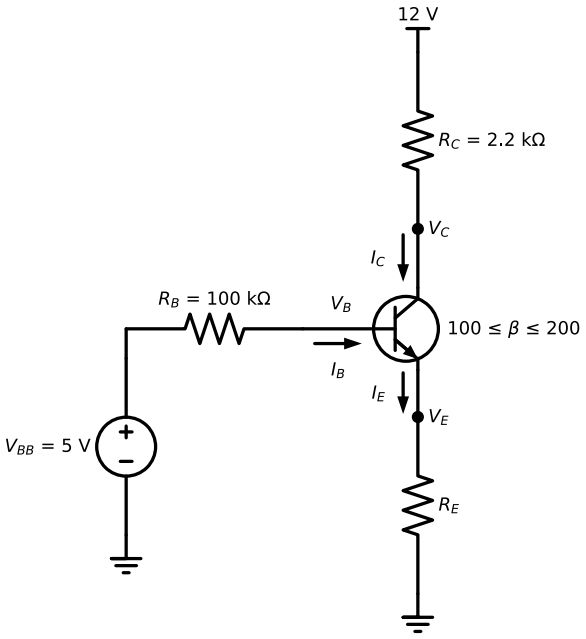
a. Mark the directions of I_B, I_C and I_E, and find I_B and I_C. [3]
b. Find V_E, V_B, V_C and V_{EC}, and confirm the state. [2]
c. What is the largest R_C that keeps the transistor active? [1]



Circuit for New Q2: pnp, emitter 2 kΩ to +10 V, base 220 kΩ to ground, collector 2.2 kΩ to ground.

New Q3 [7 marks] (sensitivity). An npn transistor has its collector connected to 12 V through R_C = 2.2 kΩ. Its base is driven from V_{BB} = 5 V through R_B = 100 kΩ. An emitter resistor R_E connects the emitter to ground. $100 \le \beta \le 200$, $V_{BE,on}$ = 0.7 V, $V_{CE,sat}$ = 0.2 V.

a. With R_E = 0, find I_C and the state at β = 100 and at β = 200. [2]
b. Find the smallest R_E that keeps the transistor active over the whole β range. [3]
c. Choose the next standard value above it (E12) and give the range of I_C and V_{CE}. [2]



Circuit for New Q3: V_BB = 5 V through 100 kΩ, R_E to be chosen.

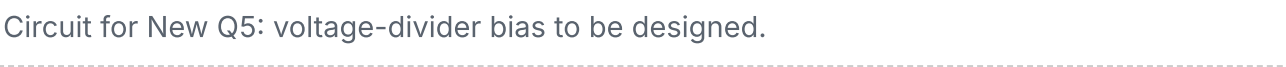
New Q4 [8 marks] (two transistors, npn into pnp). Q_1 is an npn (β1 = 150). Its base is fed by a divider from 12 V: R_1 = 56 kΩ from 12 V to the base and R_2 = 12 kΩ from base to ground. Its emitter goes through R_{E1} = 1 kΩ to ground, and its collector through R_{C1} = 4.7 kΩ to 12 V. Q_1 's collector connects directly to the base of Q_2, a pnp (β2 = 100). Q_2's emitter goes through R_{E2} = 2.2 kΩ to 12 V, and its collector, the output V_{C2}, goes through R_{C2} = 1.8 kΩ to ground. Use $|V_{BE}|$ = 0.7 V and $|V_{CE,sat}|$ = 0.2 V.

a. Test whether the approximate divider method is valid for Q_1, and find I_{C1} by the exact method. [2]
b. Ignoring I_{B2}, find V_{C1}, V_{E2} and I_{E2}. [2]
c. Repeat (b) including I_{B2} (state which way it flows), then find V_{C2} and check both transistors. [3]
d. What is the largest R_{C2} for which Q_2 stays active? [1]

Circuit for New Q4: divider-biased npn Q1 directly driving pnp Q2.

New Q5 [8 marks] (design). Design a voltage-divider bias circuit with V_{CC} = 15 V for I_{CQ} = 2 mA and V_{CEQ} = 6 V. The transistor has $100 \leq \beta \leq 300$.

a. Use $V_E = 0.1V_{CC}$ to find R_E and R_C. [2]
b. Choose R_2 so that the approximate method is valid at $\beta_{\min}$, then find R_1. [2]
c. Choose E24 standard values and use the exact method to find I_{CQ} and V_{CEQ} at β = 100 and β = 300. [4]



Circuit for New Q5: voltage-divider bias to be designed.

New Q6 [6 marks] (collector feedback, β range). An npn transistor has R_C = 3.3 kΩ from 12 V to the collector, R_F = 180 kΩ from collector to base, and R_E = 0.5 kΩ from emitter to ground. V_{BE} = 0.7 V.

a. Find I_C, V_C and V_{CE} for β = 80, 160 and 240. [4]
b. A fixed-bias circuit (R_B from 12 V to the base, the same R_C, no R_E) is designed to give the same I_C at β = 160. Find R_B and the V_{CE} it gives at β = 240. Comment. [2]



Circuits for New Q6: (a) collector feedback, (b) the fixed-bias comparison.

6. Quick recall check

1. An active-mode calculation gives $V_{CE} = -4$ V. What do you assume next, and which two equations replace $I_C = \beta I_B$?
2. You need a transistor to stay active for every β in $100 \leq \beta \leq 200$. Which β do you design for, and why?
3. In a collector-feedback circuit, what current flows through R_C?
4. A pnp has V_E = 3.0 V and is active. What is V_B, and which way do I_B and I_C flow?
5. What two rules of thumb set R_E and R_2 in voltage-divider bias design?

EDC · PAPER

EDC practice test: marking guide

Practice questions · Day 08 · Sat 31 Oct · pack EPT

How to use this: Sit the paper first ("Resources/Practice test_ENGE2003.pdf"): 90 minutes, timed, open book, calculator, no peeking at this guide. Then put your pen down and mark it question by question, opening each `<details>` block only when you are ready to compare. Be strict: award a mark only if your script shows the line that earns it. Write your score per question in the table under "After marking" and use that table to plan your revision.

The paper has 8 questions and 50 marks. The formula table is on page 2 of the paper (before Q1), not at the back. Every circuit value in this guide was read from the paper's own figures and matches the values stated in your Test 1 Reference (p. 17–21). Every number was recomputed in Python.

Before you start

Time budget. 90 min for 50 marks is **1.8 min per mark**. Leave about 5 min at the end to check.

Question	Marks	Target time	Running clock
Q1 MCQ	5	5 min	0:05
Q2 exponential model	5	7 min	0:12
Q3 three diodes, PWL	8	14 min	0:26
Q4 Zener design	5	8 min	0:34
Q5 clipper	5	8 min	0:42
Q6 load line	6	12 min	0:54
Q7 Thevenin + BJT	10	16 min	1:10
Q8 two BJTs	6	12 min	1:22
Check		8 min	1:30

Q1 is quick. Bank the time you save there for Q6, because it needs a ruler and careful graph reading.

What earns method marks on this paper.

- Write the formula in symbols first, then substitute, then the answer with units. For example, $\Delta V_D = nV_T \ln(I_2/I_1)$, then numbers, then "= 0.179 V". If the number is wrong but the formula and substitution are right, you still get most of the marks.

- State the model you are using in words: “third PWL model: each diode = ideal diode + 0.6 V + 0.2 kΩ”, “active mode: $V_{BE} = 0.7$ V, $I_C = \beta I_B$”, “ideal diode”.
- “Redraw” parts (Q3(a), Q7(b)) are worth 2 and 4 marks. Draw them neatly and label every element value and every node you later use (V_A, V_B, I_B, βI_B).
- “Verify” parts (Q3(b)) need a stated conclusion: “I = 15.3 µA > 0, so all three diodes conduct.” A number without the sentence loses the mark.
- When the question tells you to assume a state (Q3, Q7, Q8), check it at the end in one line: current > 0, $V_{CE} > V_{CE,sat}$.

Before moving on from each question, check these:

1. Units. Are kΩ and mA paired? Is µA written as µA, not mA?
2. Sign and direction. Do the current arrows on your sketch match the signs you used?
3. Is the state-assumption check written down?
4. Is the final answer underlined or boxed, with units?

BJT small-signal I

Practice questions · Day 10 · Mon 2 Nov · pack E6

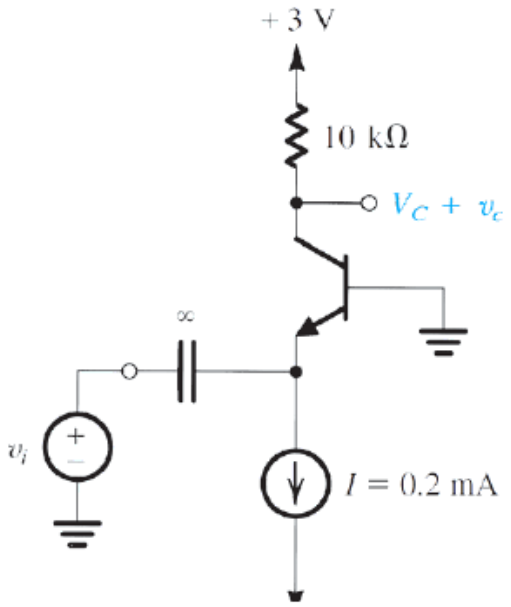
5. Practice

From your course files

WS6 Q1 (course question). A BJT having $\beta = 100$ is biased at a DC collector current of 1 mA. Find the value of g_m, r_e, and r_π at the bias point.

WS6 Q2 (course question). Given $g_m = 20$ mA/V and $\beta = 100$, find r_π and r_e. Then, draw four small-signal equivalent circuit models.

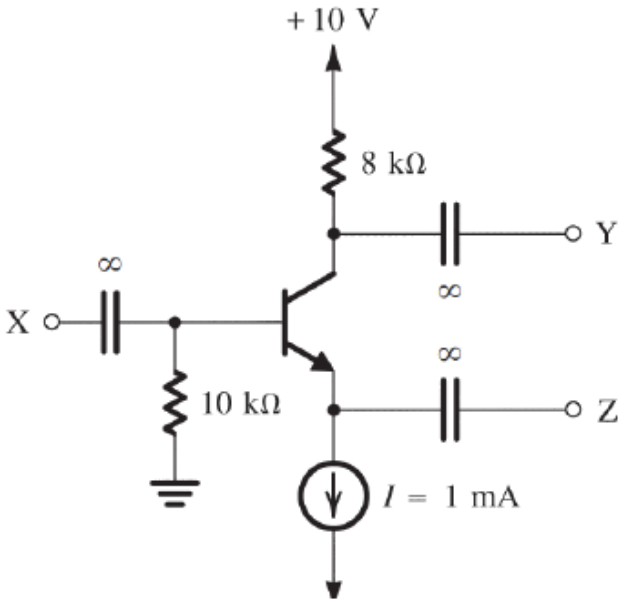
WS6 Q4 (course question). The transistor amplifier shown in the figure 9 [the figure below; the sheet labels it Figure 4] is biased with a current source $I = 0.2$ mA and has a very high β. Assuming the transistor is in the active state, the DC voltage of the collector is 1 V, and $r_e = 125\ \Omega$. Replace the transistor with the **T model** and find voltage gain, $\frac{v_c}{v_i}$.



WS6 Q4 figure (Figure 4 of the workshop sheet)

WS6 Q5 (course question). The transistor amplifier shown in the figure 10 [the figure below; the sheet labels it Figure 5] is biased with a current source $I = 1$ mA and has $\beta = 100$ and $V_A = 100$ V. Assuming that $g_m = 39.604$ mA/V, $r_\pi = 2.525\ \text{k}\Omega$ and $r_o = 101\ \text{k}\Omega$, answer the following questions:

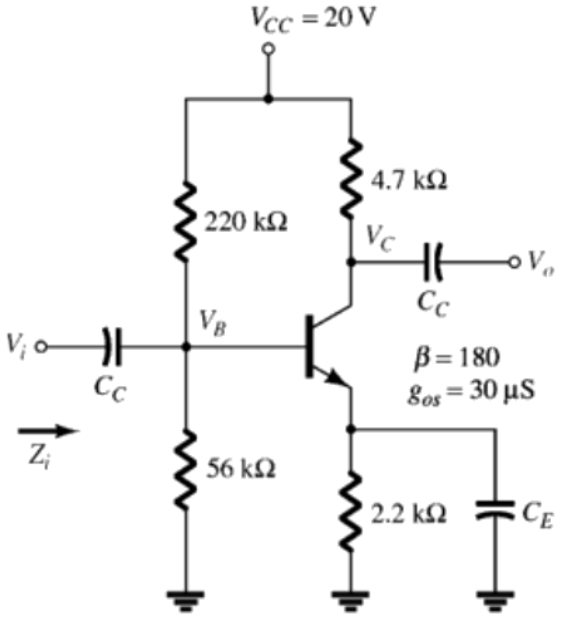
1. If terminal Z is connected to ground, X to a signal source v_{sig} with a source resistance $R_{sig} = 2\ \text{k}\Omega$, and Y to an 8-kΩ load resistance, use the hybrid-π model to draw the small-signal equivalent circuit of the amplifier.
2. Calculate the overall voltage gain, $\dfrac{v_y}{v_{sig}}$.



WS6 Q5 figure (Figure 5 of the workshop sheet)

WS6 Q6 (course question). For the network of figure 6:

1. Determine r_e.
2. Calculate V_B and V_C.
3. Calculate Z_i and $A_V = V_o/V_i$.

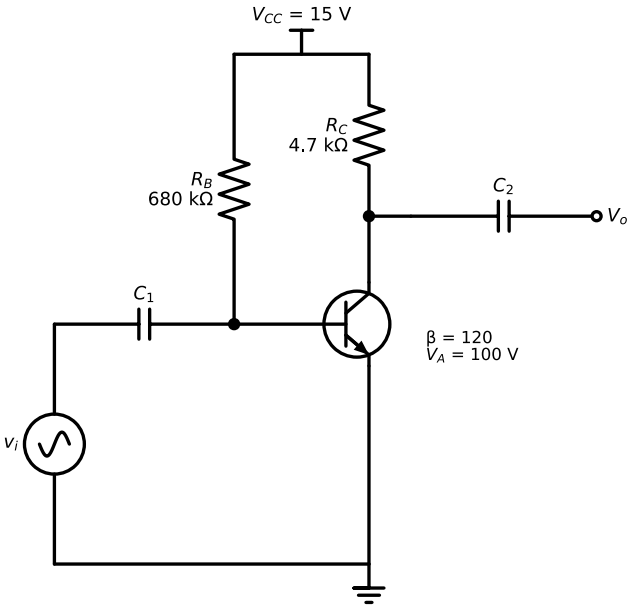


WS6 Q6 figure (Figure 6 of the workshop sheet)

New questions

N1 (easy) [4 marks]. A BJT with $\beta = 150$ and $V_A = 80$ V is biased at $I_C = 2$ mA. Take $V_T = 26$ mV. (a) Find g_m, r_π, r_e and r_o. [3] (b) Show numerically that $r_\pi = (\beta + 1) r_e$. [1]

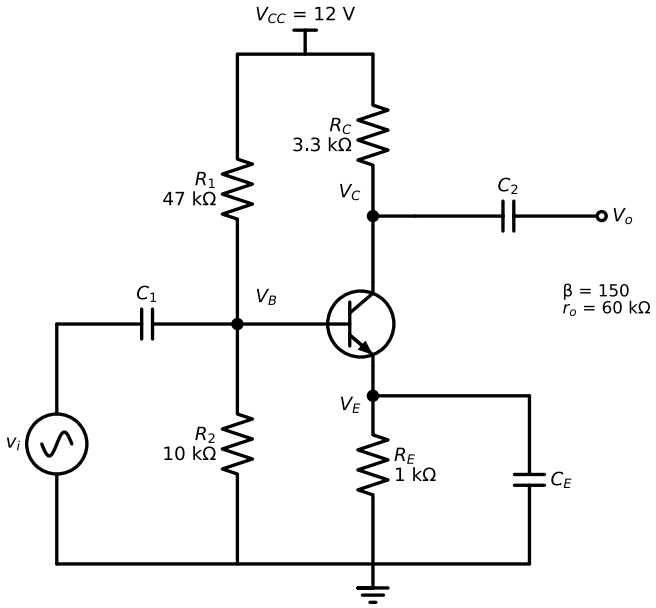
N2 (medium) [7 marks]. CE fixed-bias amplifier: $V_{CC} = 15$ V, $R_B = 680$ kΩ, $R_C = 4.7$ kΩ, $\beta = 120$, $V_A = 100$ V, ideal coupling capacitors. (a) Find the Q-point (I_B, I_C, V_{CE}) and confirm active mode. [2] (b) Find r_e and r_o. [1] (c) Find Z_i, Z_o and A_v with and without r_o. [3] (d) $v_i = 5 \sin \omega t$ mV. Write $v_o(t)$ and say whether the output will clip. [1]



N2: CE fixed-bias amplifier

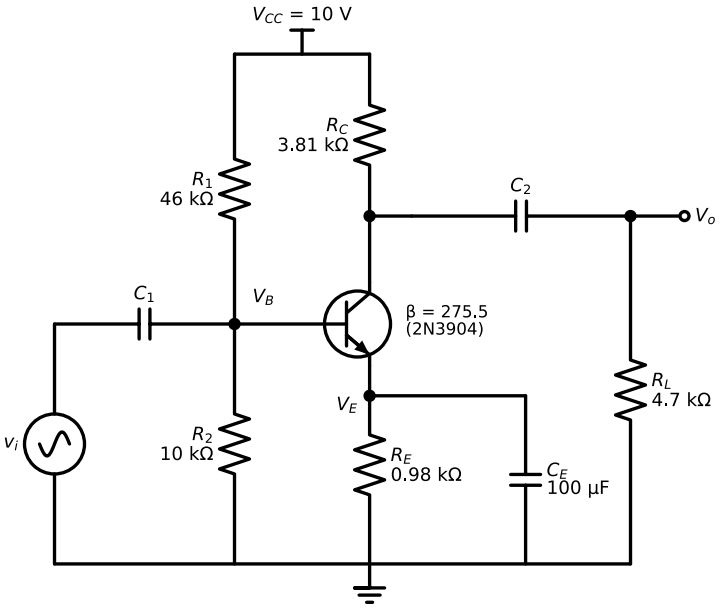
N3 (medium) [8 marks]. Voltage-divider CE with bypassed emitter: $V_{CC} = 12$ V, $R_1 = 47$ kΩ, $R_2 = 10$ kΩ, $R_C = 3.3$ kΩ, $R_E = 1$ kΩ with C_E across it, $\beta = 150$, $r_o = 60$ kΩ. (a) Test

whether the approximate method may be used, and find I_E, V_C and V_{CE}. [3] (b) Find r_e. [1] (c) Find Z_i, Z_o and A_v (include r_o). [3] (d) Write the approximate h-parameters h_{ie}, h_{fe}, h_{oe} of the transistor at this Q-point. [1]



N3: voltage-divider CE with bypassed emitter

N4 (multi-part, chains) [9 marks] — your Practical 3 amplifier. As built: $V_{CC} = 10$ V, $R_1 = 46$ kΩ, $R_2 = 10$ kΩ, $R_C = 3.81$ kΩ, $R_E = 0.98$ kΩ bypassed by 100 µF, 2N3904 with measured $\beta = 275.5$, $V_A \approx 100$ V. (a) Predict V_B, I_E with the approximate method, and compare with your measured $V_B = 1.735$ V and $I_E = 1.12$ mA. [2] (b) Using the measured $I_E = 1.12$ mA, find r_e and r_o. [1] (c) Find Z_i, Z_o and the no-load A_v, first with $r_o = \infty$ and then with r_o. [3] (d) You measured $A_v = 142.2$ (magnitude). Give one reason, other than r_o, why the measurement is lower. [1] (e) A 4.7 kΩ load is connected via C_2. It appears in parallel with $R_C \parallel r_o$. Predict the loaded gain and compare with the measured 77.1. [2]



N4: your Practical 3 amplifier as built, with the 4.7 kΩ load on C_2

N5 (hard, design) [8 marks]. (a) For any CE stage with bypassed emitter and $r_o = \infty$, show that $|A_v| = V_{R_C}/V_T$, where V_{R_C} is the DC voltage across R_C. [2] (b) A fixed-bias CE stage runs from $V_{CC} = 12$ V with $\beta = 100$. What is the largest $|A_v|$ possible if the design keeps $V_{CEQ} = 6$ V? [1] (c) Design the stage ($I_C = 1$ mA) for $A_v = -150$ (use $|A_v| = g_m R_C$): find R_C, V_{CEQ} and R_B. [3] (d) Recalculate A_v using r_e (from I_E) and with $V_A = 100$ V, and comment. [2]

6. Quick recall check

1. Write g_m, r_π, r_e and r_o in terms of the Q-point quantities.
2. In the AC equivalent of a VDB amplifier, where do R_1, R_C and the bypassed R_E go?
3. Why is $g_m v_\pi$ the same as βi_b?
4. Bypassed VDB stage: give Z_i, Z_o and A_v (with r_o).
5. A CE stage has $V_{CC} = 10$ V and $V_C = 5$ V at Q. Estimate $|A_v|$ in one line.

BJT small-signal II

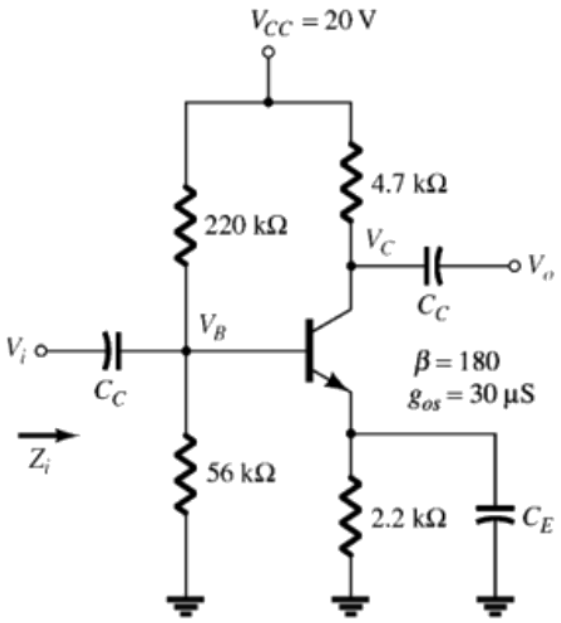
Practice questions · Day 12 · Wed 4 Nov · pack E7

5. Practice

From your course files

WS6 Q6 (course question). For the network of figure 6:

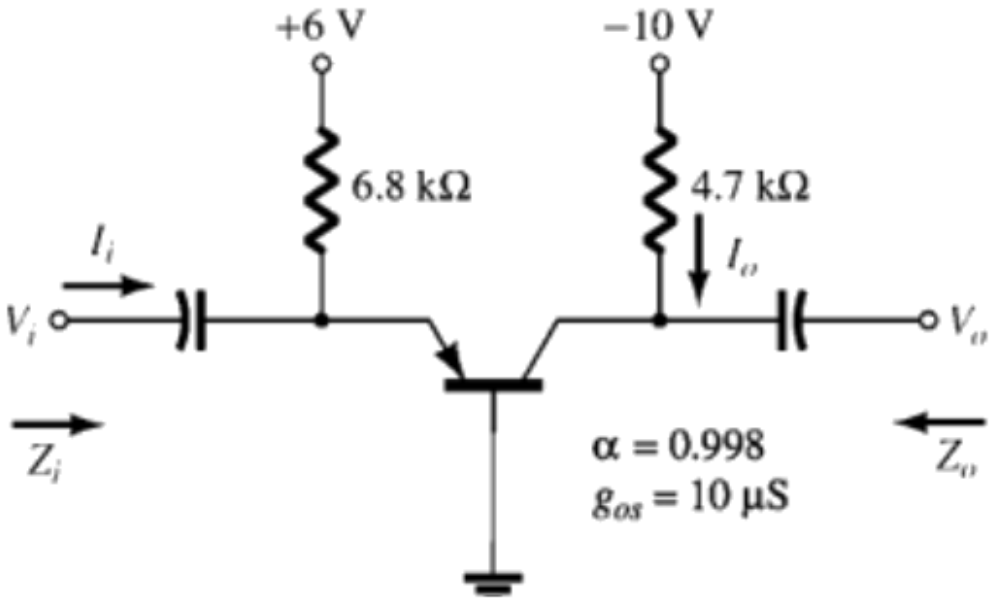
1. Determine r_e.
2. Calculate V_B and V_C.
3. Calculate Z_i and $A_V = V_o/V_i$.



WS6 Q6 figure (Figure 6 of the workshop sheet)

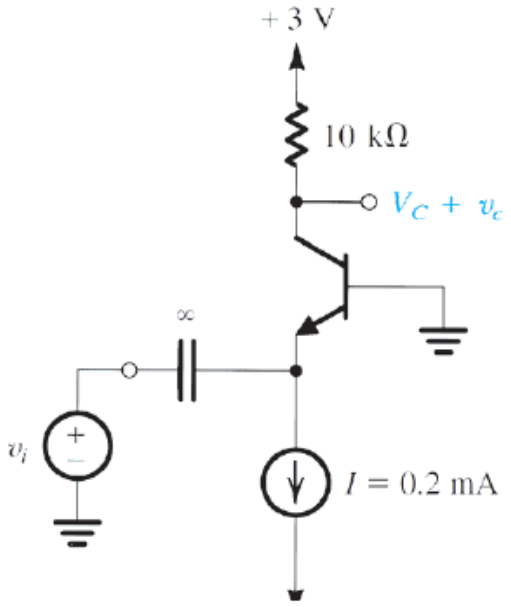
WS6 Q7 (course question). For the common-base configuration of figure 7:

1. Determine r_e.
2. Calculate Z_i and Z_o.
3. Calculate A_V.



WS6 Q7 figure (Figure 7 of the workshop sheet)

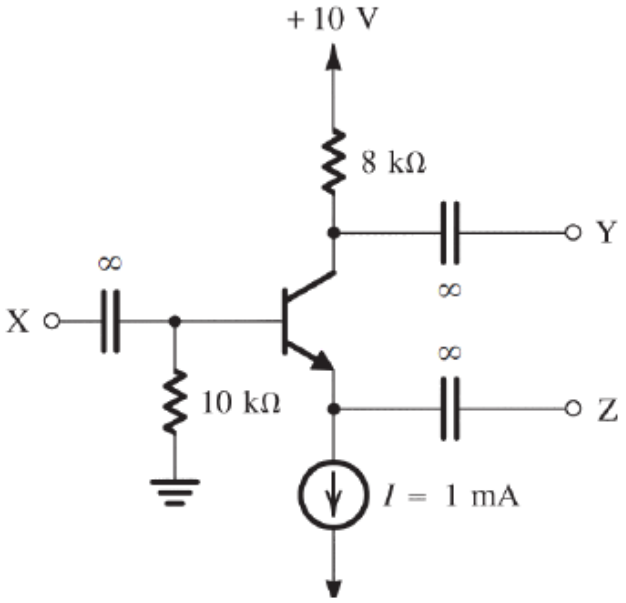
WS6 Q4 (course question). The transistor amplifier shown in the figure 9 [the figure below; the sheet labels it Figure 4] is biased with a current source $I = 0.2$ mA and has a very high β. Assuming the transistor is in the active state, the DC voltage of the collector is 1 V, and $r_e = 125\ \Omega$. Replace the transistor with the **T model** and find voltage gain, $\frac{v_c}{v_i}$.



WS6 Q4 figure (Figure 4 of the workshop sheet)

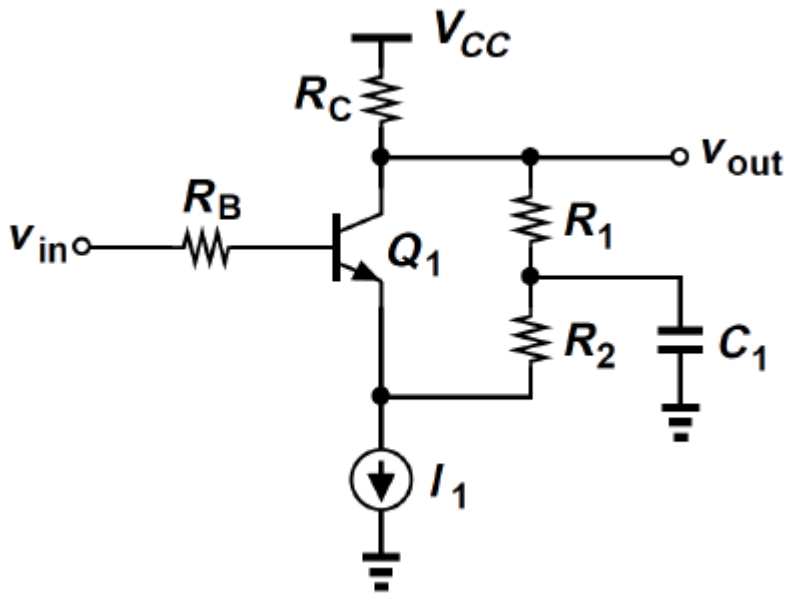
WS6 Q5 (course question). The transistor amplifier shown in the figure 10 [the figure below; the sheet labels it Figure 5] is biased with a current source $I = 1$ mA and has $\beta = 100$ and $V_A = 100$ V. Assuming that $g_m = 39.604$ mA/V, $r_\pi = 2.525\ \text{k}\Omega$ and $r_o = 101\ \text{k}\Omega$, answer the following questions:

1. If terminal Z is connected to ground, X to a signal source v_{sig} with a source resistance $R_{sig} = 2\ \text{k}\Omega$, and Y to an 8-kΩ load resistance, use the hybrid-π model to draw the small-signal equivalent circuit of the amplifier.
2. Calculate the overall voltage gain, $\dfrac{v_y}{v_{sig}}$.



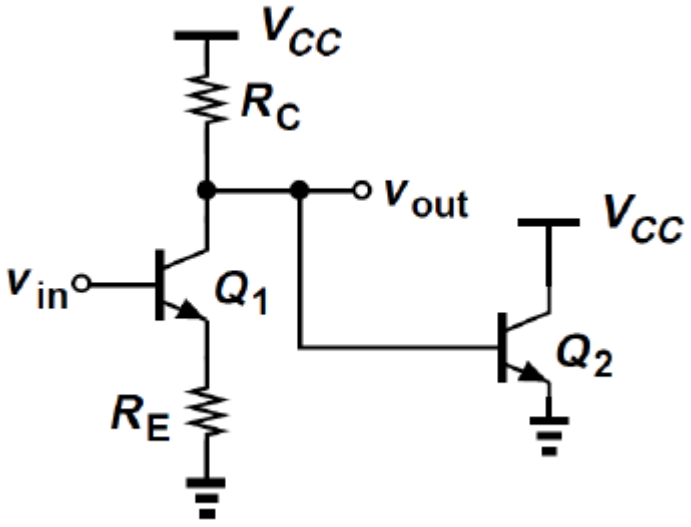
WS6 Q5 figure (Figure 5 of the workshop sheet)

WS6 Q9 (course question). The transistor amplifier shown in the figure 9 is biased with a current source I_1. Using a hybrid-π model, find expressions for AC voltage gain ($A_v = \dfrac{v_{out}}{v_{in}}$), input and output impedance. Neglect early voltage effect.



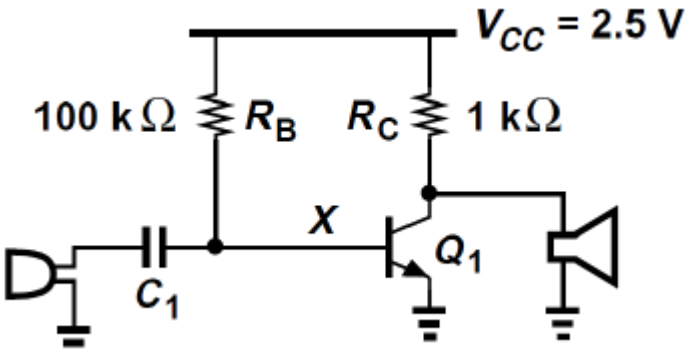
WS6 Q9 figure (Figure 9 of the workshop sheet)

WS6 Q10 (course question). The two BJTs in the amplifier of figure 10 are different. Using hybrid-π models, find an expression for AC voltage gain ($A_v = \frac{v_{out}}{v_{in}}$) of the amplifier.



WS6 Q10 figure (Figure 10 of the workshop sheet)

WS6 Q11 (course question). Typical speakers incorporate a solenoid (inductor) to actuate a membrane. The solenoid exhibits a very low DC resistance, e.g., less than 1 Ω. Having learned about AC coupling, the student in the previous Example modifies the design to that shown in figure 11 and attempts to drive a speaker. Unfortunately, the circuit still fails. Explain why. How can we resolve the issue?

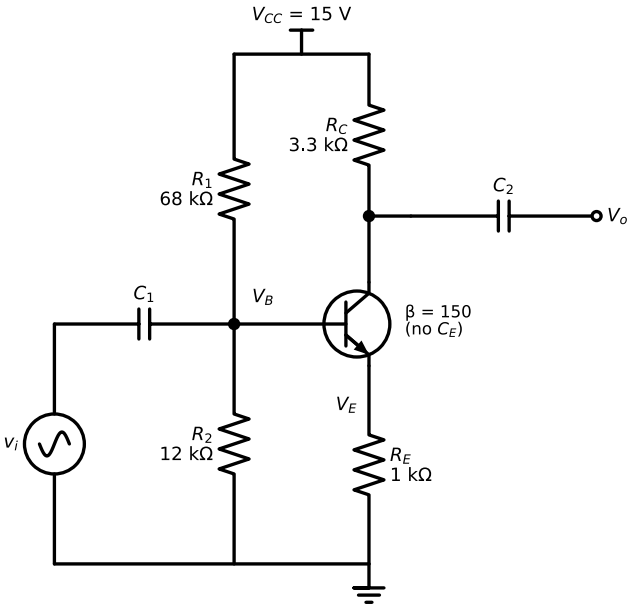


WS6 Q11 figure (Figure 11 of the workshop sheet)

New questions

N1 (easy) [5 marks]. A voltage-divider CE amplifier has $V_{CC} = 15$ V, $R_1 = 68\ \text{k}\Omega$, $R_2 = 12\ \text{k}\Omega$, $R_C = 3.3\ \text{k}\Omega$, $R_E = 1\ \text{k}\Omega$ (no bypass capacitor), $\beta = 150$.

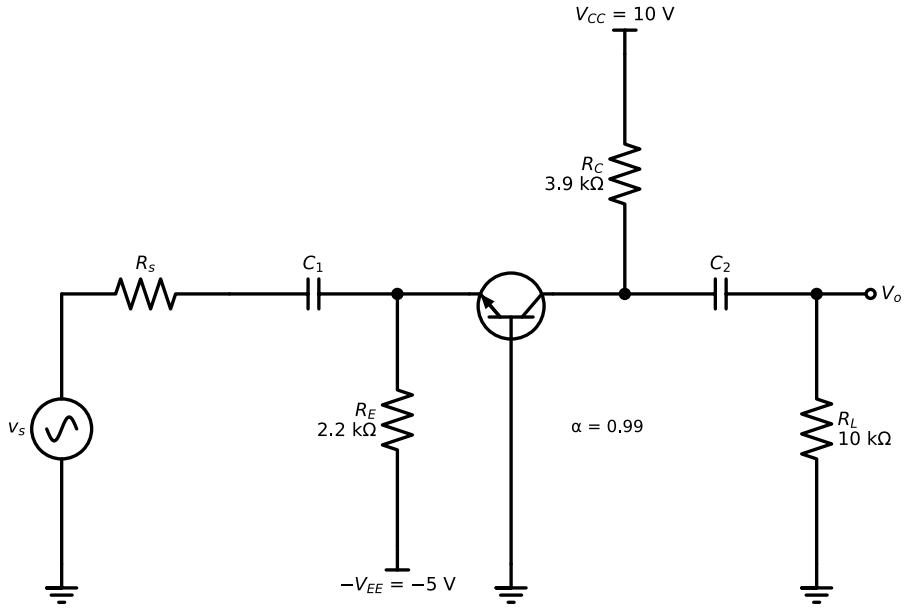
a. Find the Q-point using the approximate method, after checking that it is allowed. [2]
b. Find r_e, Z_i, Z_o and A_v. Compare A_v with $-R_C/R_E$ and with the gain if R_E were bypassed. [3]



N1: voltage-divider CE with R_E unbypassed

N2 (medium) [7 marks]. A common-base stage has $V_{EE} = 5$ V through $R_E = 2.2\ \text{k}\Omega$ to the emitter, $V_{CC} = 10$ V through $R_C = 3.9\ \text{k}\Omega$ to the collector, base grounded, $\alpha = 0.99$. A load $R_L = 10\ \text{k}\Omega$ is coupled to the collector.

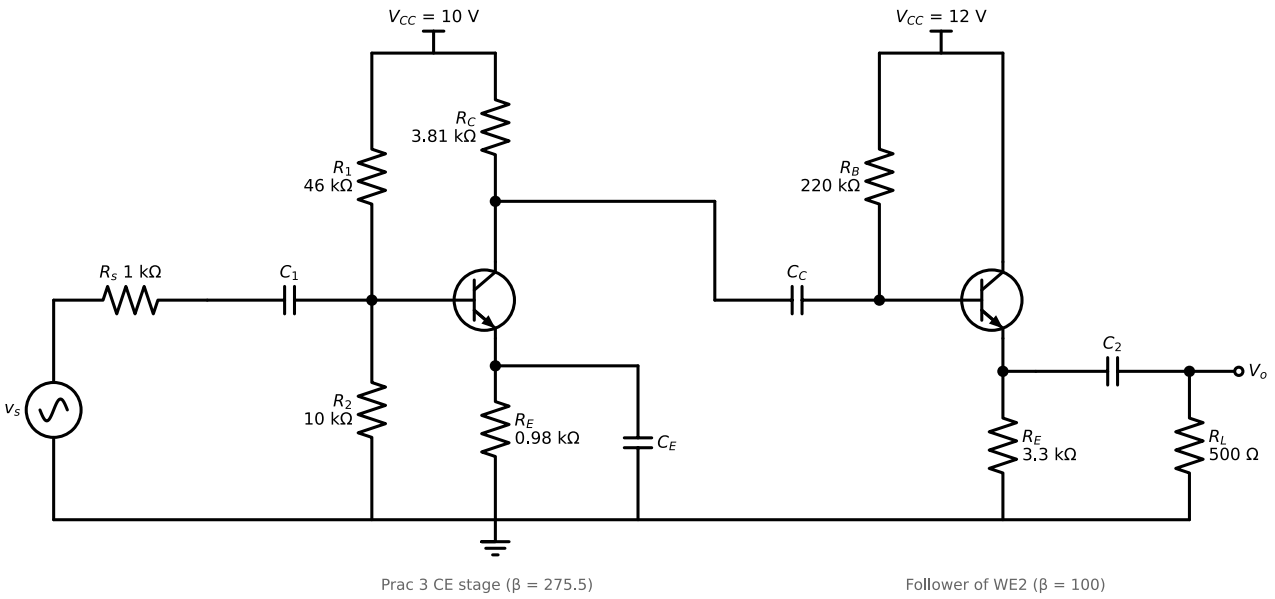
a. Find I_E, r_e and V_{CB}. [2]
b. Find Z_i, Z_o, $A_{v_{NL}}$ and A_{v_L}. [3]
c. Find A_{v_s} for $R_s = 50\ \Omega$ and for $R_s = 1\ \text{k}\Omega$. Comment. [2]



N2: common-base stage, signal into the emitter through C_1, base grounded

N3 (medium–hard, multi-part) [10 marks]. Your Prac 3 amplifier (bypassed; $Z_i = 3.60$ kΩ, $Z_o = 3.81$ kΩ, $A_{v_{NL}} = -164$, from Worked example 4) is driven from $R_s = 1$ kΩ and must drive $R_L = 500\ \Omega$.

a. Using the two-port approach, find A_{v_L} and A_{v_s} with the load connected directly. [3]
b. The emitter follower of Worked example 2 ($R_B = 220$ kΩ, $R_E = 3.3$ kΩ, β = 100, $r_e = 12.60\ \Omega$) is inserted between the CE collector and the 500 Ω load. Find the follower's Z_i with the 500 Ω load. [2]
c. Find the CE stage gain now that it is loaded by that Z_i, the follower gain, and the overall A_{v_s}. [3]
d. Find the output impedance the 500 Ω load now sees. [2]



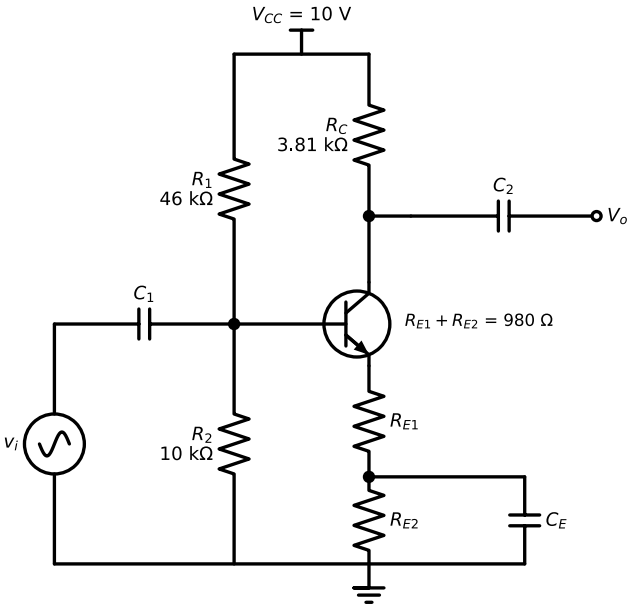
N3: the Prac 3 CE stage driving the emitter follower of Worked example 2, which drives the 500 Ω load

N4 (design) [6 marks]. In the Prac 3 amplifier ($I_E = 1.12$ mA, $r_e = 23.2\ \Omega$, $R_C = 3.81$ kΩ, $R' = 8.214$ kΩ, β = 275.5) you want a no-load gain of -10 without changing the Q-point.

a. Explain how to split $R_E = 980\ \Omega$ and choose the unbypassed part R_{E1}. Pick standard values. [3]

b. Find A_v and Z_i with your values. Repeat Z_i for β = 100. [2]

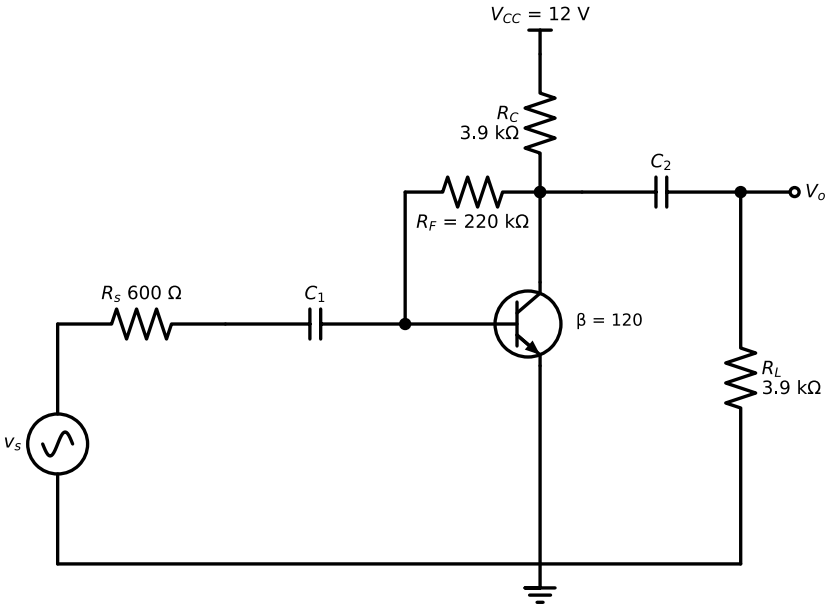
c. I_E rises 10 % with temperature. By what percentage does A_v change, bypassed versus your design? [1]



N4: the Prac 3 amplifier with R_E split into an unbypassed R_{E1} and a bypassed R_{E2}

N5 (hard) [8 marks]. The WS6 Q8 collector-feedback amplifier ($r_e = 13.08\ \Omega$, β = 120, $R_C = 3.9$ kΩ, $R_F = 220$ kΩ, ignore r_o) now drives $R_L = 3.9$ kΩ and is fed from $R_s = 600\ \Omega$.

a. Find A_{v_L}. [2]
b. Use the Miller view to find Z_i with the load connected. Explain why it differs from the unloaded 507 Ω. [3]
c. Find A_{v_s}. [1]
d. A student uses the two-port approach with the unloaded Z_i and $Z_o = R_C \parallel R_F$. What do they get, and why is it wrong? [2]



N5: the WS6 Q8 collector-feedback amplifier fed from R_s = 600 Ω and driving R_L = 3.9 kΩ

6. Quick recall check

1. A CE stage has $R_C = 4.7$ kΩ and $R_E = 1.2$ kΩ unbypassed. Roughly what is A_v, and which transistor parameter is missing from it?
2. What is Z_o of an emitter follower with $r_e = 10\ \Omega$, $R_E = 2$ kΩ, driven from $R_s = 2$ kΩ (ignore R_B), β = 100?
3. Which two configurations have Z_i that changes when R_L is connected?
4. Write A_{v_s} in terms of Z_i, R_s and A_{v_L}.
5. With $R_E = R_{E1} + R_{E2}$ and only R_{E2} bypassed, which resistance goes in the DC loop and which in A_v?

Multistage amplifiers

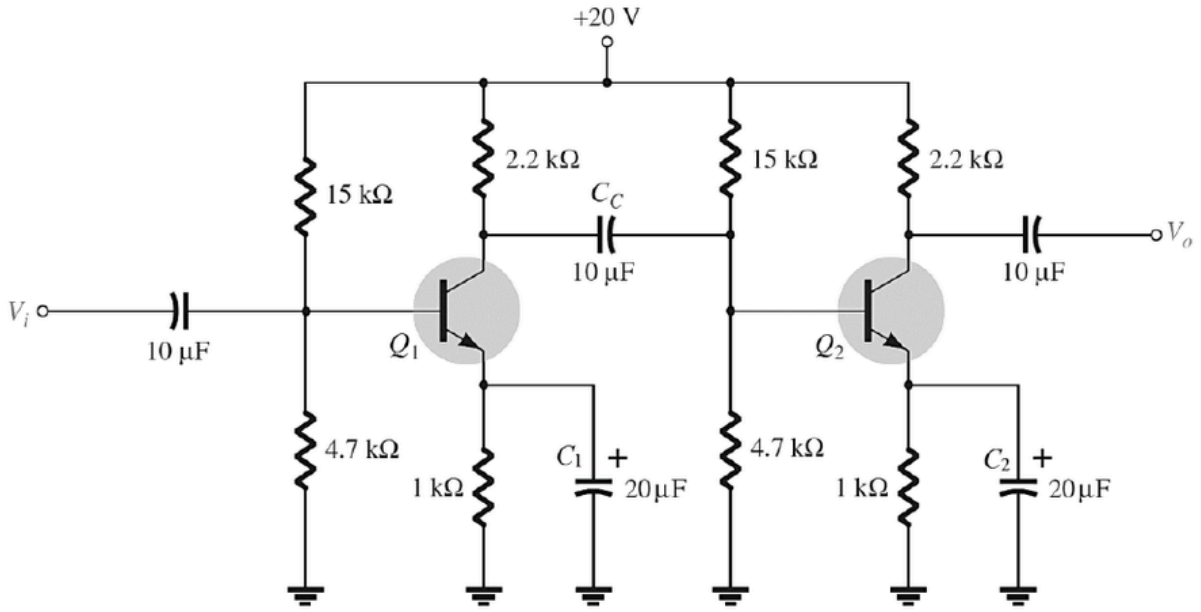
Practice questions · Day 14 · Fri 6 Nov · pack E8

5. Practice

From your course files

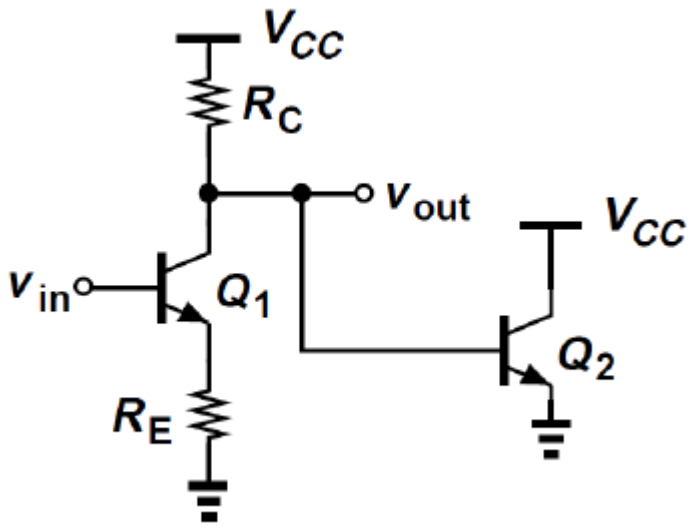
1. Prac 4 calculation section ("For the network of Figure 1: 1) Calculate the no-load voltage gain of the RC-coupled transistor amplifiers in Figure 1. 2) Calculate the overall gain and output voltage if a 4.7 kΩ load is applied to the second stage and compare it to the results of part (1). 3) Calculate the input impedance of the first stage and the output impedance of the second stage.") is written out in full as Worked Example 2 above. Redo it without looking: no-load 463, loaded 316, Z_{i_1} = 941 Ω, Z_{o_2} = 2.2 kΩ.

Prac 4 step 6 (practical). Add a bypass capacitor at the emitter of the second stage. The network should be the one in Figure 3. Calculate the voltage gain of the network of Figure 3.



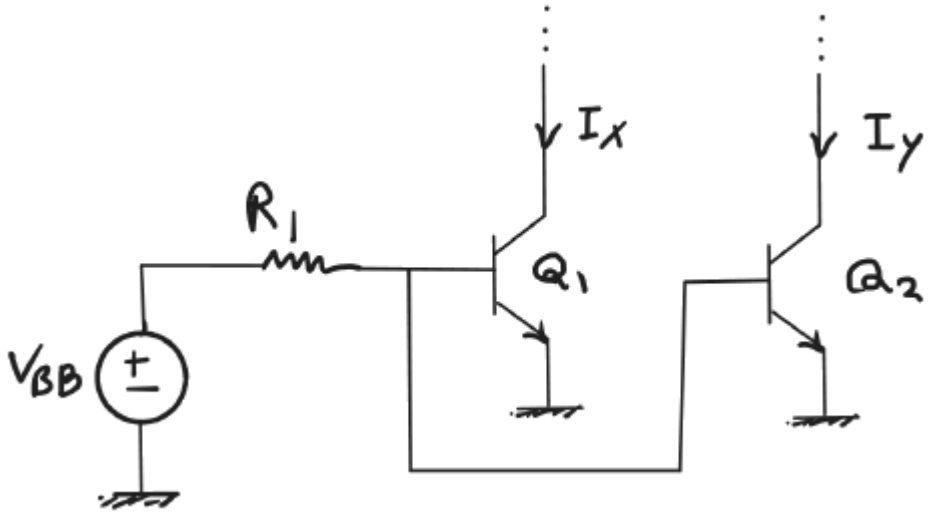
Prac 4 Figure 3 (from the practical handout)

WS6 Q10 (course question). The two BJTs in the amplifier of figure 10 are different. Using hybrid-π models, find an expression for AC voltage gain ($A_v = \frac{v_{out}}{v_{in}}$) of the amplifier.



WS6 Q10 figure (Figure 10 of the workshop sheet)

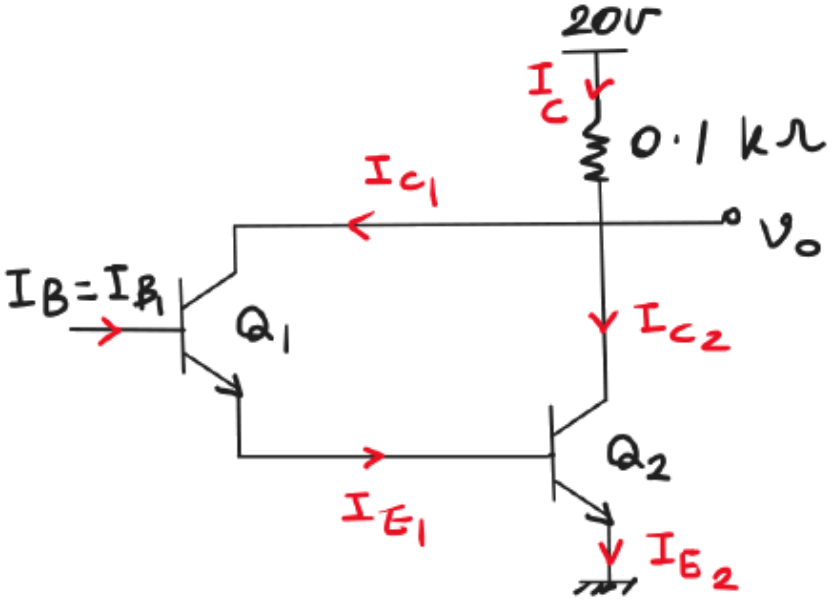
WS6 Q12 (course question). In the circuit shown in figure 12, the two BJTs are identical with $\beta = 100$, $V_{BE,on} = 0.7$ V and $V_{CE,sat} = 0.2$. If $I_X = 2I_Y = 1$ mA and $R_1 = 5\ k\Omega$, find voltage V_{BB}. Assume the two BJTs are in the active state.



WS6 Q12 figure (Figure 12 of the workshop sheet)

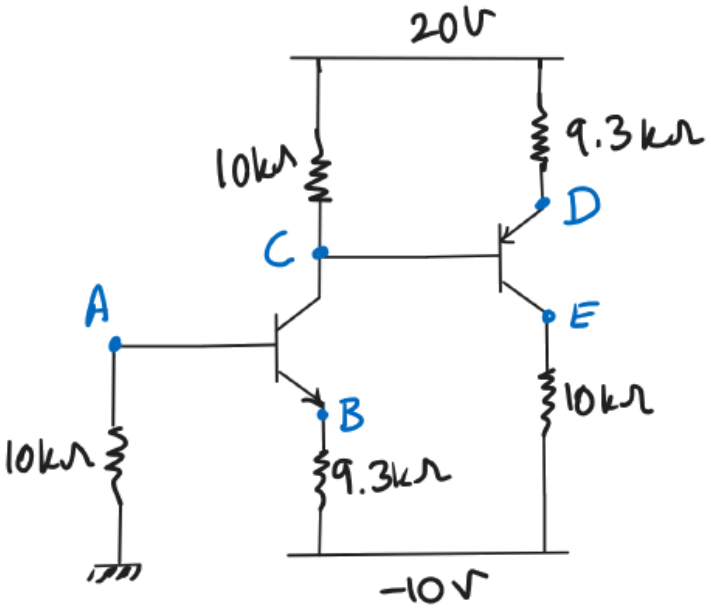
WS6 Q13 (course question). In the circuit shown in figure 13, $I_{E_2} = 120$ mA, $\alpha_1 = 0.99$, $\alpha_2 = 0.98$. Assuming that both transistors are in the **active** state, answer the following questions:

1. Find I_{C_1}, I_{B_1}, I_{E_1}, I_{C_2}, and I_C.
2. Find the output voltage, v_o.



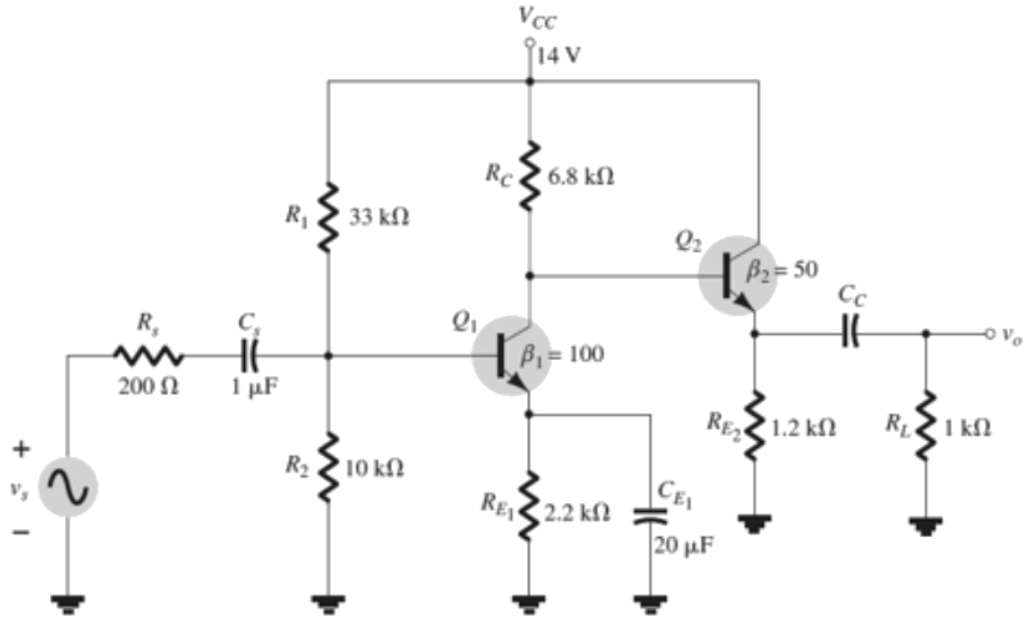
WS6 Q13 figure (Figure 13 of the workshop sheet)

WS6 Q14 (course question). In the circuit shown in Figure 14, both transistors have $V_{BE,on} = 0.7$ V. Assuming that $\beta = \infty$ and both transistors are in either **active or saturation** state, find the voltage at nodes A, B, C, D, and E.



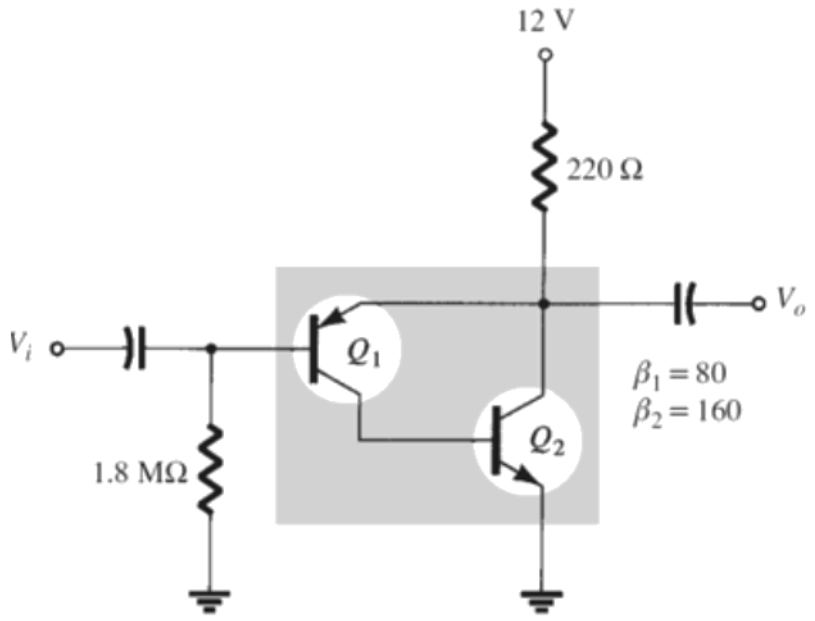
WS6 Q14 figure (Figure 14 of the workshop sheet)

WS6 Q16 (course question). Determine the dc levels for the currents and voltages of the direct-coupled amplifier of figure 16.



WS6 Q16 figure (Figure 16 of the workshop sheet)

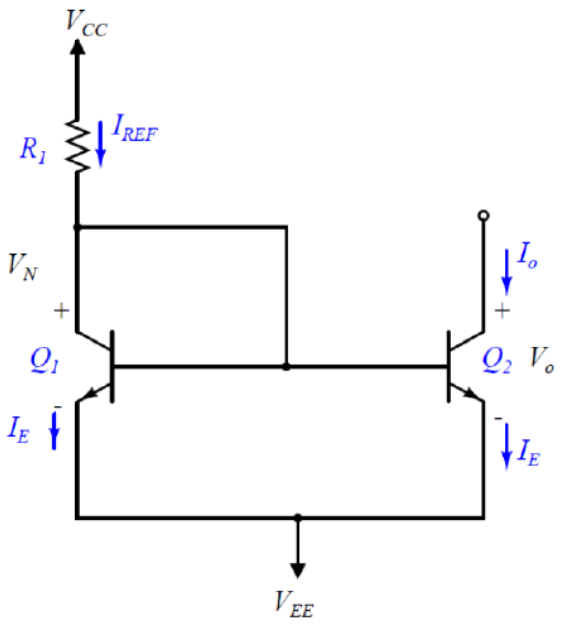
WS6 Q17 (course question). For the feedback amplifier of figure 17 determine (a) the base and collector current of each transistor and (b) the base, emitter, and collector voltages of each transistor.



WS6 Q17 figure (Figure 17 of the workshop sheet)

WS6 Q18 (course question). In the circuit below, a current mirror is shown where the transistors Q_1 and Q_2 are identical.

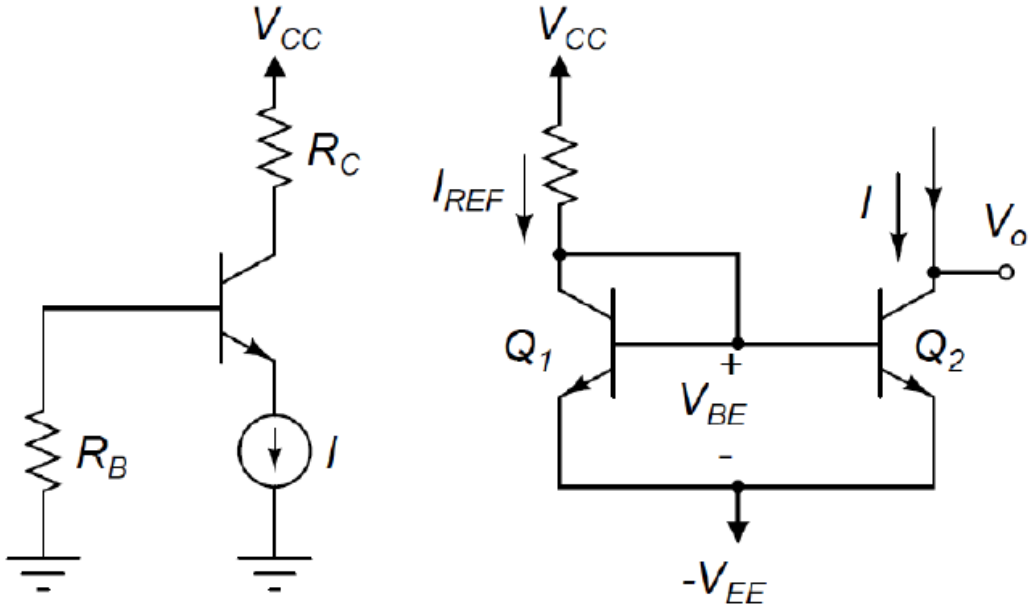
1. Assume the transistor β is not very large, derive an expression for the ratio of $\frac{I_o}{I_{REF}}$.
2. Simplify this expression if β is very large.



WS6 Q18 figure (from the workshop sheet)

WS6 Q19 (course question). In the circuit shown below, $V_{CC} = 10$ V, $I = 1$ mA, $\beta = 100$, $R_B = 100\ k\Omega$ and $R_C = 7.5\ k\Omega$

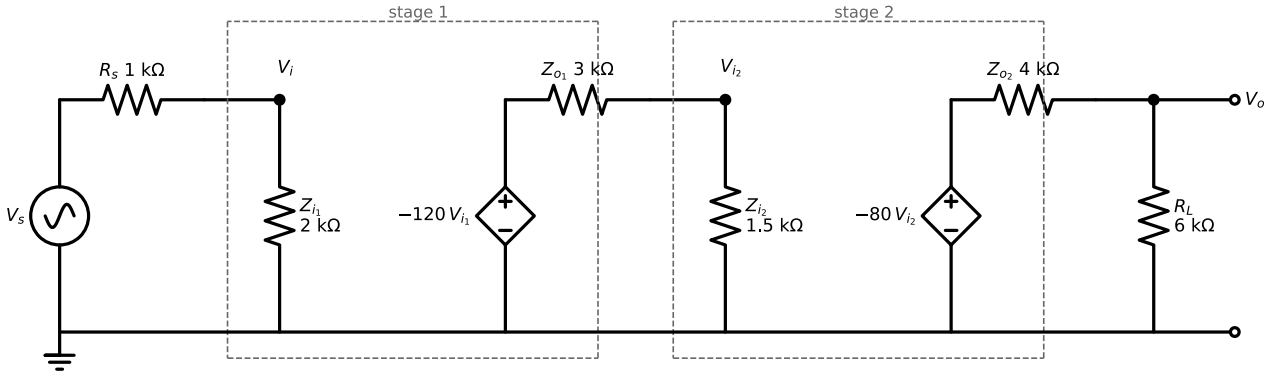
1. Find the DC voltage at the base, the emitter and the collector terminals of the left circuit.
2. Assuming $V_{EE} = 10$ V, and assuming β is very large, find the required value of the resistor in order for the right circuit to implement the current source I shown in the left circuit. Q_1 and Q_2 are identical.



WS6 Q19 figure (from the workshop sheet)

New questions

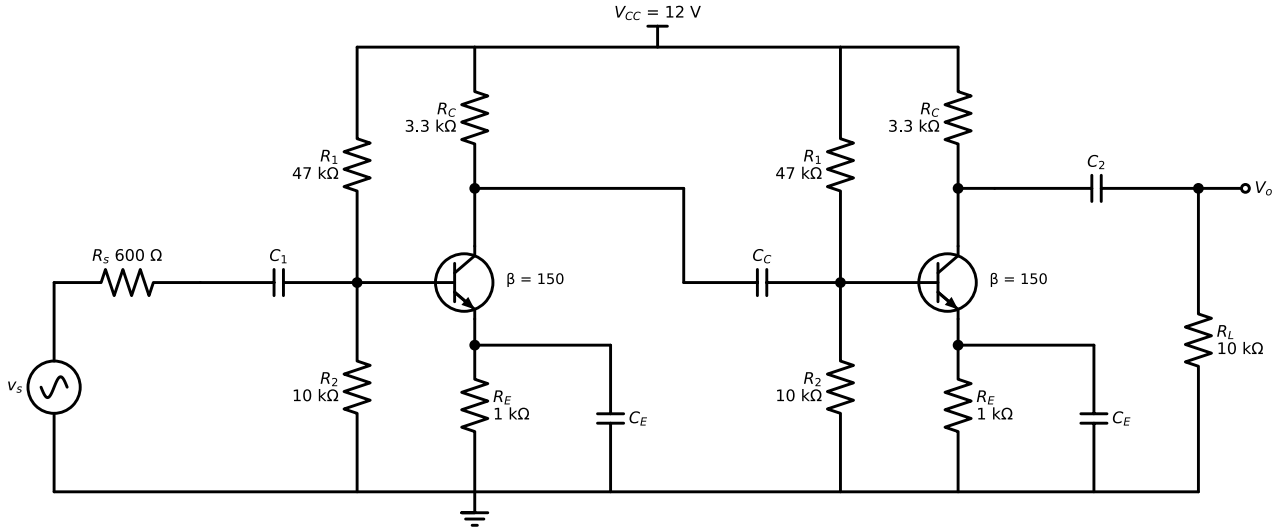
N1 [5 marks]. Two-port cascade. A two-stage amplifier is described by two-port parameters. Stage 1 has $A_{v_{NL1}} = -120$, $Z_{i_1} = 2$ kΩ, $Z_{o_1} = 3$ kΩ. Stage 2 has $A_{v_{NL2}} = -80$, $Z_{i_2} = 1.5$ kΩ, $Z_{o_2} = 4$ kΩ. The source has $R_s = 1$ kΩ and the load is $R_L = 6$ kΩ. (a) Find the loaded gain of each stage and $A_{v_T} = V_o/V_i$. [3] (b) Find $A_{v_s} = V_o/V_s$. [1] (c) By what factor is A_{v_s} below the "ideal" product $A_{v_{NL1}} A_{v_{NL2}}$? [1]



N1: the two-stage amplifier as a chain of two-port models

N2 [10 marks]. Two-stage CE–CE cascade. Two identical common-emitter stages are RC-coupled. Each stage uses voltage-divider bias from $V_{CC} = 12$ V, with $R_1 = 47$ kΩ from V_{CC} to the base and $R_2 = 10$ kΩ from the base to ground. Each has $R_C = 3.3$ kΩ from V_{CC} to the collector, and $R_E = 1$ kΩ from the emitter to ground, **fully bypassed** by a large capacitor. Both transistors have $\beta = 150$ and $r_o = \infty$. All coupling capacitors are shorts at signal frequency. A source with $R_s = 600\ \Omega$ drives the base of Q1. The collector of Q2 drives an $R_L = 10$ kΩ load through a coupling capacitor.

a. Find V_B, I_E, V_{CE} and r_e for each stage, justifying the approximate method. [3]
b. Find Z_{i_2} and the loaded gain of stage 1. [2]
c. Find the gain of stage 2 with R_L, and the overall $A_{v_T} = V_o/V_i$. [2]
d. Find Z_{i_1} and $A_{v_s} = V_o/V_s$. What is v_o for $v_s = 20$ µV peak? [3]



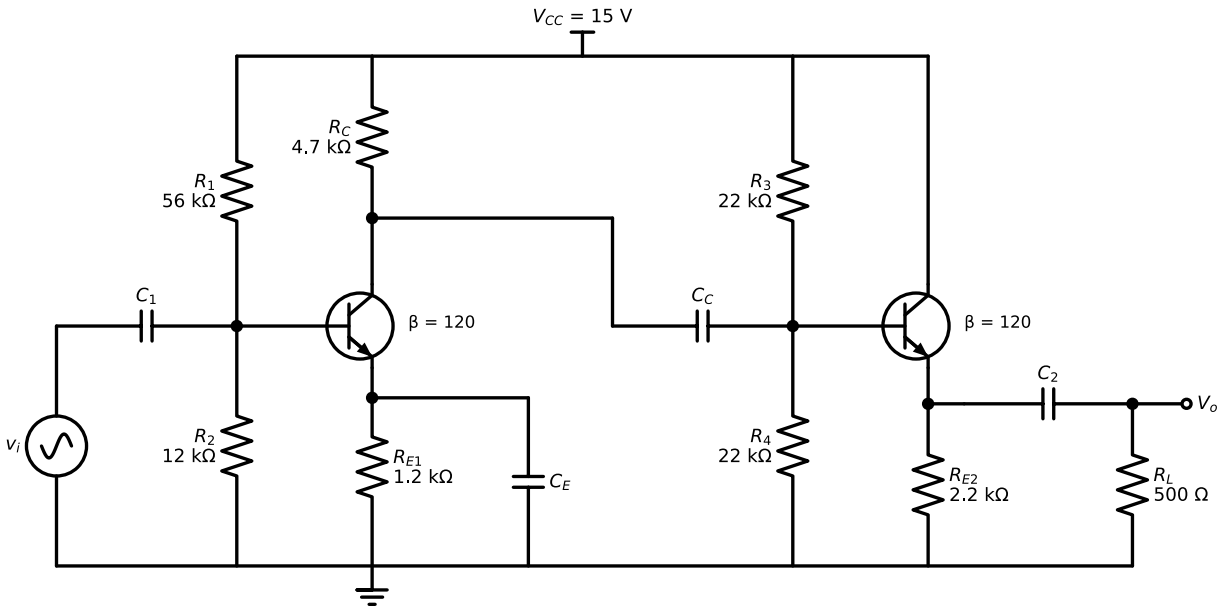
N2: two identical RC-coupled CE stages with R_s = 600 Ω and R_L = 10 kΩ

N3 [10 marks]. CE–EF cascade (buffering a low-resistance load). Stage 1 is a common-emitter amplifier on $V_{CC} = 15$ V with $R_1 = 56$ kΩ (to V_{CC}), $R_2 = 12$ kΩ (to ground), $R_C = 4.7$ kΩ, and $R_{E1} = 1.2$ kΩ fully bypassed. Its collector is capacitor-coupled to the base of stage 2. Stage 2 is an emitter follower on the same 15 V supply, with base divider $R_3 = 22$ kΩ (to V_{CC}) and $R_4 = 22$ kΩ (to ground), collector tied directly to V_{CC}, and $R_{E2} = 2.2$ kΩ from emitter to ground. The emitter of Q2 drives an $R_L = 500\ \Omega$ load through a coupling capacitor. Both transistors have $\beta = 120$ and $r_o = \infty$.

a. Find I_{E1}, r_{e1}, I_{E2} and r_{e2}. [3]
b. Find Z_{i_2}. [2]

c. Find A_{v_1}, A_{v_2} and A_{v_T}. [3]

d. Find the gain if the 500 Ω load were connected straight to Q1's collector (no follower), and explain the difference. [2]



N3: CE stage capacitor-coupled to an emitter follower that drives the 500 Ω load

N4 [7 marks]. Darlington emitter follower. A Darlington pair (Q1 emitter to Q2 base, collectors tied together to V_{CC}) is biased from $V_{CC} = 18$ V by $R_B = 3.3$ MΩ from V_{CC} to Q1's base. An emitter resistor $R_E = 390\ \Omega$ runs from Q2's emitter to ground. $\beta_1 = 100$, $\beta_2 = 80$, and the Darlington $V_{BE} = 1.6$ V. (a) Find β_D, I_B, I_E, V_E and V_{CE}. [4] (b) Find Z_i and A_i (no load), and state A_v. [3]

N4: Darlington emitter follower

N5 [6 marks]. Current mirror design. Two identical npn transistors form a basic current mirror. Both emitters are grounded and the bases are tied together. Q1 is diode-connected, and R_{REF} runs from $V_{CC} = 12$ V to Q1's collector. Q2's collector draws the output current I_o through a load resistor R_L from the same 12 V rail. $V_{BE} = 0.7$ V, $V_{CE,\text{sat}} = 0.2$ V and $\beta = 80$.

a. Choose R_{REF} for $I_o = 2$ mA, assuming β is very large. [1]
b. With that resistor and $\beta = 80$, what is the actual I_o? [2]
c. What R_{REF} gives exactly 2 mA with $\beta = 80$? [2]
d. What is the largest R_L that keeps Q2 active? [1]

N5: basic current mirror with R_{REF} to be designed and load R_L on Q_2

6. Quick recall check

1. In a two-stage cascade, which stage's gain changes when you connect stage 2 to stage 1, and what resistance causes it?
2. Write Z_{i_2} for a stage 2 that is a CE stage with an **unbypassed** R_E.
3. A Darlington has $\beta_1 = 50$ and $\beta_2 = 120$. What are β_D and the approximate Z_i (ignoring R_B) with $R_E = 1$ kΩ?
4. Why does a cascode have a wider bandwidth than a single CE stage with the same gain?
5. A current mirror has $\beta = 50$. What is I_o/I_{REF}?

FET physics

Practice questions · Day 15 · Sat 7 Nov · pack E9

5. Practice

From your course files

WS7 Q3 (course question). Given $I_{DSS} = 6$ mA and $V_P = -4.5$ V:

a. Determine I_D at $V_{GS} = -2$ V.
b. Determine V_{GS} at $I_D = 3$ mA.

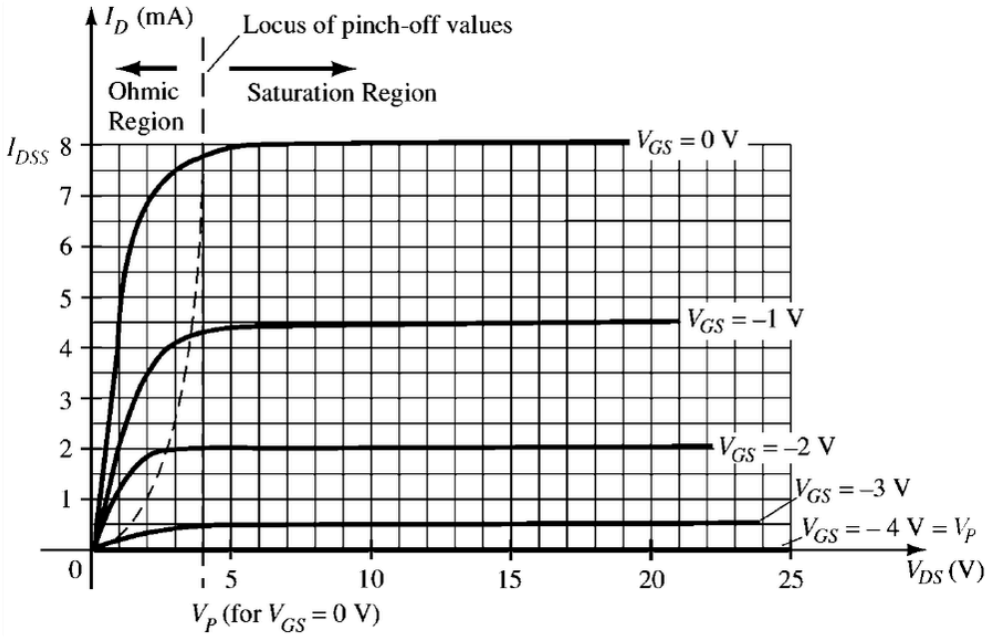
WS7 Q6 (course question). Given a Q-point of $I_{D_Q} = 3$ mA and $V_{GS} = -3$ V, determine I_{DSS} if $V_P = -6$ V.

WS7 Q5 (course question). Given $I_{DSS} = 12$ mA and $V_P = -4$ V, sketch the transfer characteristics for the JFET transistor.

WS7 Q1 (course question). Given $I_{DSS} = 12$ mA and $|V_P| = 6$ V, sketch a probable distribution of characteristic curves for the JFET.

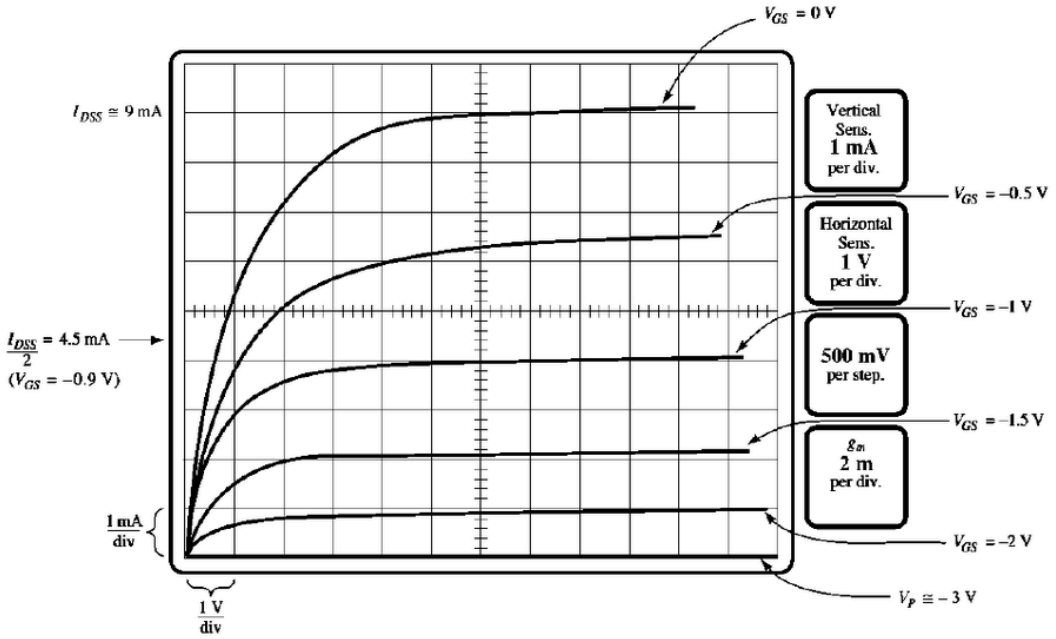
WS7 Q7 (course question). Sketch the transfer and drain characteristics of an n-channel depletion-type MOSFET with $I_{DSS} = 12$ mA and $V_P = -8$ V for a range of $V_{GS} = -V_P$ to $V_{GS} = 1$ V.

WS7 Q2 (course question). Using the characteristics of Fig. 1, determine the difference in drain current (for $V_{DS} > V_P$) between $V_{GS} = 0$ V and $V_{GS} = -1$ V.



WS7 Fig. 1 (from the workshop sheet)

WS7 Q4 (course question). Using the characteristics of Fig. 2, determine I_D at $V_{GS} = -0.7$ V and $V_{DS} = 10$ V.



WS7 Fig. 2 (from the workshop sheet)

New questions

Q1 [3 marks] A p-channel JFET has $I_{DSS} = 6$ mA and $V_P = +6$ V (Week 6 slide 13 device).

a. Find I_D at $V_{GS} = +2$ V. [1]
b. Find V_{GS} for $I_D = 1.5$ mA. [1]
c. State the direction of I_D and the sign of V_{DS}. [1]

Q2 [4 marks] An n-channel E-MOSFET datasheet gives $V_{GS(\text{Th})} = 2$ V and $I_{D(\text{on})} = 9$ mA at $V_{GS(\text{on})} = 8$ V.

a. Find k. [1]
b. Find I_D at $V_{GS} = 5$ V and at $V_{GS} = 1.5$ V. [1]
c. Find V_{GS} for $I_D = 4$ mA. [1]
d. What minimum V_{DS} keeps the device in saturation at that point? [1]

Q3 [5 marks] An n-channel depletion MOSFET has $I_{DSS} = 8$ mA and $V_P = -3$ V.

a. Find I_D at $V_{GS} = -1$ V, at -1.5 V and at $+1$ V. [3]
b. Which of these points are in depletion mode and which in enhancement mode? [1]
c. Find the exact V_{GS} for $I_D = 4$ mA, and compare it with the shorthand value. [1]

Q4 [8 marks] (chained) An n-channel JFET in saturation is measured at two bias points: $I_D = 4.5$ mA at $V_{GS} = -1$ V, and $I_D = 0.5$ mA at $V_{GS} = -3$ V.

a. Find V_P. [3]
b. Find I_{DSS}. [1]
c. Find I_D at $V_{GS} = -2$ V, and say which shorthand point this is. [1]
d. At $V_{GS} = -2$ V, find the V_{DS} at which the device enters saturation. [1]
e. In the ohmic region the channel resistance at $V_{GS} = 0$ is $r_o = 250\ \Omega$. Find r_d at $V_{GS} = -2$ V. [2]

Q5 [6 marks] A CMOS inverter runs from $V_{DD} = 5$ V. The nMOS has $V_T = +1$ V and the pMOS has $V_T = -1$ V. Its VTC gives $V_{OL} = 0.1$ V, $V_{OH} = 4.9$ V, $V_{IL} = 2.1$ V and $V_{IH} = 2.9$ V.

a. For $V_i = 0$, 2.5 V and 5 V, find each transistor's V_{GS} and whether it is on or off. Give V_o for the two logic inputs. [3]
b. Find NM_L and NM_H, and compare them with the ideal inverter. [2]
c. With $t_{PHL} = 12$ ns and $t_{PLH} = 18$ ns, find t_P. [1]

Q6 [6 marks] (hard, chained) The pMOS in a CMOS inverter has $V_{GS(\mathrm{Th})} = -1.5$ V and $I_{D(\mathrm{on})} = 5$ mA at $V_{GS(\mathrm{on})} = -6.5$ V.

a. Find k. [1]
b. Find $|I_D|$ at $V_{GS} = -4$ V, assuming saturation. [1]
c. Find V_{GS} for $|I_D| = 1.8$ mA. Show both roots and justify your choice. [2]
d. With $V_{DD} = 5$ V and $V_i = 0.5$ V, find the pMOS V_{GS} and its saturation current. State whether the nMOS ($V_T = +1.5$ V) conducts. [2]

6. Quick recall check

1. What two conditions define I_{DSS}?
2. For $V_P = -5$ V, what is I_D (as a fraction of I_{DSS}) at $V_{GS} = -2.5$ V, and what V_{GS} gives $I_{DSS}/2$?
3. Which FET type has no I_{DSS}, and what equation replaces Shockley's?
4. For a pMOS in a CMOS inverter with $V_{DD} = 5$ V and $V_i = 0$, what is V_{GS}, and is the pMOS on?
5. Write NM_L and NM_H in terms of the VTC levels.

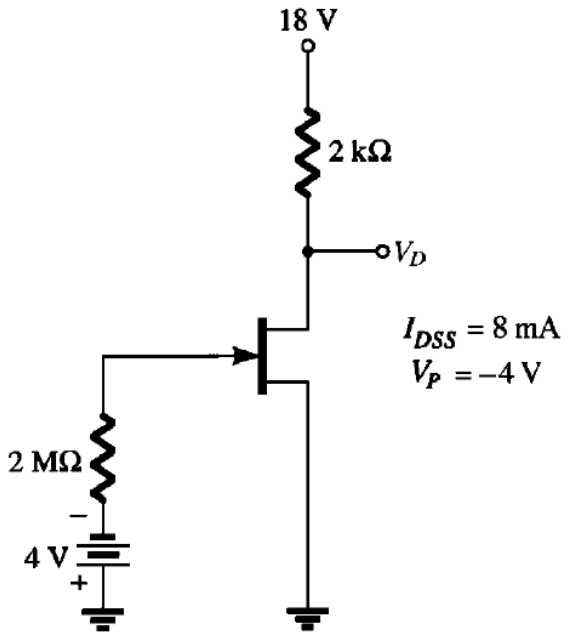
FET DC bias

Practice questions · Day 17 · Mon 9 Nov · pack E10

5. Practice

From your course files

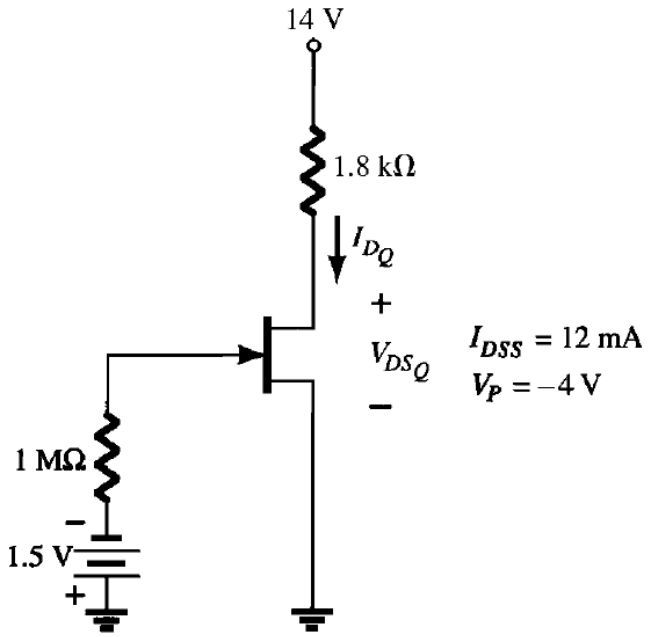
WS8 Q2 (course question). Determine V_D and V_{GS} for the fixed-bias configuration of Fig. 2.



WS8 Fig. 2 (from the workshop sheet)

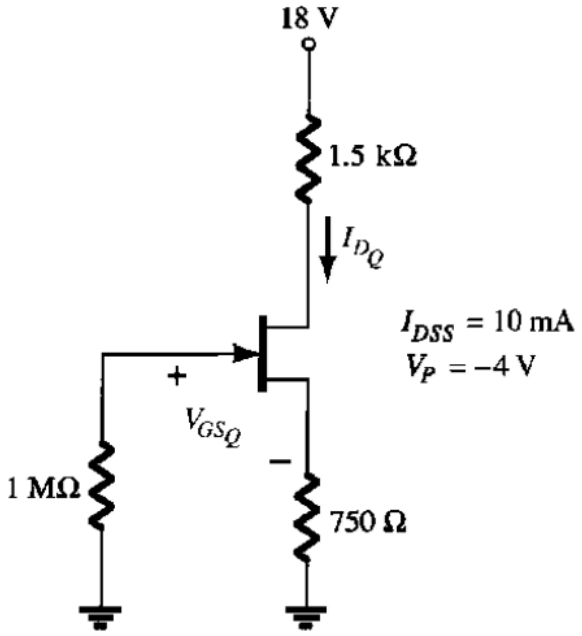
WS8 Q1 (course question). For the fixed-bias configuration of Fig. 1:

a. Sketch the transfer characteristics of the device.
b. Determine I_{D_Q} and V_{DS_Q}.
c. Using Shockley's equation, solve for I_{D_Q} and then find V_{DS_Q}. Compare with the solutions of part (b).



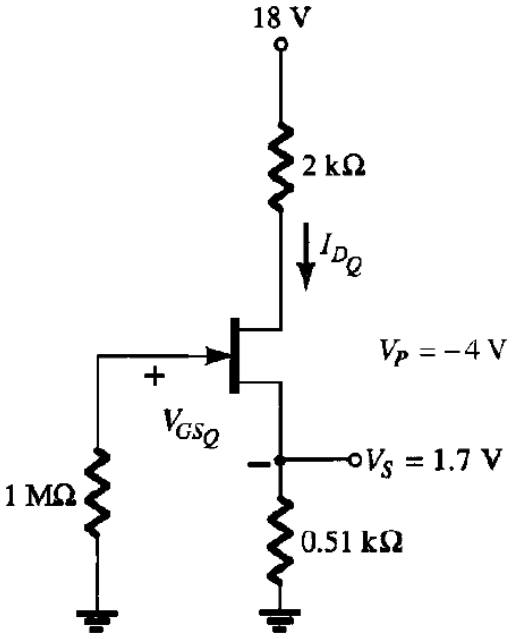
WS8 Fig. 1 (from the workshop sheet)

WS8 Q3 (course question). Determine I_{D_Q} for the network of Fig. 3 using a purely mathematical approach.



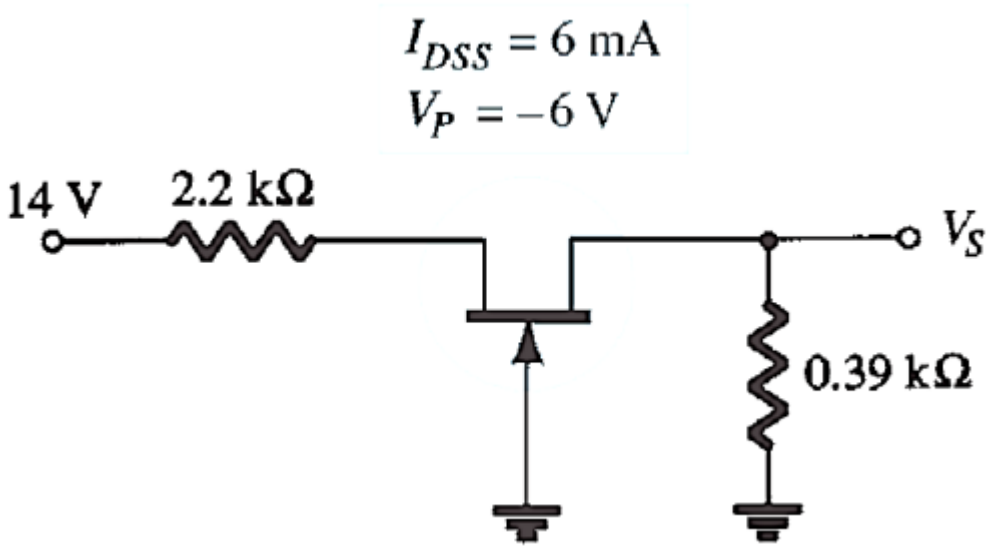
WS8 Fig. 3 (from the workshop sheet)

WS8 Q4 (course question). Given the measurement $V_S = 1.7$ V for the network of Fig. 4, determine: (a) I_{D_Q}, (b) V_{GS_Q}, (c) I_{DSS}, (d) V_D and (e) V_{DS}.



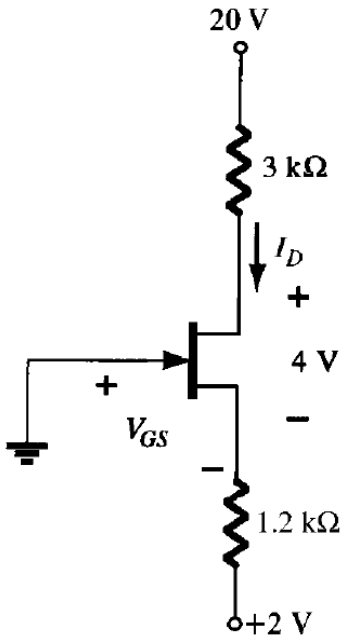
WS8 Fig. 4 (from the workshop sheet)

WS9 Q1 (course question). Find V_S for the network of Fig. 1.



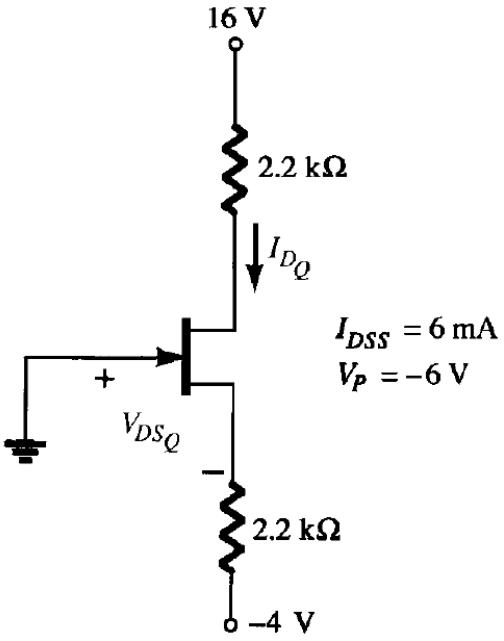
WS9 Fig. 1 (from the workshop sheet)

WS9 Q3 (course question). Given $V_{DS} = 4$ V for the network of Fig. 3, determine: I_D, V_D, V_S and V_{GS}.



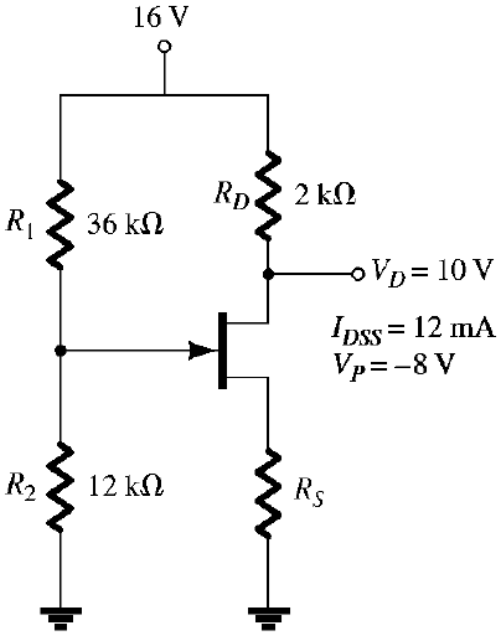
WS9 Fig. 3 (from the workshop sheet)

WS9 Q2 (course question). For the network of Fig. 2, determine: (a) I_{D_Q} and V_{GS_Q} and (b) V_{DS} and V_S.



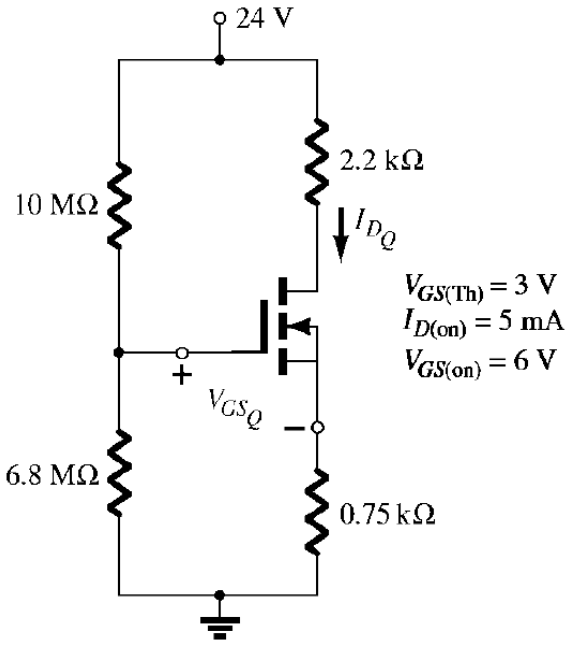
WS9 Fig. 2 (from the workshop sheet)

WS8 Q5 (course question). Determine the value of R_S the network of Fig. 5 to establish $V_D = 10$ V.



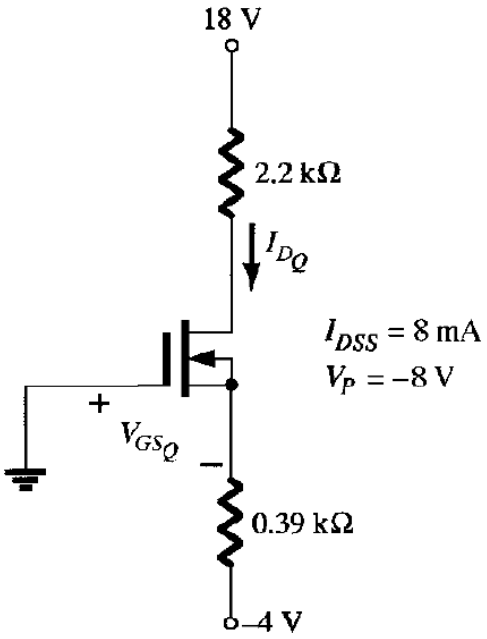
WS8 Fig. 5 (from the workshop sheet)

WS8 Q6 (course question). For the voltage-divider configuration of Fig. 6, determine: (a) I_{D_Q} and V_{GS_Q} and (b) V_D and V_S.



WS8 Fig. 6 (from the workshop sheet)

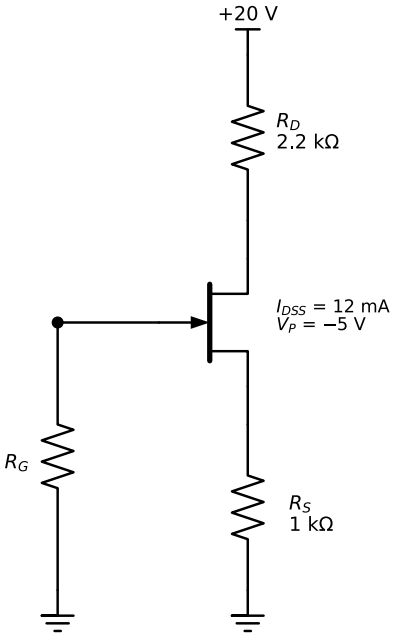
WS9 Q4 (course question). For the network of Fig. 4, determine: (a) I_{D_Q} and V_{GS_Q} and (b) V_{DS} and V_S.



WS9 Fig. 4 (from the workshop sheet)

New questions

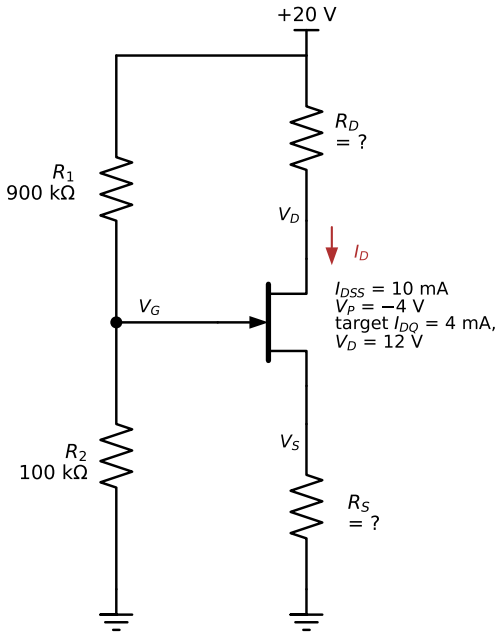
N1. [4 marks] A JFET with $I_{DSS} = 8$ mA and $V_P = -6$ V is fixed-biased with $V_{GG} = 2$ V (so $V_{GS} = -2$ V), $R_D = 1.5$ kΩ and $V_{DD} = 15$ V. Find I_{DQ} and V_{DSQ}, and confirm saturation.



Circuit for N2: self-bias JFET

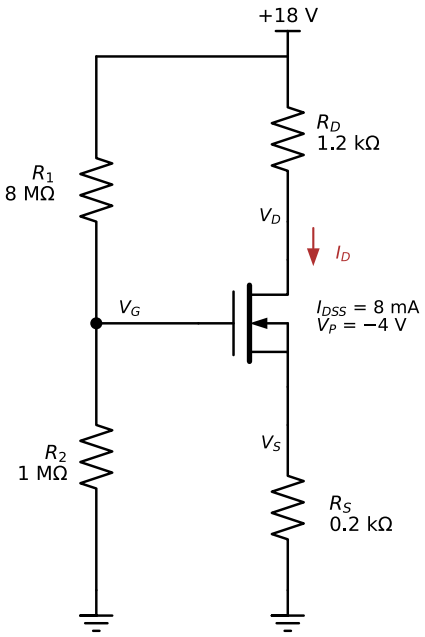
N2. [6 marks] A self-bias JFET has $I_{DSS} = 12$ mA, $V_P = -5$ V, $R_S = 1$ kΩ, $R_D = 2.2$ kΩ and $V_{DD} = 20$ V.

a. Write the bias line and solve for I_{DQ} and V_{GSQ}, justifying your choice of root. [4]
b. Find V_{DS} and check saturation. [2]



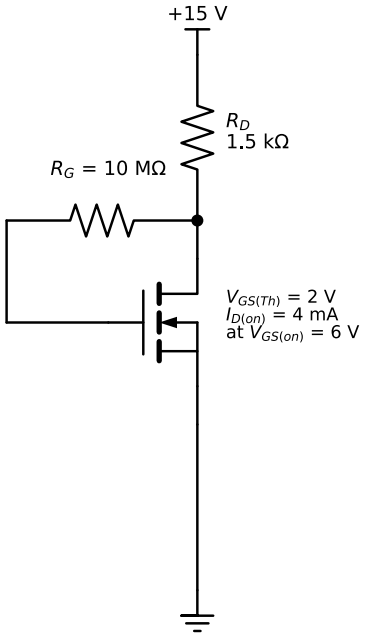
Circuit for N3: voltage-divider JFET, R_S and R_D to be designed

N3. [6 marks] Design. A JFET with $I_{DSS} = 10$ mA and $V_P = -4$ V uses voltage-divider bias with $V_{DD} = 20$ V, $R_1 = 900$ kΩ and $R_2 = 100$ kΩ. Choose R_S for $I_{DQ} = 4$ mA, and R_D for $V_D = 12$ V. Then check saturation.



Circuit for N4: D-MOSFET voltage-divider bias

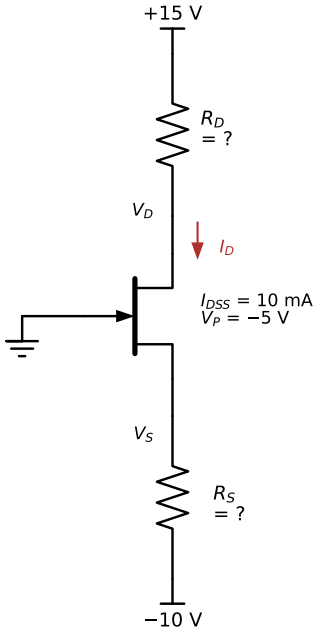
N4. [6 marks] An n-channel D-MOSFET with $I_{DSS} = 8$ mA and $V_P = -4$ V has $V_{DD} = 18$ V, $R_1 = 8$ MΩ, $R_2 = 1$ MΩ, $R_D = 1.2$ kΩ and $R_S = 0.2$ kΩ. Find I_{DQ}, V_{GSQ}, V_D, V_S and V_{DS}. Comment on the sign of V_{GSQ} and on saturation.



Circuit for N5: E-MOSFET drain-feedback bias

N5. [8 marks] Chained. An n-channel E-MOSFET has $V_{GS(\text{Th})} = 2$ V and $I_{D(\text{on})} = 4$ mA at $V_{GS(\text{on})} = 6$ V. It is used in drain-feedback bias with $V_{DD} = 15$ V, $R_D = 1.5$ kΩ and $R_G = 10$ MΩ.

a. Find k. [1]
b. Find I_{DQ} and V_{GSQ}. [4]
c. State V_{DSQ} and explain why no saturation check is needed. [1]
d. Find the new R_D that would give $V_{DSQ} = 8$ V. [2]



Circuit for N6: common-gate JFET with R_S returned to -10 V

N6. [8 marks] Common-gate design, hard. A JFET with $I_{DSS} = 10$ mA and $V_P = -5$ V has its gate grounded. R_D goes to $V_{DD} = +15$ V, and R_S goes to $-V_{SS} = -10$ V.

a. Find R_S for $I_{DQ} = 5$ mA. [3]
b. Find R_D for $V_D = 6$ V, then find V_S and V_{DS} and check saturation. [3]
c. Someone fits $R_D = 2.2$ kΩ instead. Is the JFET still saturated? What is the largest R_D that keeps it saturated, assuming I_D stays at 5 mA? [2]

6. Quick recall check

1. Write the self-bias line for a JFET and say where it crosses the axes.
2. For a JFET quadratic, which root do you keep?
3. Can I_{DQ} exceed I_{DSS}? For which device, and when?
4. State the saturation condition for a JFET and for an E-MOSFET.
5. For an E-MOSFET with drain feedback, what is the relation between V_{DS} and V_{GS}, and why?

FET small-signal I

Practice questions · Day 18 · Tue 10 Nov · pack E11

5. Practice

From your course files

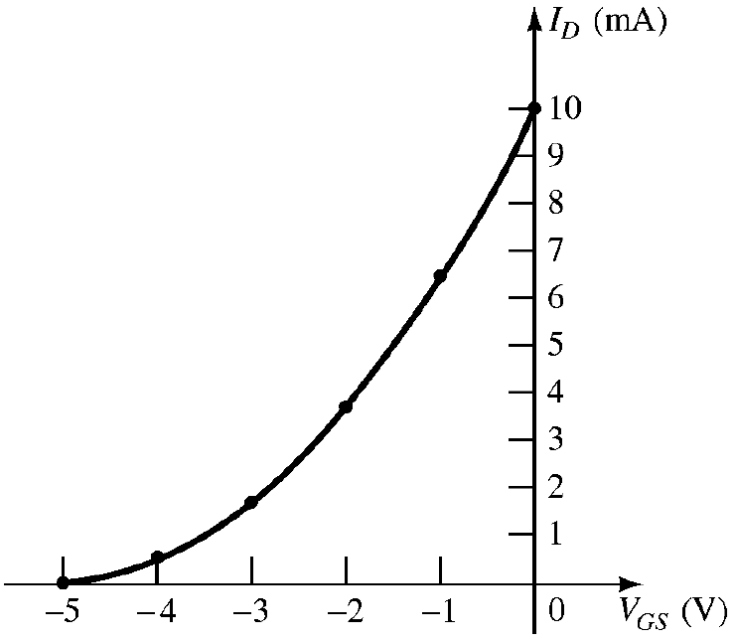
WS8 Q7 (course question). Calculate g_{m0} for a JFET having device parameters $I_{DSS} = 12$ mA and $V_P = -4$ V.

WS8 Q8 (course question). For a JFET having device parameters $g_{m0} = 5$ mS and $V_P = -4$ V, what is the device current at $V_{GS} = 0$ V?

WS8 Q9 (course question). For a JFET having specified values of $g_{fs} = 4.5$ mS and $g_{os} = 25$ μS, determine the device output impedance Z_o (FET) and device ideal voltage gain A_v (FET).

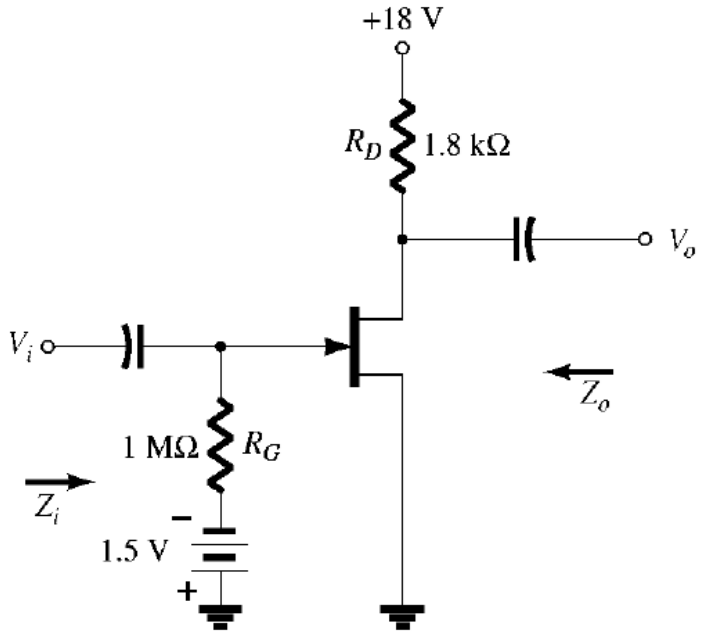
WS8 Q10 (course question). Using the transfer characteristic of Fig. 7:

a. What is the value of g_{m0}?
b. Determine g_m at $V_{GS} = -0.5$ V graphically.
c. Calculate the value of g_m at $V_{GS} = -0.5$ V and compare with the solution to (b).



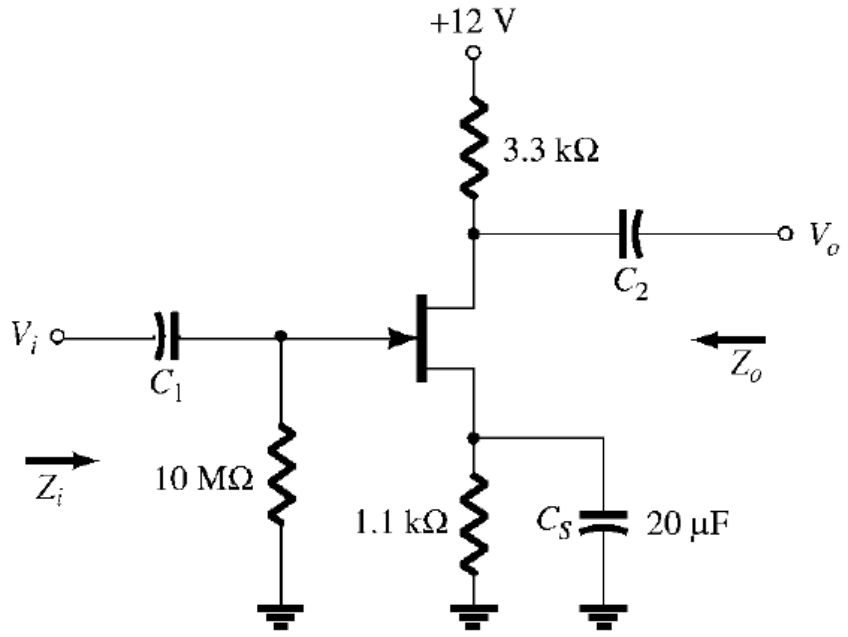
WS8 Fig. 7 (from the workshop sheet)

WS8 Q11 (course question). Determine Z_i, Z_o, and A_v for the network of Fig. 8 if $I_{DSS} = 10$ mA, $V_P = -6$ V, and $r_d = 40$ kΩ.



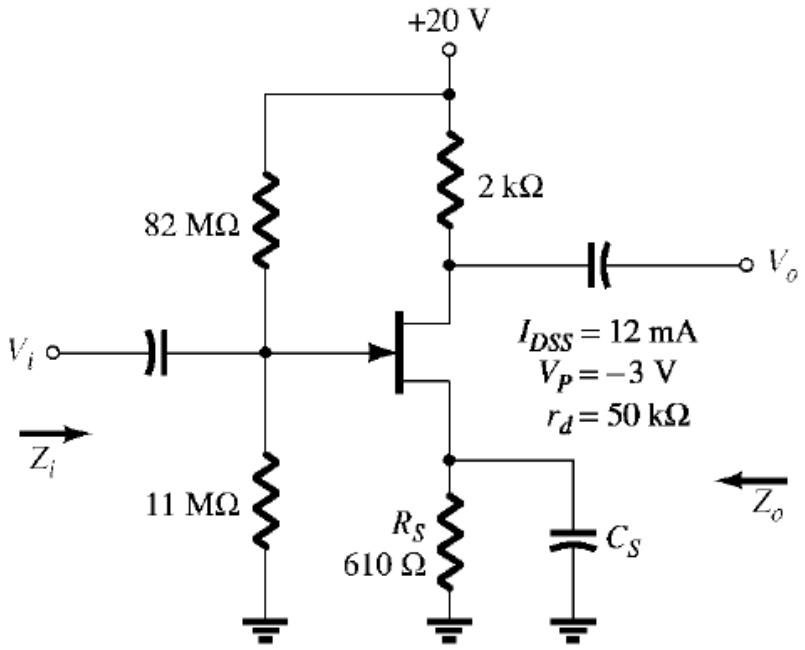
WS8 Fig. 8 (from the workshop sheet)

WS8 Q12 (course question). Determine Z_i and Z_o for the network of Fig. 9. Given: $g_{fs} = 3000\ \mu$S and $g_{os} = 50\ \mu$S.



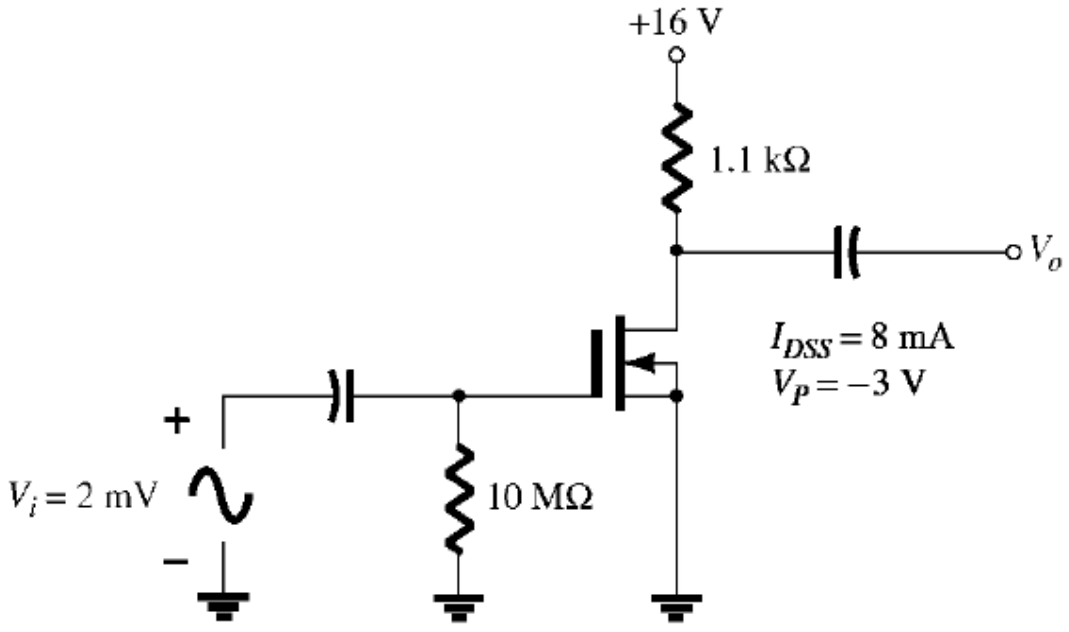
WS8 Fig. 9 (from the workshop sheet)

WS8 Q13 (course question). Determine Z_i, Z_o, and V_o for the network of Fig. 10 if $V_i = 20$ mV.



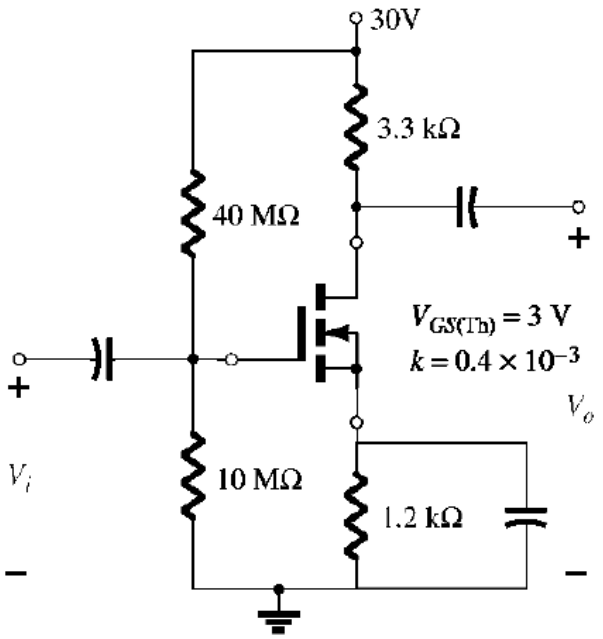
WS8 Fig. 10 (from the workshop sheet)

WS8 Q14 (course question). Determine V_o for the network of Fig. 11 if $g_{os} = 20\ \mu$S.



WS8 Fig. 11 (from the workshop sheet)

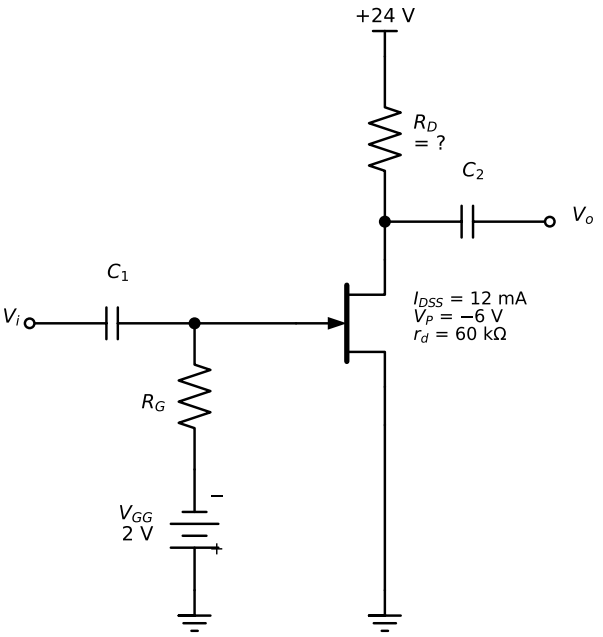
WS8 Q15 (course question). Determine the output voltage for the network of Fig. 12 if $V_i = 0.8$ mV and $r_d = 40$ kΩ.



WS8 Fig. 12 (from the workshop sheet)

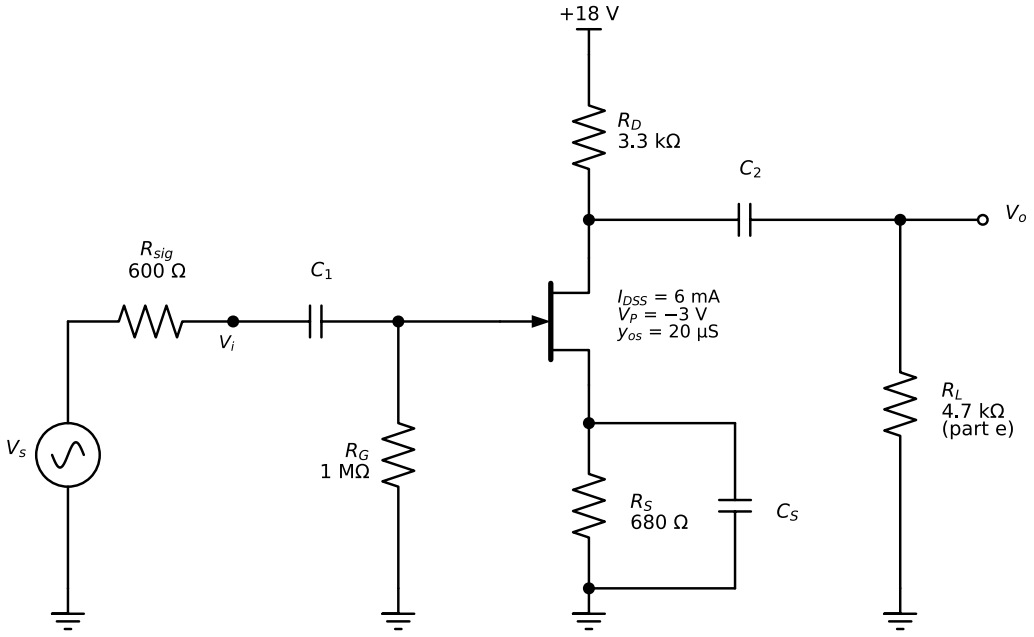
New questions

N1 [3 marks] (easy). A JFET has $I_{DSS} = 8$ mA and $V_P = -4$ V. (a) Find g_{m0}. (b) Find g_m at $V_{GS_Q} = -1.5$ V. (c) Find g_m at $I_{D_Q} = 2$ mA without first finding V_{GS}.



Circuit for N2: fixed-bias JFET amplifier, R_D to be chosen

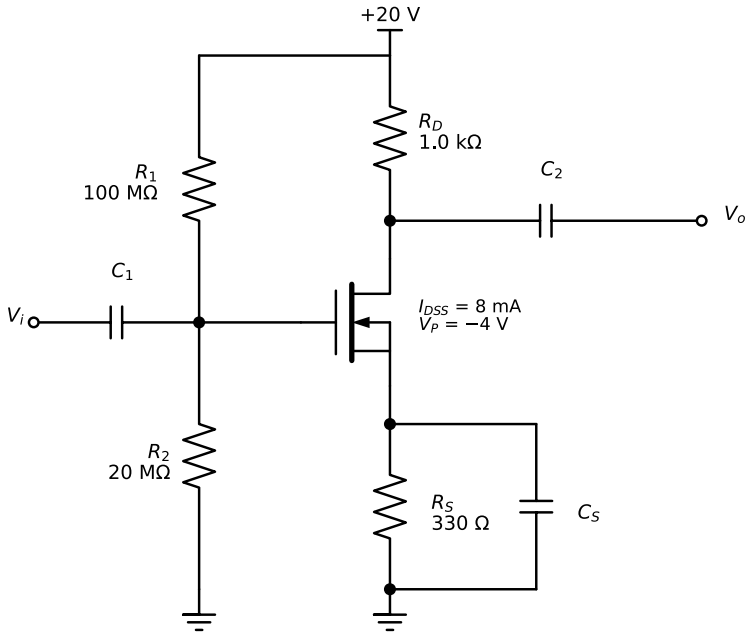
N2 [4 marks] (design). A fixed-bias JFET amplifier has $V_{DD} = 24$ V and gate bias $V_{GG} = 2$ V (so $V_{GS_Q} = -2$ V). The device has $I_{DSS} = 12$ mA, $V_P = -6$ V and $r_d = 60$ kΩ. (a) Find R_D for $A_v = -6$. (b) Find V_{DS_Q} with that R_D and confirm saturation.



Circuit for N3: self-bias JFET stage with source resistance and load

N3 [8 marks] (multi-part, chained). A self-bias JFET stage has $V_{DD} = 18$ V, $R_G = 1$ MΩ, $R_D = 3.3$ kΩ, $R_S = 680\ \Omega$. The device has $I_{DSS} = 6$ mA, $V_P = -3$ V, $y_{os} = 20$ µS.

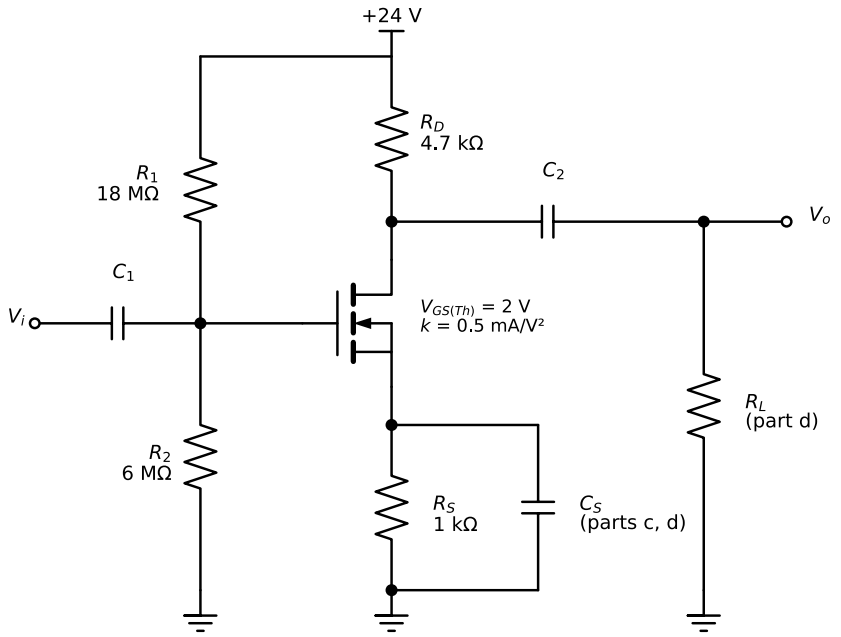
a. Find V_{GS_Q}, I_{D_Q} and V_{DS_Q}. [3]
b. Find g_m and r_d. [1]
c. With R_S fully bypassed, find Z_i, Z_o and A_v. [2]
d. The bypass capacitor fails open. Find A_v, ignoring r_d. [1]
e. With C_S working, a 4.7 kΩ load is connected and the source has $R_{sig} = 600\ \Omega$. Find $A_{vs} = V_o/V_s$. [1]



Circuit for N4: D-MOSFET voltage-divider stage

N4 [5 marks]. An n-channel D-MOSFET has $I_{DSS} = 8$ mA and $V_P = -4$ V, with r_d negligible. It is in a voltage-divider stage: $V_{DD} = 20$ V, $R_1 = 100$ MΩ, $R_2 = 20$ MΩ, $R_S = 330\ \Omega$ (bypassed), $R_D = 1.0$ kΩ.

a. Find the Q-point and check saturation.
b. Find g_m and compare it to g_{m0}.
c. Find Z_i and A_v.



Circuit for N5: E-MOSFET voltage-divider stage

N5 [6 marks] (hard). An n-channel E-MOSFET has $V_{GS(Th)} = 2$ V and $k = 0.5$ mA/V², with r_d ignored. It is in a voltage-divider stage: $V_{DD} = 24$ V, $R_1 = 18$ MΩ, $R_2 = 6$ MΩ, $R_S = 1$ kΩ, $R_D = 4.7$ kΩ.

a. Find V_{GS_Q}, I_{D_Q} and V_{DS_Q}. [2]
b. Find g_m. [1]
c. Find A_v with R_S unbypassed and with R_S bypassed. [2]
d. With R_S bypassed, what load R_L reduces $|A_v|$ to 5? [1]

6. Quick recall check

1. Write g_{m0} and g_m for a JFET. Which quantity takes an absolute value?
2. What fraction of g_{m0} is g_m when $I_D = I_{DSS}/4$?
3. Give Z_i, Z_o and A_v for a voltage-divider JFET with R_S bypassed.
4. Give A_v for self bias with R_S unbypassed, ignoring r_d.
5. Why can g_m exceed g_{m0} for a D-MOSFET but not for a JFET?

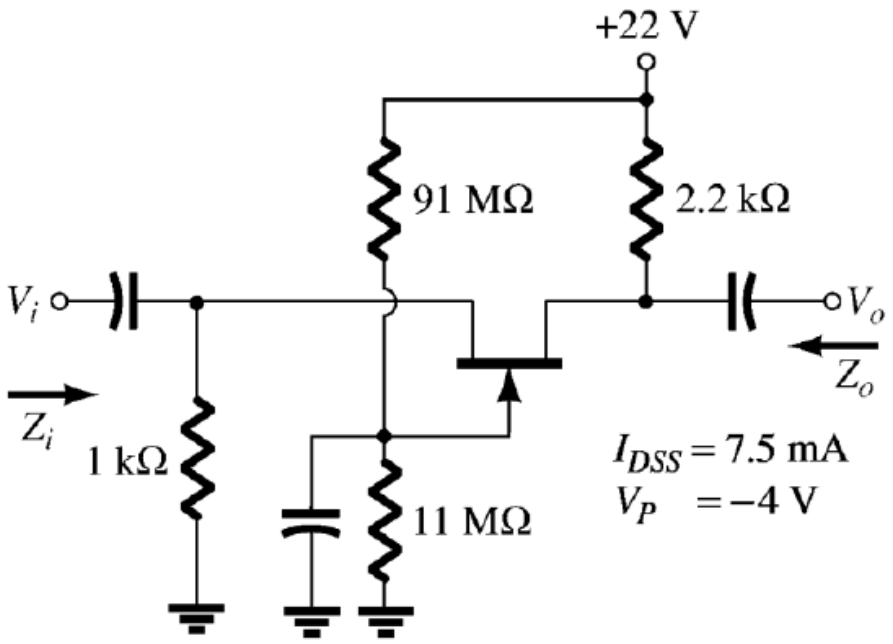
FET small-signal II

Practice questions · Day 20 · Thu 12 Nov · pack E12

5. Practice

From your course files

WS9 Q5 (course question). Determine Z_i, Z_o, and A_v for the network of Fig. 5 if $r_d = 30$ kΩ.

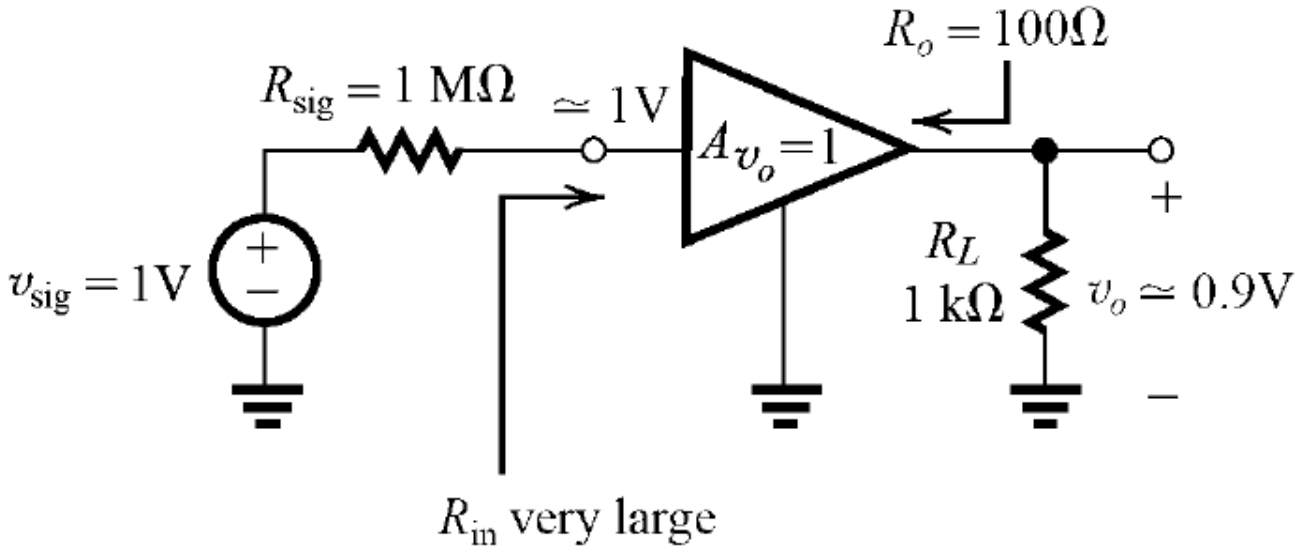


WS9 Fig. 5 (from the workshop sheet)

WS9 Q6 (course question). A common-gate (CG) amplifier is required to match a signal source with $R_{sig} = 100\ \Omega$. At what current should the MOSFET be biased if it is operated at an overdrive voltage of 0.20 V? If the total resistance in the drain circuit is 2 kΩ, what overall voltage gain is realized?

WS9 Q7 (course question). It is required to design a source follower that implements the buffer amplifier shown in Fig. 6. If the MOSFET is operated with an overdrive voltage

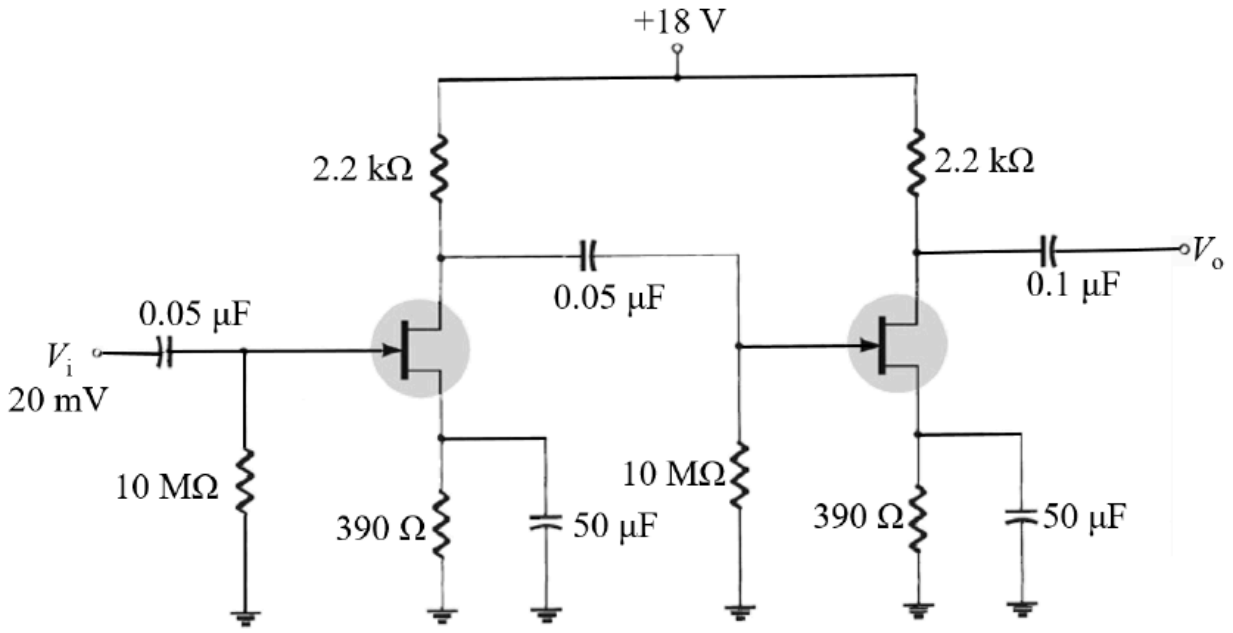
$V_{ov} = 0.25$ V, at what drain current should it be biased? Find the output signal amplitude and the signal amplitude between gate and source.



WS9 Fig. 6 (from the workshop sheet)

WS9 Q8 (course question). A MOSFET is connected in the source-follower configuration and employed as the output stage of a cascade amplifier. It is required to provide an output resistance of 200 Ω. If the MOSFET has $k = 0.4$ mA/V^2 and is operated at $V_{ov} = 0.25$ V, find the required W/L ratio. Also specify the dc bias current If the amplifier load resistance varies over the range 1 kΩ to 10 kΩ what is the range of G_v of the source follower?

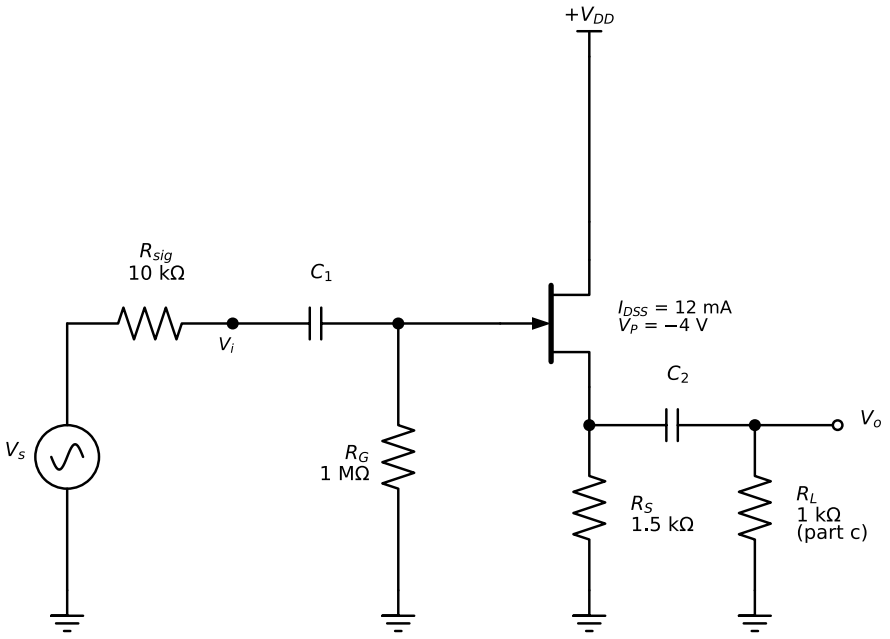
WS9 Q9 (course question). For the JFET cascade amplifier of Fig. 7, using identical JFETs with $I_{DSS} = 8$ mA and $V_P = -4.5$ V, calculate the voltage gain of each stage, the overall gain of the amplifier, and the output voltage V_o.



WS9 Fig. 7 (from the workshop sheet)

New questions

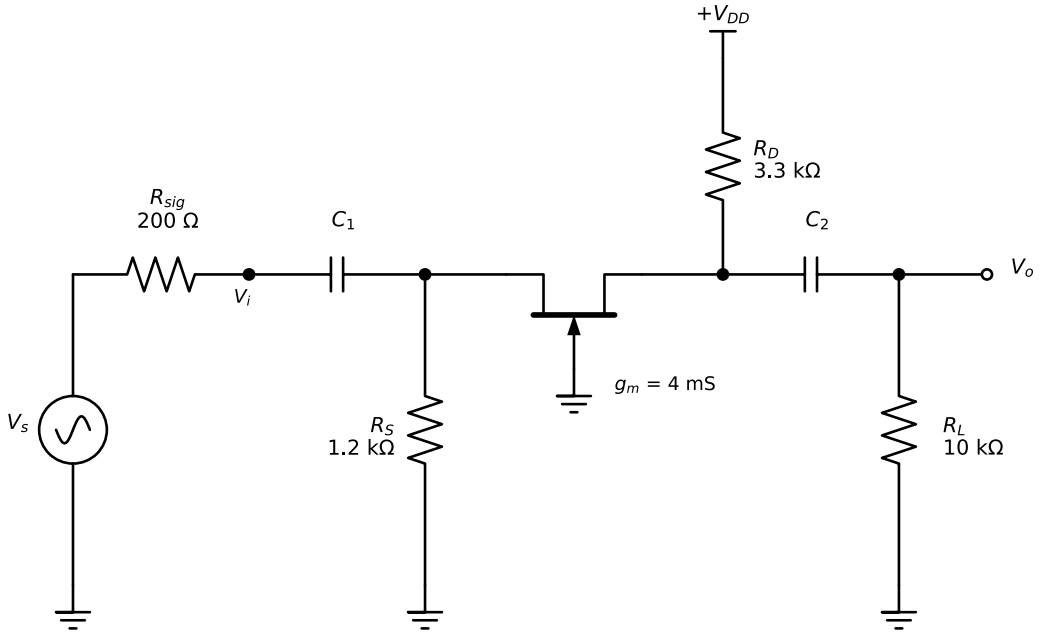
N1. [3 marks] An n-channel E-MOSFET has $k = 0.3$ mA/V² and $V_{GS(\mathrm{Th})} = 2$ V, biased at $V_{GSQ} = 5$ V. Find I_{DQ} and g_m, and confirm g_m with the overdrive form.



Circuit for N2: JFET source follower with source resistance and load

N2. [7 marks] A JFET source follower has $I_{DSS} = 12$ mA, $V_P = -4$ V, $R_G = 1$ MΩ (gate to ground), $R_S = 1.5$ kΩ (source to ground), drain to V_{DD}, $r_d = \infty$.

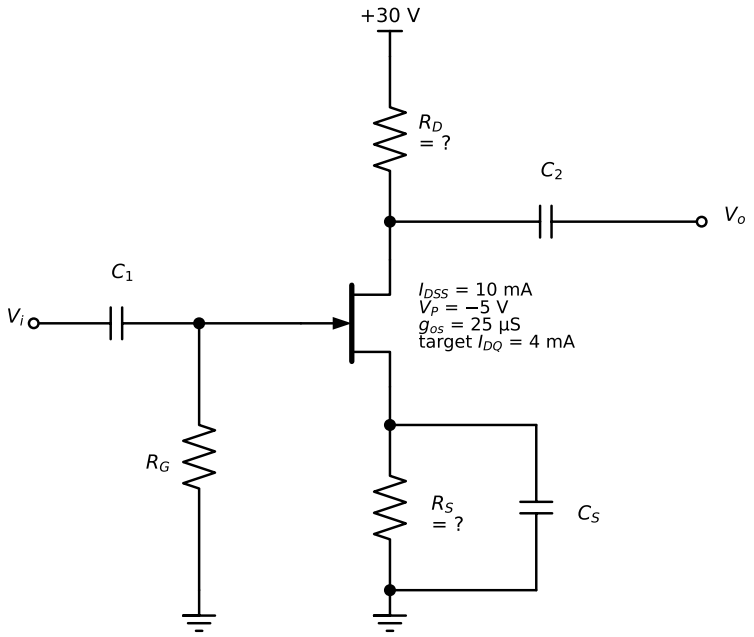
a. Find I_{DQ}, V_{GSQ} and g_m. [3]
b. Find Z_i, Z_o and the no-load A_v. [2]
c. The stage drives $R_L = 1$ kΩ from a source with $R_{sig} = 10$ kΩ. Find A_{v_L} and A_{v_s}. [2]



Circuit for N3: common-gate stage fed from R_sig and driving R_L

N3. [6 marks] A common-gate stage has $g_m = 4$ mS, $R_S = 1.2$ kΩ, $R_D = 3.3$ kΩ, $r_d = \infty$. It is fed from $R_{sig} = 200\ \Omega$ and drives $R_L = 10$ kΩ.

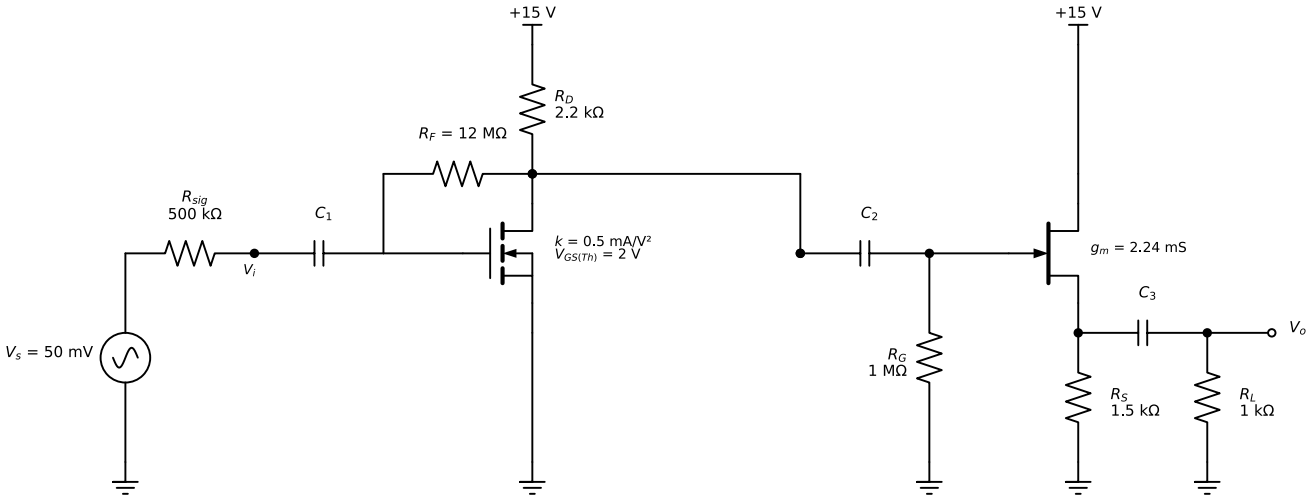
a. Find Z_i and Z_o. [2]
b. Find the no-load gain, A_{v_L} and A_{v_s}. [3]
c. In one sentence, why is the overall gain about half of A_{v_L}? [1]



Circuit for N4: self-bias JFET stage to be designed

N4. [8 marks] Design a self-bias JFET stage with R_S fully bypassed. $I_{DSS} = 10$ mA, $V_P = -5$ V, $g_{os} = 25$ µS, $V_{DD} = 30$ V, target $I_{DQ} = 4$ mA and $A_v = -10$.

a. Find V_{GSQ} and R_S. [2]
b. Find g_m. [1]
c. Find R_D for $A_v = -10$, then the gain with the standard values 4.3 kΩ and 4.7 kΩ. [3]
d. Check the FET stays in saturation with the exact R_D. [1]
e. If C_S is removed, what is A_v (exact R_D, include r_d)? [1]



Circuit for N5: E-MOSFET drain-feedback stage driving a JFET source follower

N5. [10 marks] Two-stage cascade. Stage 1 is an E-MOSFET drain-feedback amplifier: $V_{DD} = 15$ V, $R_D = 2.2$ kΩ, $R_F = 12$ MΩ, $k = 0.5$ mA/V², $V_{GS(\text{Th})} = 2$ V. Stage 2 is the source follower of N2 ($R_G = 1$ MΩ, $g_m = 2.24$ mS, $R_S = 1.5$ kΩ), driving $R_L = 1$ kΩ. The source is $V_s = 50$ mV with $R_{sig} = 500$ kΩ. $r_d = \infty$ for both.

a. Find I_{DQ}, V_{GSQ} and g_m for stage 1. [3]
b. Find A_{v1} (loaded by stage 2) and Z_{i1}. [3]
c. Find A_{v2}, the overall $A_v = V_o/V_i$, and V_o. [3]
d. State Z_o of the cascade. [1]

N6. [5 marks] (Sedra form.) An NMOS has $\mu_n C_{ox} = 0.2$ mA/V², $W/L = 40$, $V_t = 0.6$ V and $I_D = 0.5$ mA.

a. Find V_{ov}, V_{GS} and g_m. [3]
b. Give the course k and confirm $g_m = 2kV_{ov}$. [1]
c. Find the CS gain with $R_D = 5$ kΩ and $R_L = 10$ kΩ. [1]

6. Quick recall check

1. Write Z_o for a source follower with $r_d = \infty$.
2. What is Z_i (approximate) for an E-MOSFET drain-feedback amplifier?
3. A CG stage has $g_m = 5$ mS. What is its approximate Z_i if $R_S \gg 200\ \Omega$?
4. In Sedra notation, $I_D = 2$ mA and $V_{ov} = 0.4$ V. What is g_m?
5. Two identical CS stages each have $A_v = -4$. What is the overall gain, and is the output in phase with the input?

Op-amps, differential amps, power amps

Practice questions · Day 21 · Fri 13 Nov · pack E13

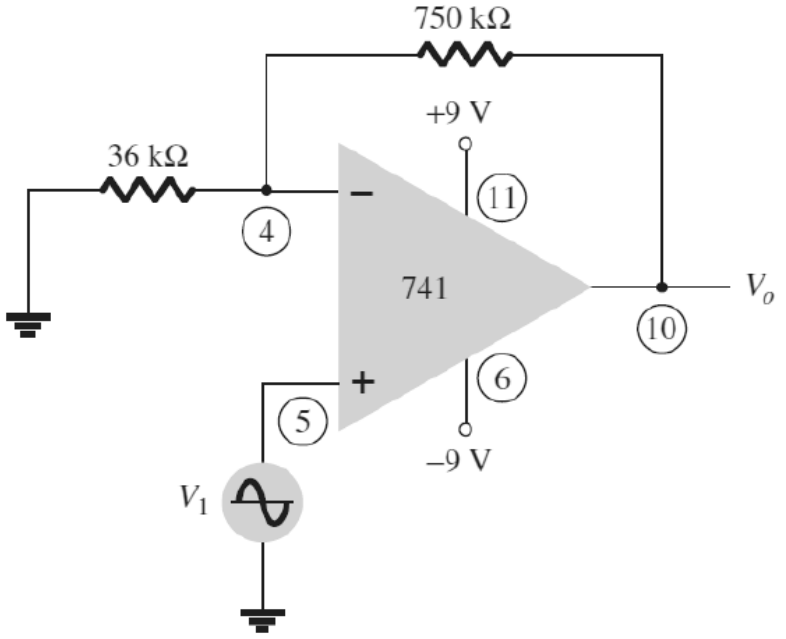
5. Practice

From your course files

WS10 Q1 (course question). Calculate the CMRR (in dB) for the circuit measurements of $V_d = 1$ mV, $V_o = 120$ mV, $V_C = 1$ mV, and $V_o = 20$ mV.

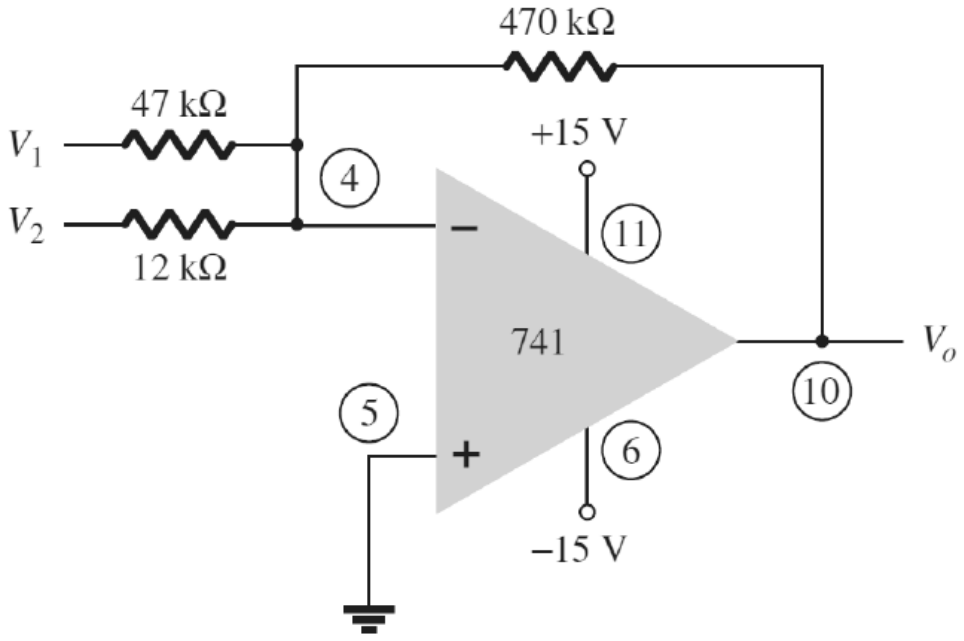
WS10 Q2 (course question). Determine the output voltage of an op-amp for input voltages of $V_{i1} = 200$ mV and $V_{i2} = 140$ mV. The amplifier has a differential gain of $A_d = 6000$ and the value of CMRR is: (a) 200 and (b) 10^5.

WS10 Q3 (course question). Calculate the output voltage of the circuit of Fig. 1 for an input of 150 mV rms.



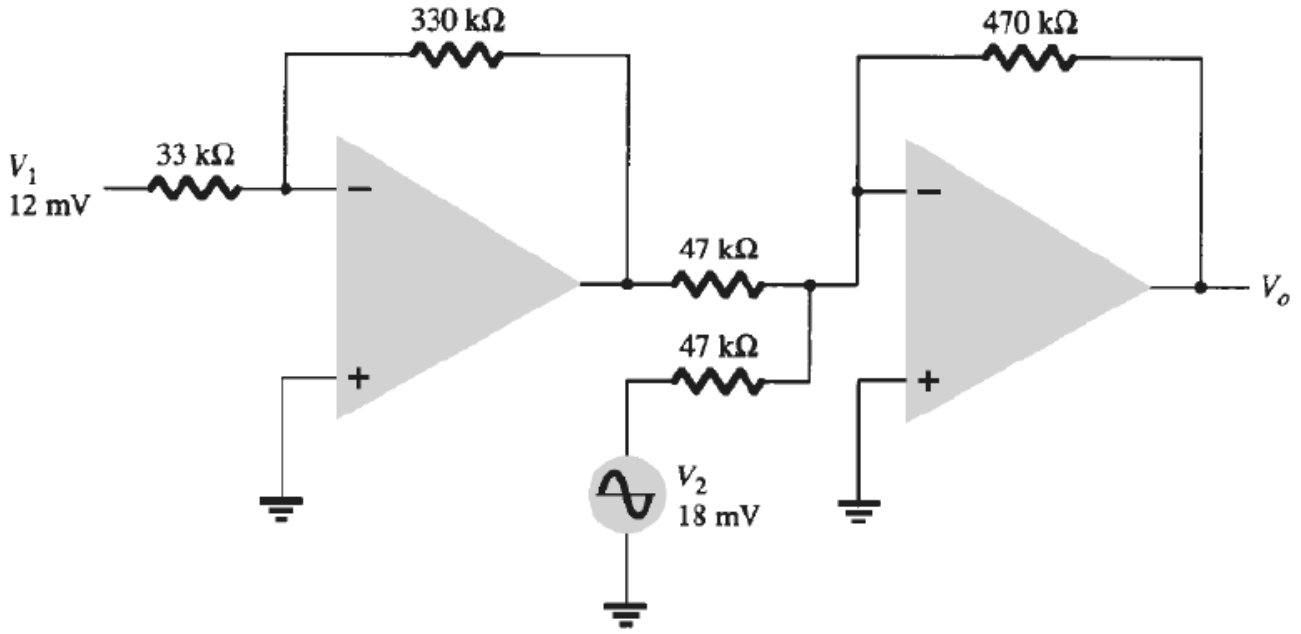
WS10 Fig. 1, page 2 (from the workshop sheet)

WS10 Q4 (course question). Calculate the output voltage for the circuit of Fig. 2 with inputs of $V_1 = 40$ mV rms and $V_2 = 20$ mV rms.



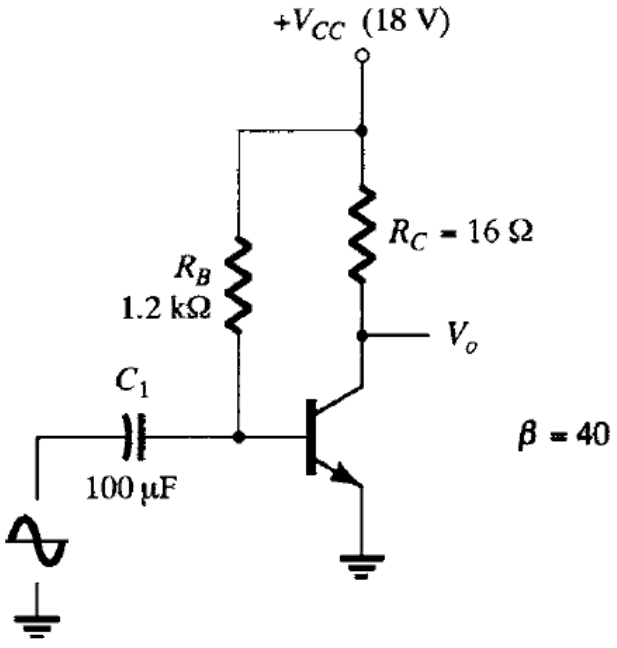
WS10 Fig. 2 (from the workshop sheet)

WS10 Q5 (course question). Determine the output voltage for the circuit of Fig. 3.



WS10 Fig. 3 (from the workshop sheet)

WS10 Q6 (course question). Calculate the input and output power for the circuit of Fig. 1. The input signal results in a base current of 5 mA rms.



WS10 Fig. 1, page 3 (from the workshop sheet)

WS10 Q7 (course question). Calculate the input power dissipated by the circuit of Fig. 1 if R_B is changed to 1.5 kΩ. *(Fig. 1 here is the class A circuit shown under WS10 Q6.)*

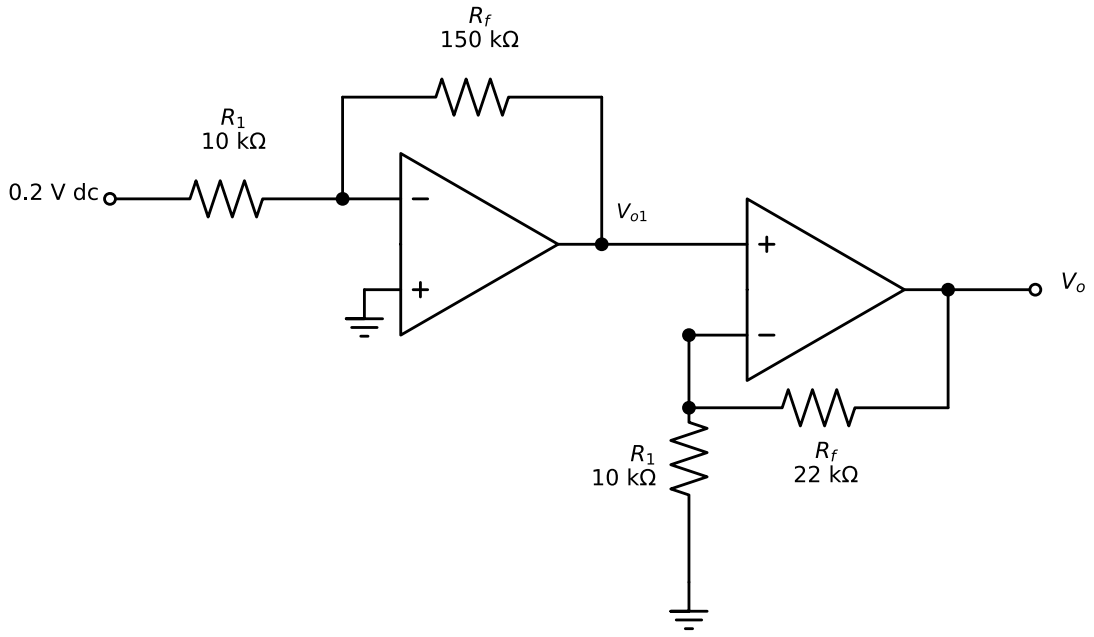
WS10 Q8 (course question). What maximum output power can be delivered by the circuit of Fig. 1 if R_B is changed to 1.5 kΩ? *(Same class A circuit as WS10 Q6.)*

WS10 Q9 (course question). Calculate the transformer turns ratio required to connect four parallel 16-Ω speakers so that they appear as an 8-kΩ effective load.

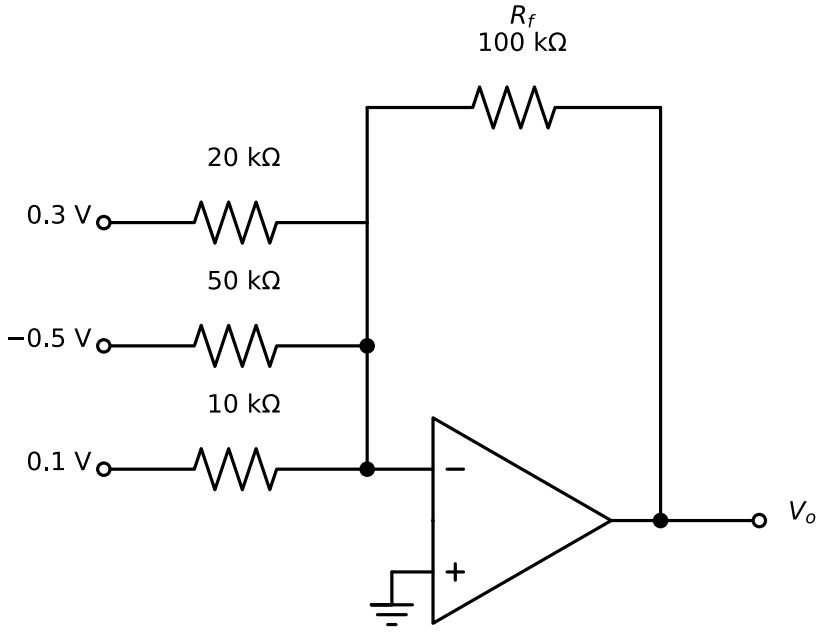
WS10 Q10 (course question). For a class B amplifier with $V_{CC} = 25$ V driving an 8-Ω load, determine: (a) Maximum input power, (b) Maximum output power, and (c) Maximum circuit efficiency.

New questions

N1. [3 marks] An op-amp has $A_d = 5000$ and CMRR = 80 dB. (a) Find A_c. (b) Find V_o for $V_{i1} = 1.25$ mV and $V_{i2} = 0.75$ mV.

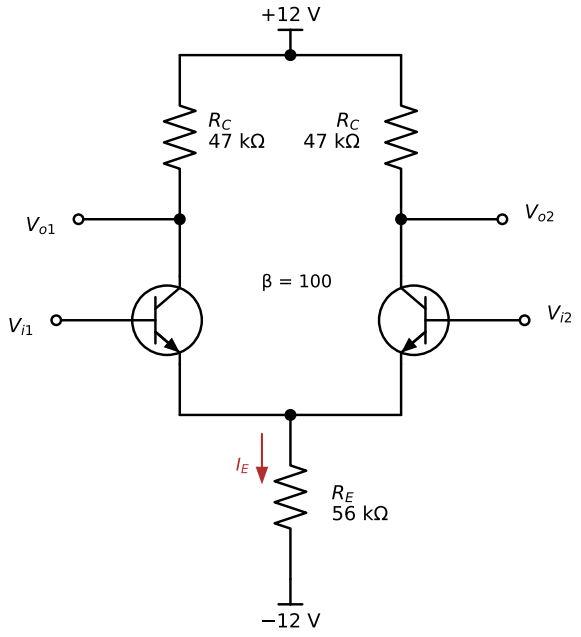


Circuit for N2(a): inverting stage into a non-inverting stage



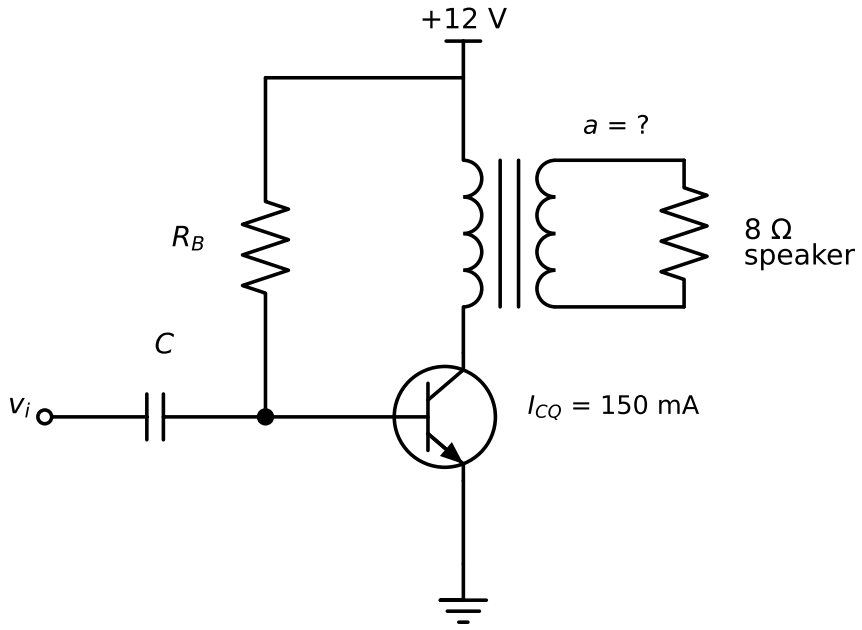
Circuit for N2(b): three-input inverting summer

N2. [5 marks] (a) A 0.2 V dc input drives an inverting stage with $R_1 = 10$ kΩ and $R_f = 150$ kΩ. That output drives a non-inverting stage with $R_1 = 10$ kΩ and $R_f = 22$ kΩ. Find both stage outputs. (b) An inverting summer has $R_f = 100$ kΩ and inputs $V_1 = 0.3$ V through 20 kΩ, $V_2 = -0.5$ V through 50 kΩ, and $V_3 = 0.1$ V through 10 kΩ. Find V_o.



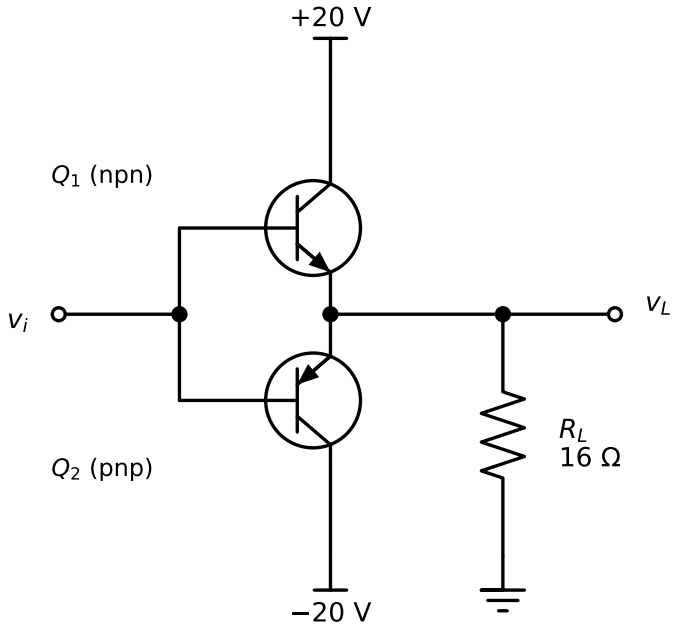
Circuit for N3: BJT differential amplifier

N3. [6 marks] A BJT differential amplifier has $V_{CC} = V_{EE} = 12$ V, tail resistor $R_E = 56$ kΩ, $R_C = 47$ kΩ on each collector, $\beta = 100$, and both bases at 0 V dc. Find (a) I_E (tail), I_C per transistor and V_C; (b) r_e; (c) the single-ended gain with one input driven, and the double-ended differential gain.



Circuit for N4: transformer-coupled class A amplifier

N4. [6 marks] A transformer-coupled class A amplifier runs from $V_{CC} = 12$ V with $I_{CQ} = 150$ mA. The signal swings V_{CE} between 2 V and 22 V and I_C between 10 mA and 290 mA. The speaker is 8 Ω. Find (a) P_o, (b) P_i and η, (c) the transistor dissipation, (d) the effective load seen by the collector and the turns ratio.



Circuit for N5: class B push-pull, V_CC = 20 V, R_L = 16 ohm

N5. [8 marks, chained] A class B push-pull amplifier uses $V_{CC} = 20$ V into $R_L = 16\ \Omega$. (a) For a peak output of 16 V, find P_i, P_o, η, and the dissipation per transistor. (b) Find the maximum possible P_o. (c) Find the worst-case total transistor dissipation and the output peak voltage at which it occurs. (d) Explain in one sentence why the transistors need heatsinks sized for (c), not for (b).

6. Quick recall check

1. $V_{i1} = 3$ mV, $V_{i2} = 1$ mV. What are V_d and V_c?
2. CMRR = 100 dB as a ratio?
3. Gain of a non-inverting amp with $R_f = 90$ kΩ, $R_1 = 10$ kΩ?
4. Maximum efficiency of series-fed class A, transformer-coupled class A and class B?
5. A 10 Ω load must look like 1 kΩ. What is the turns ratio?

EDC · MOCK

EDC mock A

Practice questions · Day 27 · Thu 19 Nov · pack EX1

How to use this: Sit this paper two days before the EDC exam, timed (2 hours), with your hard-copy notes and a calculator, and do not look at the marking guide. Then mark it strictly with the guide below. The paper is modelled on the practice test (open book, formula table first, MCQ concepts and then workshop-style calculations). It has been scaled up to a full 2-hour final covering the whole course. About 43% of the marks are on diodes and BJTs and about 57% on FETs, multistage amplifiers, op-amps and power amplifiers. Every circuit is described in words, so you do not need a figure. **Draw each circuit yourself before you start the question.** That takes about a minute and it earns the “redraw” marks.

Instructions to candidates

- This is an open-book examination. No internet access.
- Duration: **2 hours**. There are **10 questions**. Answer all of them. Total: **82 marks**.
- The numbers in square brackets are the marks for each part.
- Show detailed calculations: write the formula in symbols, then substitute, then give the answer with units.
- Unless a question says otherwise: Si BJTs have $V_{BE,on} = 0.7$ V and $V_{CE,sat} = 0.2$ V, $V_T = 26$ mV at room temperature, and all coupling and bypass capacitors are short circuits at signal frequency and open circuits at DC.
- A sensible pace is about **1.4 minutes per mark**. Keep 5 minutes at the end for checking.

> **FORMULA TABLE**

Topic	Formulas
Diodes	$I_D = I_S\left(e^{V_D/nV_T}-1\right) \approx I_S e^{V_D/nV_T}$; $V_T = kT/q$; $\Delta V_D = nV_T \ln(I_2/I_1) = 2.3\,nV_T \log_{10}(I_2/I_1)$; $r_d = nV_T/I_D$; Si knee 0.7 V, Ge 0.3 V
Rectifier and Zener	$V_r = V_P/(fCR)$; $V.R. = \dfrac{V_{NL}-V_{FL}}{V_{FL}} \times 100\%$; Zener shunt regulator: $I_S = I_Z + I_L$, $I_S = (V_i - V_Z)/R_S$, $P_Z = V_Z I_Z$
BJT DC	$I_C = \beta I_B$, $I_E = (\beta+1)I_B$, $\alpha = \beta/(\beta+1)$; voltage divider: $V_{Th} = \dfrac{R_2}{R_1+R_2}V_{CC}$, $R_{Th} = R_1 \parallel R_2$, $I_B = \dfrac{V_{Th}-V_{BE}}{R_{Th}+(\beta+1)R_E}$; approximate method valid if $\beta R_E \geq 10R_2$
BJT AC (r_e model)	$r_e = 26\ \text{mV}/I_E$; $g_m = I_C/V_T$; $r_\pi = \beta r_e$; CE (bypassed): $Z_i = R_1 \parallel R_2 \parallel \beta r_e$, $Z_o = R_C \parallel r_o$, $A_v = -(R_C \parallel r_o)/r_e$; emitter follower: $Z_b = \beta r_e + (\beta+1)R_E$, $Z_o = R_E \parallel (r_e + R_s'/\beta)$, $A_v = R_E/(R_E+r_e)$
Loading	$A_{v_L} = A_{v_{NL}}\dfrac{R_L}{R_L+Z_o}$; $A_{v_s} = \dfrac{Z_i}{Z_i+R_s}A_{v_L}$; cascade: $A_{v_T} = A_{v1}A_{v2}\cdots$ with each stage loaded by the next stage's Z_i
JFET and D-MOSFET	$I_D = I_{DSS}\left(1-\dfrac{V_{GS}}{V_P}\right)^2$; $V_{GS} = V_P\left(1-\sqrt{I_D/I_{DSS}}\right)$; $g_{m0} = \dfrac{2I_{DSS}}{\lvert V_P\rvert}$; $g_m = g_{m0}\left(1-\dfrac{V_{GS}}{V_P}\right)$; $r_d = 1/y_{os}$
E-MOSFET	$I_D = k\big(V_{GS}-V_{GS(Th)}\big)^2$; $k = \dfrac{I_{D(on)}}{\big(V_{GS(on)}-V_{GS(Th)}\big)^2}$; $g_m = 2k\left(V_{GSQ}-V_{GS(Th)}\right)$; saturation needs $V_{DS} \geq V_{GS}-V_{GS(Th)}$

Topic	Formulas
FET amplifiers	CS (R_S bypassed): $Z_i = R_G$ (or $R_1 \parallel R_2$), $Z_o = r_d \parallel R_D$, $A_v = -g_m(r_d \parallel R_D)$; source follower: $A_v = \dfrac{g_m(r_d \parallel R_S)}{1 + g_m(r_d \parallel R_S)}$
Differential and op-amps	$V_d = V_{i1} - V_{i2}$, $V_c = \frac{1}{2}(V_{i1} + V_{i2})$; $V_o = A_d V_d + A_c V_c$; $\mathrm{CMRR} = A_d/A_c$, $\mathrm{CMRR(dB)} = 20\log_{10}(A_d/A_c)$; BJT pair (single-ended): $A_d = \dfrac{R_C}{2r_e}$, $A_c = \dfrac{\beta R_C}{r_i + 2(\beta+1)R_E}$; inverting: $V_o = -\dfrac{R_f}{R_1}V_1$; non-inverting: $V_o = \left(1 + \dfrac{R_f}{R_1}\right)V_1$; summing: $V_o = -\sum \dfrac{R_f}{R_k}V_k$
Power amplifiers	Class A series-fed: $P_i = V_{CC}I_{CQ}$, $P_o = I_{c(rms)}^2 R_C = \dfrac{I_{c(p)}^2 R_C}{2}$, $\eta = P_o/P_i$; class B push-pull: $P_i = V_{CC}\dfrac{2}{\pi}I_{(p)}$, $P_o = \dfrac{V_{L(p)}^2}{2R_L}$, $P_{o(max)} = \dfrac{V_{CC}^2}{2R_L}$, $P_{i(max)} = \dfrac{2V_{CC}^2}{\pi R_L}$, $P_{2Q(max)} = \dfrac{2V_{CC}^2}{\pi^2 R_L}$; transformer: $R'_L = a^2 R_L$

Question 1 [10 marks]

Circle the correct answer in each multiple-choice question (I to X). [1 mark each]

I. When the reverse bias across a pn junction is increased, the depletion region:

- a. widens, and the junction capacitance decreases
- b. narrows, and the junction capacitance increases
- c. widens, and the junction capacitance increases
- d. does not change, because no current flows in reverse bias

II. A bridge rectifier with a capacitor filter feeds a fixed load. If the filter capacitance is doubled, the peak-to-peak ripple voltage:

- a. doubles
- b. roughly halves
- c. falls to zero
- d. does not change, because ripple depends only on frequency

III. In a Zener shunt regulator with a fixed series resistor R_S, the Zener is most likely to drop out of regulation when:

- a. the supply voltage is at its maximum and the load current is zero
- b. the supply voltage is at its maximum and the load current is at its maximum
- c. the supply voltage is at its minimum and the load current is at its maximum
- d. the load is disconnected

IV. An NPN BJT is measured with $V_{BE} = 0.7$ V and $V_{BC} = +0.5$ V. The transistor is in:

- a. the active region
- b. cut-off
- c. saturation
- d. the reverse-active region

V. A bypass capacitor is connected across R_E in a voltage-divider CE amplifier. Compared with the unbypassed amplifier:

- a. the Q-point moves and the gain falls
- b. the Q-point is unchanged and $|A_v|$ rises from about R_C/R_E to about R_C/r_e
- c. the Q-point is unchanged and the input impedance rises
- d. the Q-point moves to saturation

VI. An n-channel JFET has $V_{GS} = V_P$. Its drain current is:

- a. approximately zero for any V_{DS}
- b. equal to I_{DSS}
- c. equal to $I_{DSS}/4$
- d. set only by the external drain resistor

VII. Which device can be operated with either a positive or a negative gate-to-source voltage and still conduct?

- a. n-channel JFET
- b. n-channel enhancement-type MOSFET
- c. n-channel depletion-type MOSFET
- d. p-channel enhancement-type MOSFET

VIII. For a JFET, the transconductance g_m is largest when:

- a. $V_{GS} = V_P$
- b. $V_{GS} = 0$ V
- c. $V_{GS} = V_P/2$
- d. $I_D = I_{DSS}/4$

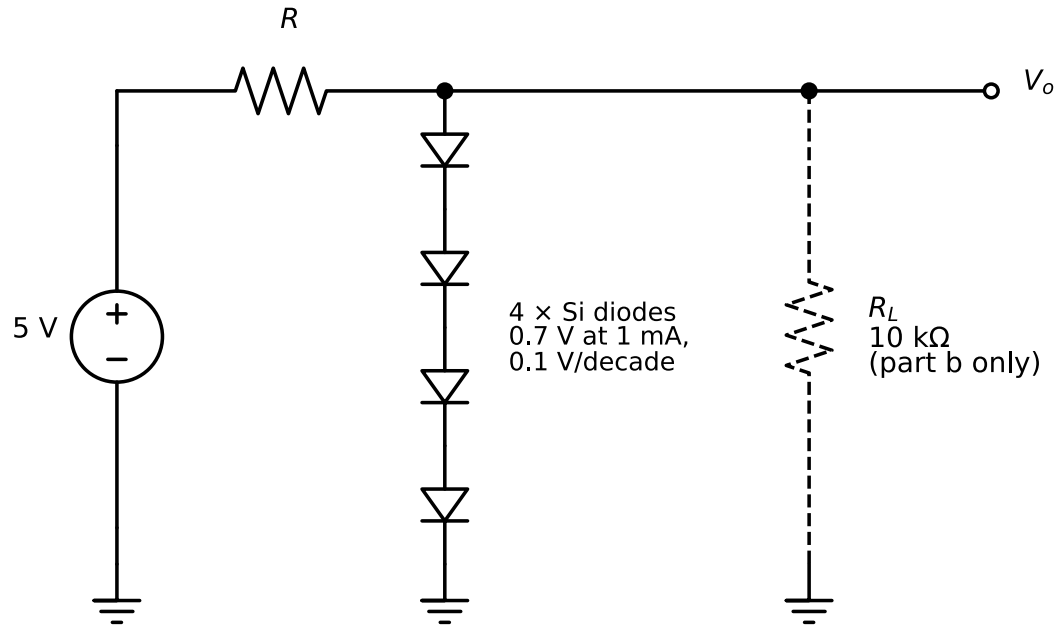
IX. In an ideal op-amp inverting amplifier with negative feedback, the inverting input terminal:

- a. is at virtually the same voltage as the non-inverting input, which is grounded, and it draws no current
- b. is at the output voltage
- c. carries the full input current into the op-amp
- d. is at the input voltage V_1

X. Crossover distortion in a class B push-pull output stage is caused by:

- a. the output transformer saturating
- b. the transistors' β being unequal
- c. the efficiency exceeding 78.5%
- d. both transistors being off while $|v_{in}|$ is below about 0.7 V

Question 2 [8 marks]



Question 2 circuit: 5 V supply, R, four series diodes, optional 10 kOhm load

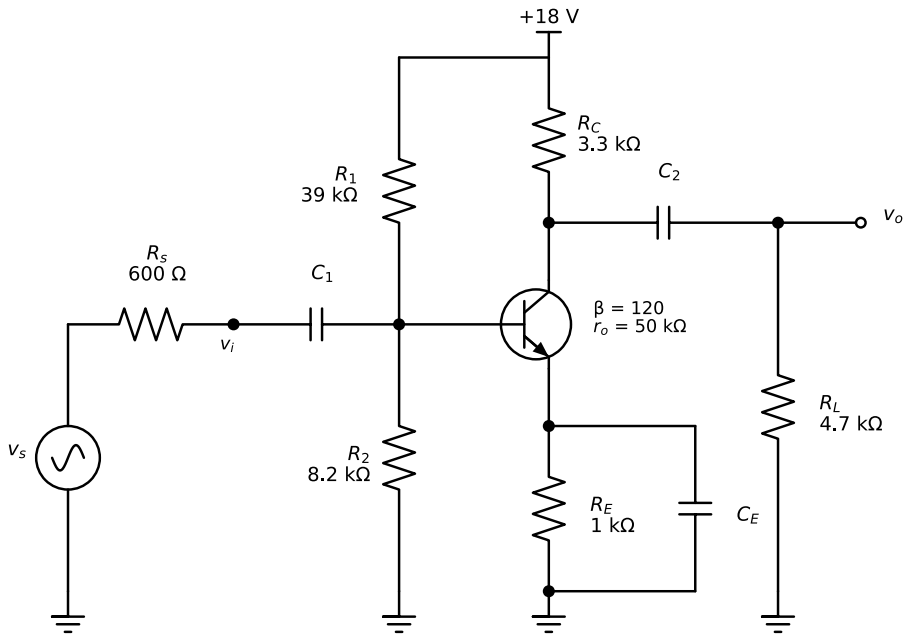
A 5 V DC supply feeds a resistor R. The other end of R is the output node. Between the output node and ground there is a string of **four identical silicon diodes in series**, all forward biased (anode towards the output node). The output V_o is the voltage across the whole string. Each diode has a forward drop of 0.7 V at 1 mA, and this drop changes by 0.1 V for every decade change in current.

a. Design the circuit for $V_o = 2.6$ V with no load: find the diode current and the value of R. [3 marks]

b. A load resistor $R_L = 10\ \text{k}\Omega$ is now connected from the output node to ground, in parallel with the diode string. Keep R at your value from (a). Find the new V_o and the new diode current by iteration. [3 marks]

c. With no load connected, find the incremental resistance of one diode at the design point. Use it to estimate the change in V_o when the supply changes by ± 0.5 V. [2 marks]

Question 3 [13 marks]



Question 3 circuit: voltage-divider CE amplifier with bypassed R_E, source R_s and load R_L

A voltage-divider-biased common-emitter amplifier uses an NPN BJT with $\beta = 120$ and $r_o = 50\ \text{k}\Omega$.

- $V_{CC} = 18$ V. $R_1 = 39\ \text{k}\Omega$ from V_{CC} to the base, and $R_2 = 8.2\ \text{k}\Omega$ from the base to ground.
- $R_C = 3.3\ \text{k}\Omega$ from V_{CC} to the collector, and $R_E = 1\ \text{k}\Omega$ from the emitter to ground. R_E is fully bypassed by a capacitor C_E.
- The signal source v_s has internal resistance $R_s = 600\ \Omega$ and is capacitively coupled to the base.
- The output is taken from the collector through a coupling capacitor to a load $R_L = 4.7\ \text{k}\Omega$.

a. State whether the approximate method may be used. Then use the **exact (Thevenin) method** to find V_{Th}, R_{Th}, I_B, I_C, I_E and V_{CE}. Confirm that the BJT is in the active region. [5 marks]

b. Find r_e. Then find Z_i, Z_o and the no-load voltage gain $A_v = v_o/v_i$, including r_o. [4 marks]

c. Find the loaded gain $A_{v_L} = v_o/v_i$ with R_L connected, and the overall gain $A_{v_s} = v_o/v_s$. Find the peak output voltage for $v_s = 10$ mV peak. State which of R_s and R_L costs more gain in this circuit, and why. [4 marks]

Question 4 [9 marks]



An emitter follower uses an NPN BJT with $\beta = 100$ (r_o may be ignored).

- $V_{CC} = 12$ V is connected directly to the collector.
- $R_B = 220\ \text{k}\Omega$ is connected from V_{CC} to the base.
- $R_E = 2.2\ \text{k}\Omega$ is connected from the emitter to ground.
- The source v_s, with $R_s = 1\ \text{k}\Omega$, is capacitively coupled to the base.
- The output is taken from the emitter through a coupling capacitor to a load $R_L = 1\ \text{k}\Omega$.

a. Find I_B, I_E, V_E and V_{CE}. [2 marks]

b. Find r_e. Find Z_i with no load, and again with R_L connected. [2 marks]

c. Find the output impedance Z_o seen by R_L, including the effect of R_s. [2 marks]

d. Find $A_{v_L} = v_o/v_i$ and $A_{v_s} = v_o/v_s$. [2 marks]

e. In one sentence, explain why this stage is useful even though its gain is less than 1. [1 mark]

Question 5 [8 marks]



An n-channel JFET with $I_{DSS} = 10$ mA, $V_P = -4$ V and $y_{os} = 20\ \mu\text{S}$ is used in a **self-bias** common-source amplifier.

- $V_{DD} = 20$ V. $R_D = 2.2\ \text{k}\Omega$ from V_{DD} to the drain.
- $R_S = 680\ \Omega$ from the source to ground, bypassed by a capacitor C_S.
- $R_G = 1\ \text{M}\Omega$ from the gate to ground. The input is capacitively coupled to the gate. The output is taken from the drain through a capacitor (no load).

a. Write the equation for I_D by combining Shockley's equation with the bias line. Solve the quadratic, show **both** roots, and choose the correct one with a reason. State I_{DQ} and V_{GSQ}. [3 marks]

b. Find V_S, V_D and V_{DS}. Check that the JFET is in the saturation (pinch-off) region. [2 marks]

c. Find g_m, r_d, Z_i, Z_o and A_v. [2 marks]

d. C_S is removed. Taking r_d as very large, derive and evaluate the new A_v. [1 mark]

Question 6 [7 marks]

Question 6 circuit: E-MOSFET voltage-divider bias

An n-channel enhancement-type MOSFET has $V_{GS(Th)} = 3$ V and $I_{D(on)} = 5$ mA at $V_{GS(on)} = 8$ V. It is biased with a voltage divider:

- $V_{DD} = 24$ V. $R_1 = 10\ \text{M}\Omega$ from V_{DD} to the gate, and $R_2 = 6.8\ \text{M}\Omega$ from the gate to ground.
- $R_D = 2\ \text{k}\Omega$ from V_{DD} to the drain, and $R_S = 1\ \text{k}\Omega$ from the source to ground.

a. Find the device constant k. [1 mark]

b. Find V_G. Then find I_{DQ} and V_{GSQ} by solving the quadratic, and justify your choice of root. [3 marks]

c. Find V_{DS} and confirm that the MOSFET is in saturation. [2 marks]

d. Find g_m at the Q-point. If R_S were bypassed and r_d is very large, what would A_v be? [1 mark]

Question 7 [8 marks]

Question 7 circuit: JFET common-source stage RC-coupled to a BJT common-emitter stage

A two-stage RC-coupled amplifier runs from a single supply, $V_{DD} = V_{CC} = 15$ V.

- **Stage 1 (JFET common source):** n-channel JFET with $I_{DSS} = 8$ mA and $V_P = -4$ V (r_d may be ignored). $R_G = 1\ \text{M}\Omega$ from the gate to ground. $R_{S1} = 480\ \Omega$ from the source to ground, fully bypassed. $R_{D1} = 3.3\ \text{k}\Omega$ from the supply to the drain. A DC analysis gives $V_{GSQ} = -1.5$ V (you may use this value).
- **Coupling:** the drain of stage 1 is coupled by a capacitor to the base of stage 2.
- **Stage 2 (BJT common emitter):** NPN BJT with $\beta = 150$ (r_o may be ignored). $R_1 = 33\ \text{k}\Omega$ from the supply to the base, and $R_2 = 10\ \text{k}\Omega$ from the base to ground. $R_{C2} = 2.2\ \text{k}\Omega$ from the supply to the collector. $R_{E2} = 1.06\ \text{k}\Omega$ from the emitter to ground, fully bypassed. A DC analysis gives $I_{E2} = 2.5$ mA (you may use this value).
- The source has $R_{sig} = 10\ \text{k}\Omega$ and drives the gate of stage 1. The collector of stage 2 drives $R_L = 4.7\ \text{k}\Omega$ through a capacitor.

a. Find g_{m1}, r_{e2} and the input impedance of stage 2, Z_{i2}. [2 marks]

b. Find the loaded gain of each stage, A_{v1} and A_{v2}, and the total gain $A_{vT} = v_o/v_i$. [3 marks]

c. Find Z_i and Z_o of the whole amplifier, the overall gain $A_{v_s} = v_o/v_s$, and v_o for $v_s = 2$ mV peak. State the phase of v_o relative to v_s. [2 marks]

d. Stage 1 has a no-load gain of about -8, but its loaded gain is much smaller. Explain why in one sentence, and name one change that would fix it. [1 mark]

Question 8 [7 marks]

a. An op-amp has a differential gain $A_d = 4000$. The input voltages are $V_{i1} = 1.05$ mV and $V_{i2} = 0.95$ mV. Determine the output voltage when the CMRR is (i) 80 dB and (ii) 40 dB. [3 marks]

b. A BJT differential amplifier uses two matched NPN transistors with $\beta = 100$. Each collector connects to $+V_{CC} = +12$ V through $R_C = 10\ \text{k}\Omega$. The two emitters are joined and connected to $-V_{EE} = -12$ V through a single tail resistor $R_E = 15\ \text{k}\Omega$. Both bases are at 0 V DC (the input sources are ground-referenced). The output is taken single-ended from one collector.

Question 8(b) circuit: BJT differential pair with a 15 kOhm tail resistor

Find the DC emitter current of each transistor, r_e and V_C. Then find the single-ended differential gain A_d, the common-mode gain A_c, and the CMRR as a ratio and in dB. [4 marks]

Question 9 [5 marks]

Question 9 circuit: non-inverting amplifier feeding a two-input inverting summer

Two ideal op-amps are powered from ± 15 V.

- **Op-amp 1** is a non-inverting amplifier: $V_1 = 20$ mV rms is applied to its non-inverting input, $R_1 = 4.7\ \text{k}\Omega$ goes from the inverting input to ground, and $R_f = 47\ \text{k}\Omega$ goes from the output to the inverting input.
- **Op-amp 2** is an inverting summing amplifier with its non-inverting input grounded. The output of op-amp 1 connects to op-amp 2's inverting input through $33\ \text{k}\Omega$. A second signal, $V_2 = 50$ mV rms (in phase with V_1), connects to the same inverting input through $10\ \text{k}\Omega$. The feedback resistor of op-amp 2 is $100\ \text{k}\Omega$.

a. Find the output of op-amp 1. [2 marks]

b. Find the output V_o of op-amp 2, with its phase. [2 marks]

c. What input impedance does the source V_2 see? Explain in one line. [1 mark]

Question 10 [7 marks]

a. A series-fed class A amplifier has $V_{CC} = 18$ V, $R_B = 1.2\ \text{k}\Omega$ from V_{CC} to the base, and $R_C = 20\ \Omega$ from V_{CC} to the collector. The emitter is grounded. The transistor has $\beta = 40$. The input signal produces a sinusoidal base current of 8 mA **peak**.

Question 10(a) circuit: series-fed class A amplifier

Find I_{CQ} and V_{CEQ}, the AC output power, the DC input power and the efficiency. [4 marks]

Question 10(b) circuit: class B push-pull, V_CC = 30 V, 16 ohm load

b. A class B push-pull amplifier has $V_{CC} = 30$ V and drives a 16 Ω load. Determine (i) the maximum input power, (ii) the maximum output power and (iii) the maximum circuit efficiency. [3 marks]

END OF PAPER

EDC · MOCK

EDC mock B

Practice questions · Day 28 · Fri 20 Nov · pack EX2

Instructions

- Time allowed: **2 hours**, strictly timed. Sit it in one go, the day before the exam.
- Open book (hard copy notes, your reference sheet, textbooks). Calculator allowed. No internet.
- **11 questions, 80 marks. Answer all of them.** The marks for each part are in square brackets.
- Show your working. Write the formula first, then the numbers, then the answer with units. Most marks are for method.
- The paper is about 45 % diodes and BJTs (Q1 part, Q2–Q6) and about 55 % FETs, multistage, op-amp and power (Q1 part, Q7–Q11).
- Budget about **1.5 minutes per mark**. If a part has taken twice its budget, write down what you would do next and move on.
- Unless a question says otherwise: Si diodes and BJTs have $V_D = V_{BE} = 0.7\ \text{V}$, $V_{CE,\text{sat}} = 0.2\ \text{V}$, $r_e = 26\ \text{mV}/I_E$, capacitors are short circuits at signal frequency, and r_o is ignored for BJTs.
- Every circuit is described in words. Draw it yourself before you start each question. The drawing is part of the method.

Question	Topic	Marks	Time budget
1	Multiple choice (8 parts)	8	10 min
2	Semiconductor physics; diode r_d and temperature	7	10 min
3	Two-level clipper and biased clamper	5	8 min
4	Bridge rectifier with capacitor filter (design)	5	8 min
5	pnp voltage-divider bias, β sensitivity, saturation	8	12 min
6	Collector-feedback CE and common-base amplifiers	8	12 min
7	BJT current mirror	5	7 min
8	D-MOSFET with positive V_{GS}	8	12 min
9	JFET source follower and E-MOSFET drain-feedback amplifier	12	18 min
10	Two-stage JFET cascade	7	11 min
11	Power amplifiers: transformer-coupled class A and class B	7	10 min
	Total	**80**	**118 min**

Formula table

Semiconductors and diodes

Resistance, conductivity	$R = \dfrac{\rho L}{A}$, $\rho = \dfrac{1}{\sigma}$, $\sigma = q(n\mu_n + p\mu_p)$
Mass action	$n_0 p_0 = n_i^2$
Intrinsic concentration	$n_i = BT^{3/2} e^{-E_g/(2kT)}$; Si: $B = 7.3 \times 10^{15}\ \mathrm{cm^{-3}K^{-3/2}}$, $E_g = 1.12$ eV, $k = 8.62 \times 10^{-5}$ eV/K
Thermal voltage	$V_T = \dfrac{kT_K}{q}$, $k = 1.38 \times 10^{-23}$ J/K, $q = 1.6 \times 10^{-19}$ C
Shockley	$I_D = I_S(e^{V_D/nV_T} - 1)$
Diode small-signal resistance	$r_d = \dfrac{nV_T}{I_D}$
Si temperature rules	V_D falls about 2.5 mV/°C at fixed current; I_S doubles every 10 °C
Full-wave ripple (p-p)	$V_r = \dfrac{V_P}{2fCR} = \dfrac{I_L}{2fC}$, $I_L \approx \dfrac{V_P}{R}$, $V_{dc} = V_P - \frac{1}{2}V_r$
Conduction angle	$\omega \Delta t \approx \sqrt{2V_r/V_P}$
Full-wave diode currents	$i_{Dav} = I_L\left(1 + \pi\sqrt{\dfrac{V_P}{2V_r}}\right)$, $i_{Dmax} = I_L\left(1 + 2\pi\sqrt{\dfrac{V_P}{2V_r}}\right)$
Bridge PIV	$\mathrm{PIV} = V_{s,\mathrm{peak}} - V_D$

BJT

Currents	$I_C = \beta I_B$, $I_E = (\beta+1) I_B$, $\alpha = \dfrac{\beta}{\beta+1}$
Voltage-divider bias (exact)	$I_B = \dfrac{V_{Th} - V_{BE}}{R_{Th} + (\beta+1)R_E}$ (npn); pnp: $I_B = \dfrac{V_{CC} - V_{EB} - V_{Th}}{R_{Th} + (\beta+1)R_E}$
Collector-feedback bias	$I_B = \dfrac{V_{CC} - V_{BE}}{R_F + \beta R_C}$ (no R_E)
Collector feedback (AC)	$Z_i = \dfrac{r_e}{\frac{1}{\beta} + \frac{R_C}{R_C + R_F}}$, $Z_o \approx R_C \parallel R_F$, $A_v = -\dfrac{R_C \parallel R_F}{r_e}$
Common base	$Z_i = R_E \parallel r_e$, $Z_o = R_C$, $A_v = \dfrac{\alpha (R_C \parallel R_L)}{r_e}$
Source loading	$A_{v_s} = \dfrac{Z_i}{Z_i + R_s} A_{v_L}$
Current mirror	$I_o = \dfrac{I_{REF}}{1 + 2/\beta}$

FETs

JFET and D-MOSFET	$I_D = I_{DSS}\left(1 - \dfrac{V_{GS}}{V_P}\right)^2$
E-MOSFET	$I_D = k(V_{GS} - V_{GS(Th)})^2$, $g_m = 2k(V_{GSQ} - V_{GS(Th)})$
Transconductance	$g_{m0} = \dfrac{2I_{DSS}}{\lvert V_P \rvert}$, $g_m = g_{m0}\left(1 - \dfrac{V_{GSQ}}{V_P}\right) = g_{m0}\sqrt{\dfrac{I_{DQ}}{I_{DSS}}}$, $r_d = \dfrac{1}{y_{os}} = \dfrac{1}{g_{os}}$
CS (bypassed R_S)	$Z_i = R_G$ or $R_1 \parallel R_2$, $Z_o = r_d \parallel R_D$, $A_v = -g_m(r_d \parallel R_D)$
Source follower	$Z_i = R_G$, $Z_o = r_d \parallel R_S \parallel \dfrac{1}{g_m}$, $A_v = \dfrac{g_m(r_d \parallel R_S)}{1 + g_m(r_d \parallel R_S)}$
Drain feedback	$Z_i = \dfrac{R_F + r_d \parallel R_D}{1 + g_m(r_d \parallel R_D)}$, $Z_o = R_F \parallel r_d \parallel R_D$, $A_v = -g_m(R_F \parallel r_d \parallel R_D)$
Cascade	$A_{vT} = A_{v1}A_{v2}$, each stage loaded by the next stage's Z_i

Op-amp, digital and power

CMRR	$\mathrm{CMRR} = \dfrac{A_d}{A_c}$, $\mathrm{CMRR(dB)} = 20\log_{10}\dfrac{A_d}{A_c}$
Noise margins	$NM_L = V_{IL} - V_{OL}$, $NM_H = V_{OH} - V_{IH}$
Transformer	$a = \dfrac{N_1}{N_2}$, $R'_L = a^2 R_L$
Class A (transformer-coupled)	$P_i = V_{CC} I_{CQ}$; maximum ideal $P_o = \frac{1}{2} V_{CC} I_{CQ}$, $\eta_{max} = 50\%$
Class B push-pull	$P_i = \dfrac{2V_{CC}V_{L(p)}}{\pi R_L}$, $P_o = \dfrac{V_{L(p)}^2}{2R_L}$, $P_{2Q} = P_i - P_o$, $\eta_{max} = 78.5\%$

Question 1 [8 marks]

Circle the correct answer in each part (I to VIII). [1 mark each]

I. Silicon at 300 K ($n_i = 1.5 \times 10^{10}$ cm^{-3}) is doped with donors at $N_D = 10^{16}$ cm^{-3}. The hole concentration is about:

a. 1.5×10^{10} cm^{-3} b) 2.25×10^{4} cm^{-3} c) 10^{16} cm^{-3} d) zero

II. A bridge rectifier and a centre-tapped full-wave rectifier are each designed to give the same peak output V_m. Compared with the centre-tapped circuit, each diode in the bridge must withstand a peak inverse voltage of about:

a. twice as much b) the same c) half as much d) zero, because the bridge diodes never reverse-bias

III. A BJT is in saturation. Which statement is true?

a. Only the base–emitter junction is forward biased, and $I_C = \beta I_B$
b. Both junctions are forward biased, and $I_C < \beta I_B$
c. Both junctions are reverse biased, and $I_C \approx 0$
d. Only the collector–base junction is forward biased, and $I_C > \beta I_B$

IV. An n-channel depletion-type MOSFET with $I_{DSS} = 6$ mA and $V_P = -3$ V is biased at $V_{GS} = +1$ V, with V_{DS} large. Then:

a. the device is cut off
b. $I_D < I_{DSS}$ because the gate is positive

c. the device is in enhancement mode and $I_D > I_{DSS}$
d. the gate draws a large current because the gate junction is forward biased

V. A CMOS inverter running from 5 V has $V_{OH} = 4.8$ V, $V_{OL} = 0.2$ V, $V_{IL} = 2.1$ V and $V_{IH} = 2.9$ V. Its noise margins are:

a. $NM_L = 2.1$ V, $NM_H = 2.9$ V b) $NM_L = NM_H = 1.9$ V c) $NM_L = NM_H = 0.8$ V d) $NM_L = 4.6$ V, $NM_H = 0.8$ V

VI. A differential amplifier has differential gain $A_d = 4000$ and common-mode gain $A_c = 0.4$. Its CMRR is:

a. 40 dB b) 60 dB c) 80 dB d) 10 000 dB

VII. The main cause of crossover distortion in a class B push-pull output stage is that:

a. the output transformer saturates at high current
b. neither transistor conducts until its base–emitter voltage reaches about 0.7 V, so there is a dead band near zero
c. both transistors conduct for the full cycle
d. the supply voltage is too high

VIII. A basic two-transistor BJT current mirror uses matched transistors with $\beta = 50$. The ratio I_o/I_{REF} is about:

a. 0.50 b) 0.962 c) 0.980 d) 1.04

Question 2 [7 marks]

(a) A silicon bar is 2 mm long with a square cross-section of 0.1 mm × 0.1 mm. It is doped with boron at $N_A = 5 \times 10^{15}$ cm^{-3}. Use $\mu_n = 1350$ cm²/Vs, $\mu_p = 500$ cm²/Vs and, at 300 K, $n_i = 1.5 \times 10^{10}$ cm^{-3}.

1. Find the hole and electron concentrations at 300 K. [1]
2. Find the conductivity and the resistance of the bar between its two end faces at 300 K. [2]
3. Use the n_i formula in the table to find n_i at 200 °C. Is the bar still extrinsic (that is, is $p \approx N_A$ still a good assumption)? Justify with a number. [1]

Question 2(b) circuit: 5 V supply and 20 mV signal in series with 4.3 kOhm and a diode

(b) A silicon diode with $n = 1$ is connected in series with a resistor $R = 4.3$ kΩ across a DC supply $V_{DD} = 5$ V (anode towards the supply, cathode to ground). A small sinusoidal signal of 20 mV peak is connected in series with the 5 V supply. Use $V_D = 0.7$ V for the DC analysis at 27 °C.

1. At 27 °C, find the DC diode current, the small-signal resistance r_d, and the peak ac voltage across the diode. [2]
2. The temperature rises to 77 °C. Using the course temperature rule for V_D, find the new DC current, the new r_d and the new peak ac voltage across the diode. Explain in one sentence why r_d went up even though I_D went up. [1]

Question 3 [5 marks]

Use the constant-voltage diode model with $V_D = 0.7$ V.

Question 3(a) circuit: two-level parallel clipper with 4 V and 6 V batteries

(a) Two-level parallel clipper. The input $v_i = 12 \sin \omega t$ V is applied to one end of a 1 kΩ series resistor. The other end of the resistor is the output node v_o, which has no load. Two branches go from the output node to ground:

- Branch 1: diode D1 with its **anode at the output node** and its cathode at the **positive** terminal of a 4 V battery. The battery's negative terminal is grounded.
- Branch 2: diode D2 with its **cathode at the output node** and its anode at the **negative** terminal of a 6 V battery. The battery's positive terminal is grounded.

Find the two clipping levels and the phase angles (ωt in degrees) at which each clipping starts and stops. Describe the output waveform over one cycle in words, as you would label a sketch. [3]

Question 3(b) circuit: biased clamper with a 2 V battery

(b) Biased clamper. A ±8 V, 1 kHz square wave is applied to a 1 µF capacitor in series. The other plate of the capacitor is the output node. From the output node to ground there is a 100 kΩ resistor, and also a diode with its **anode at the output node** and its cathode at the **positive** terminal of a 2 V battery, whose negative terminal is grounded. Find the two output levels and the peak-to-peak output. Describe the output waveform in words. State why the 100 kΩ resistor does not upset the result. [2]

Question 4 [5 marks]

Question 4 circuit: bridge rectifier with capacitor filter and 150 ohm load

A bridge rectifier (four Si diodes, 0.7 V each) is fed from a transformer secondary of 12 V rms at 50 Hz. The rectifier output drives a filter capacitor C in parallel with a 150 Ω load.

1. Find the peak output voltage V_P. [1]
2. Find the minimum C that keeps the peak-to-peak ripple at or below 0.8 V. [1]
3. You fit the next standard value, 1500 µF. Find the actual ripple and the DC output voltage. [1]
4. Find the peak diode current i_{Dmax} (use $I_L \approx V_P/R$). [1]
5. Find the peak inverse voltage each diode must withstand. [1]

Question 5 [8 marks]

Question 5 circuit: pnp voltage-divider bias from +12 V

A **pnp** BJT is biased with a voltage divider from a single +12 V supply. The emitter connects through $R_E = 1.5$ kΩ up to the +12 V rail. The collector connects through $R_C = 2.7$ kΩ down to ground. $R_1 = 15$ kΩ runs from the +12 V rail to the base, and $R_2 = 47$ kΩ runs from the base to ground. Take $V_{EB} = 0.7$ V, $V_{EC,\text{sat}} = 0.2$ V and $\beta = 100$.

(a) Find the Thevenin equivalent seen from the base, then find I_C, V_C and V_{EC} using the exact (Thevenin) method. Confirm the transistor is active. [4]

(b) The transistor is replaced by others with $\beta = 50$ and $\beta = 200$. Find I_C for each and the percentage change from the $\beta = 100$ value. Comment on the stability of this bias. [2]

(c) With $\beta = 100$, R_C is changed to 8.2 kΩ. Show that the active assumption now fails, and find the actual collector current. [2]

Question 6 [8 marks]

Question 6(a) circuit: collector-feedback CE amplifier

(a) Collector feedback. An npn BJT has its emitter grounded. $R_C = 2.2$ kΩ runs from $V_{CC} = 10$ V to the collector, and $R_F = 180$ kΩ runs from the collector to the base. The input signal is capacitively coupled to the base and the output is taken from the collector through a coupling capacitor. $\beta = 150$. Find I_E, r_e, Z_i, Z_o and A_v (no load). [4]

Question 6(b) circuit: common-base amplifier with R_E to -4 V

(b) Common base. An npn BJT has its base grounded. The emitter connects through $R_E = 1.2$ kΩ to a -4 V supply. The collector connects through $R_C = 3.3$ kΩ to +12 V. A signal source v_s with internal resistance $R_s = 50\ \Omega$ is capacitively coupled to the emitter. The collector is capacitively coupled to a load $R_L = 6.8$ kΩ. $\alpha = 0.99$. Find I_E, r_e, Z_i, the loaded gain $A_{v_L} = v_o/v_i$ and the overall gain $A_{v_s} = v_o/v_s$. State the phase of v_o relative to v_s. [4]

Question 7 [5 marks]

Question 7 circuit: two-transistor current mirror with emitters on -9 V

A current mirror uses two matched npn transistors, Q1 and Q2, with $\beta = 100$. Both emitters connect to a -9 V rail. The bases are tied together and Q1 is diode-connected (its collector is tied to its base). A resistor R runs from ground (0 V) to Q1's collector. Q2's collector draws the output current I_o from ground through a load resistor R_L.

1. Find R so that $I_o = 2.00$ mA. [2]
2. Find the largest R_L for which Q2 stays active. [2]
3. With R kept at your value from part 1, the transistors are replaced by a matched pair with $\beta = 50$. Find the new I_o. [1]

Question 8 [8 marks]

Question 8 circuit: D-MOSFET voltage-divider common-source amplifier

An n-channel **depletion-type MOSFET** has $I_{DSS} = 8$ mA, $V_P = -4$ V and $y_{os} = 10$ µS. It is used in a voltage-divider common-source amplifier with $V_{DD} = 18$ V. $R_1 = 100$ MΩ runs from V_{DD} to the gate and $R_2 = 20$ MΩ from the gate to ground. $R_D = 1.0$ kΩ runs from V_{DD} to the drain. $R_S = 300\ \Omega$ runs from the source to ground and is fully bypassed by a capacitor C_S. The input is capacitively coupled to the gate and the output is taken from the drain through a coupling capacitor.

(a) Find V_G, then solve for I_{DQ} and V_{GSQ}. Say which root you reject and why. [3]

(b) Find V_D, V_S and V_{DS}, and confirm the device is in saturation (the pinch-off region). [1]

(c) Find g_{m0} and g_m. Explain in one sentence why $g_m > g_{m0}$ here. [1]

(d) Find Z_i, Z_o and A_v. Then find the gain when a 3.3 kΩ load is capacitively coupled to the drain. [3]

Question 9 [12 marks]

Question 9(a) circuit: JFET source follower

(a) JFET source follower. An n-channel JFET has $I_{DSS} = 10$ mA, $V_P = -4$ V and $y_{os} = 20$ µS. The drain connects directly to $V_{DD} = 15$ V. The gate connects to ground through $R_G = 1$

MΩ, and the input is capacitively coupled to the gate. The source connects to ground through $R_S = 2$ kΩ, and the output is taken from the source through a coupling capacitor.

1. Find I_{DQ} and V_{GSQ}. [2]
2. Find g_m. [1]
3. Find Z_i and Z_o. [1]
4. Find A_v with no load, and with a 1 kΩ load coupled to the source. [1]
5. With the 1 kΩ load and $v_i = 0.5$ V peak, find the peak output voltage and the peak v_{gs}. [1]

Question 9(b) circuit: E-MOSFET drain-feedback amplifier

(b) E-MOSFET drain-feedback amplifier. An n-channel enhancement MOSFET has $k = 0.5$ mA/V², $V_{GS(Th)} = 2$ V and $r_d = 50$ kΩ. The source is grounded. $R_D = 2$ kΩ runs from $V_{DD} = 12$ V to the drain, and $R_F = 10$ MΩ runs from the drain to the gate. The input is capacitively coupled to the gate and the output is taken from the drain through a coupling capacitor.

1. Find V_{GSQ}, I_{DQ} and V_{DSQ}. Say why this device is guaranteed to be in saturation. [2]
2. Find g_m. [1]
3. Find Z_i, Z_o and A_v. [3]

Question 10 [7 marks]

Question 10 circuit: two identical self-bias JFET stages, RC coupled

A two-stage RC-coupled amplifier uses two identical n-channel JFETs with $I_{DSS} = 8$ mA, $V_P = -4.5$ V and $r_d = 40$ kΩ. Each stage is an identical self-bias common-source stage on $V_{DD} = 18$ V: $R_G = 2.2$ MΩ from gate to ground, $R_S = 1$ kΩ from source to ground (fully bypassed), and $R_D = 3.3$ kΩ from V_{DD} to drain. The drain of stage 1 is capacitively coupled to the gate of stage 2. The drain of stage 2 is capacitively coupled to a load $R_L = 10$ kΩ. The input is a source of 20 mV rms with internal resistance $R_{sig} = 10$ kΩ, coupled to the gate of stage 1.

1. Find I_{DQ}, V_{GSQ} and V_{DSQ} (the same for both stages). [2]
2. Find g_m. [1]
3. Find the loaded gain of each stage, A_{v1} and A_{v2}. [2]
4. Find the overall gain v_o/v_{sig} and the rms output voltage, plus the amplifier's Z_i and Z_o. [2]

Question 11 [7 marks]

Question 11(a) circuit: transformer-coupled class A into two 8 ohm speakers in parallel

(a) A transformer-coupled class A amplifier runs from $V_{CC} = 12$ V with a quiescent collector current $I_{CQ} = 150$ mA. The load is two 8 Ω loudspeakers in parallel on the transformer secondary. For the largest possible symmetric collector swing, the reflected load must be $R'_L = V_{CC}/I_{CQ}$.

1. Find the required turns ratio $N_1 : N_2$. [2]
2. Find the maximum ideal output power and the efficiency at that point. [1]

Question 11(b) circuit: class B push-pull, V_CC = 20 V, 8 ohm load

(b) A class B push-pull amplifier runs from $V_{CC} = 20$ V (each transistor sees the full 20 V) and drives an 8 Ω load directly.

1. Find the maximum output power and the efficiency at that output. [2]
2. Find the maximum total transistor dissipation, the dissipation in each transistor, and the peak load voltage $V_{L(p)}$ at which it occurs. What is the efficiency at that point? [2]

End of paper.